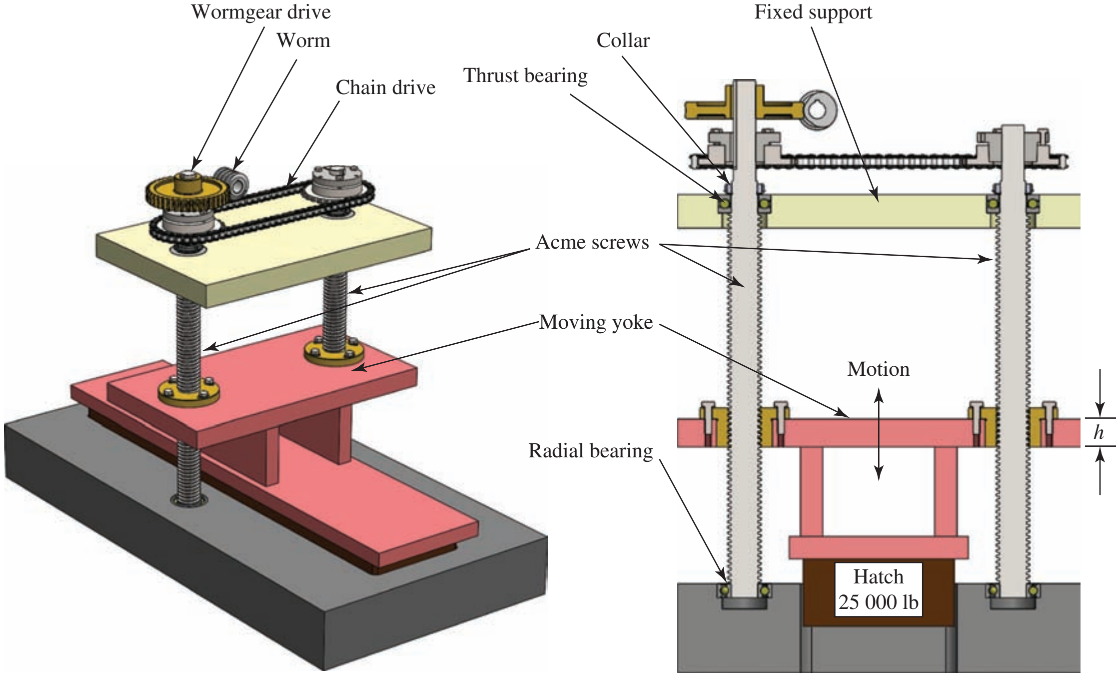


(*a*) Pictorial view (*b*) Section view through the center of the Acme screws

FIGURE 17–4 An Acme screw-driven system for raising a hatch

size should be used? The sketch suggests the Acme thread form. What other styles are available? At what speed must the screws be rotated to raise the hatch in 12.0 s or less? How much power is required to drive the screws? What safety concerns exist while the system is handling this heavy load? What advantage would there be to using a ball screw rather than a power screw?

The material in this chapter will help you make these decisions, along with providing methods of computing stresses, torques, and efficiencies. ■

17–1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Describe the operation of a power screw and the general form of *square threads, Acme threads,* and *buttress threads* as they are applied to power screws.
2. Compute the torque that must be applied to a power screw to raise or lower a load.
3. Compute the efficiency of power screws.
4. Compute the power required to drive a power screw.
5. Describe the design of a ball screw and its mating nut.
6. Specify suitable ball screws for a given set of requirements of load, speed, and life.
7. Compute the torque required to drive a ball screw, and compute its efficiency.

17–2 POWER SCREWS

Figure 17–2 shows three types of power screw threads: the square thread, the Acme thread, and the buttress thread. Of these, the square and buttress threads are the most efficient. That is, they require the least torque to move a given load along the screw. However, the Acme thread is not greatly less efficient, and it is easier to machine. The buttress thread is desirable when force is to be transmitted in only one direction.

Table 17–1 gives the preferred combinations of basic major diameter, D, and number of threads per inch, n, for Acme screw threads in U.S. units. The pitch, p, is the distance from a point on one thread to the corresponding point on the adjacent thread, and $p = 1/n$. More data for power screws can be found in References 1 and 4.

Other pertinent dimensions listed in Table 17–1 include the minimum minor diameter and the minimum pitch diameter of a screw with an external thread. When you are performing stress analyses on the screw, the safest approach is to compute the area corresponding to the minor diameter for tensile or compressive stresses. However, a more accurate stress computation results from using the *tensile stress area* (listed in Table 17–1), found from

➭ Tensile Stress Area for Screw Threads

$$A_t = \frac{\pi}{4}\left[\frac{D_r + D_p}{2}\right]^2 \qquad (17\text{–}1)$$

This is the area corresponding to the average of the minor (or root) diameter, D_r, and the pitch diameter, D_p.

TABLE 17–1 Preferred Acme Screw Threads

Nominal major diameter, D (in)	Threads per in, n	Pitch, $p = 1/n$ (in)	Minimum minor diameter, D_r (in)	Minimum pitch diameter, D_p (in)	Tensile stress area, A_t (in^2)	Shear stress area, A_s (in^2)[a]
1/4	16	0.0625	0.1618	0.2043	0.026 32	0.3355
5/16	14	0.0714	0.2140	0.2614	0.044 38	0.4344
3/8	12	0.0833	0.2632	0.3161	0.065 89	0.5276
7/16	12	0.0833	0.3253	0.3783	0.097 20	0.6396
1/2	10	0.1000	0.3594	0.4306	0.1225	0.7278
5/8	8	0.1250	0.4570	0.5408	0.1955	0.9180
3/4	6	0.1667	0.5371	0.6424	0.2732	1.084
7/8	6	0.1667	0.6615	0.7663	0.4003	1.313
1	5	0.2000	0.7509	0.8726	0.5175	1.493
$1\frac{1}{8}$	5	0.2000	0.8753	0.9967	0.6881	1.722
$1\frac{1}{4}$	5	0.2000	0.9998	1.1210	0.8831	1.952
$1\frac{3}{8}$	4	0.2500	1.0719	1.2188	1.030	2.110
$1\frac{1}{2}$	4	0.2500	1.1965	1.3429	1.266	2.341
$1\frac{3}{4}$	4	0.2500	1.4456	1.5916	1.811	2.803
2	4	0.2500	1.6948	1.8402	2.454	3.262
$2\frac{1}{4}$	3	0.3333	1.8572	2.0450	2.982	3.610
$2\frac{1}{2}$	3	0.3333	2.1065	2.2939	3.802	4.075
$2\frac{3}{4}$	3	0.3333	2.3558	2.5427	4.711	4.538
3	2	0.5000	2.4326	2.7044	5.181	4.757
$3\frac{1}{2}$	2	0.5000	2.9314	3.2026	7.388	5.700
4	2	0.5000	3.4302	3.7008	9.985	6.640
$4\frac{1}{2}$	2	0.5000	3.9291	4.1991	12.972	7.577
5	2	0.5000	4.4281	4.6973	16.351	8.511

[a]Per inch of length of engagement.

The data reflect the minimums for commercially available screws.

Another failure mode for a power screw is the shearing of the threads in the axial direction, cutting them away from the main shaft near the pitch diameter. The shear stress is computed from the direct stress formula,

$$\tau = F/A_s$$

The shear stress area, A_s, listed in Table 17–1 is also found in published data and represents the area in shear approximately at the pitch line of the threads for a 1.0-in length of engagement. Other lengths would require that the area be modified by the ratio of the actual length to 1.0 in.

Metric Power Screws

Metric power screws use a trapezoidal thread system that is similar to the Acme thread, but with slightly different thread form and made to metric dimensions. One notable difference between the Acme and ISO trapezoidal metric thread is that the angle $\phi = 15°$ instead of $14\,^1/_2°$ shown for the Acme design in Figure 17–2(b). Table 17–1M shows a selected list of sizes from the larger list of ISO trapezoidal screws. Each of these is a single-thread design. Many more diameters and pitches are available, along with many other styles with two or more threads that produce longer leads that are multiples of the pitch.

Torque Required to Move a Load

When using a power screw to exert a force, as with a jack raising a load, you need to know how much torque must be applied to the nut of the screw to move the load. The parameters involved are the force to be moved, F; the size of the screw, as indicated by its pitch diameter, D_p; the lead of the screw, L; and the coefficient of friction, f. Note that the *lead* is defined as the axial distance that the screw would move in one complete revolution. For the usual case of a single-threaded screw, the lead is equal to the pitch and can be read from Table 17–1 or computed from $L = p = 1/n$ for U.S. Acme Designs.

TABLE 17–1M Examples of Power Screws with Metric Trapezoidal Screw Thread

ISO thread system—External threads

Major diameter, D (mm)	Pitch, p (mm)	Pitch diameter, D_p (mm)	Minor diameter, D_r (mm)	Tensile stress area (mm^2)
8	1.5	7.25	6.2	35.52
10	2	9.0	7.5	53.46
12	3	10.5	8.5	70.88
14	3	12.5	10.5	103.9
16	3	14.5	12.5	143.1
20	4	18.0	15.5	220.4
22	5	19.5	16.5	254.5
24	5	21.5	18.5	314.2
28	5	25.5	22.5	452.4
30	6	27.0	23.0	490.9
32	6	29.0	33.0	754.8
36	6	33.0	29.0	754.8
40	7	36.5	32.0	921.3
46	8	42.0	37.0	1225
50	8	46.0	41.0	1486
55	9	50.5	45.0	1791
60	9	55.5	50.0	2185
70	10	65.0	59.0	3019
80	10	75.0	69.0	4072
90	12	84.0	77.0	5090
100	12	94.0	87.0	6433
120	14	113.0	104.0	9246
125	14	122.0	109.0	10 477

In the development of Equation (17–2) for the torque required to turn the screw, Figure 17–5(a), which depicts a load being pushed up an inclined plane against a friction force, is used. This is a reasonable representation for a square thread if you think of the thread as being unwrapped from the screw and laid flat. The torque for an Acme thread is slightly different from this due to the thread angle. The revised equation for the Acme thread will be shown later.

The torque computed from Equation (17–2) is called T_u, implying that the force is applied to move a load up the plane, that is, to raise the load. This observation is

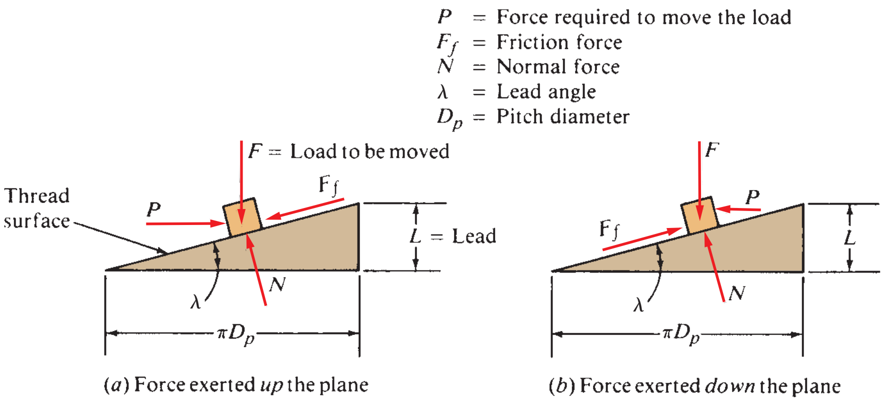


FIGURE 17–5 Screw thread force analysis

completely appropriate if the load is raised vertically, as with a jack. If, however, the load is horizontal or at some angle, Equation (17–2) is still valid if the load is to be advanced along the screw "up the thread." Equation (17–4) shows the required torque, T_d, to lower a load or move a load "down the thread."

The torque to move a load up the thread is

➪ Torque Required to Move a Load Up a Power Screw with a Square Thread

$$T_u = \frac{FD_p}{2}\left[\frac{L + \pi f D_p}{\pi D_p - fL}\right] \tag{17–2}$$

This equation accounts for the force required to overcome friction between the screw and the nut in addition to the force required just to move the load. If the screw or the nut bears against a stationary surface while rotating, there will be an additional friction torque developed at that surface. For this reason, many jacks and similar devices incorporate antifriction bearings at such points.

The coefficient of friction for use in Equation (17–2) depends on the materials used and the manner of lubricating the screw. For well-lubricated steel screws acting in steel nuts, $f = 0.15$ should be conservative.

An important factor in the analysis for torque is the angle of inclination of the plane. In a screw thread, the angle of inclination is referred to as the *lead angle*, λ. It is the angle between the tangent to the helix of the thread and the plane transverse to the axis of the screw. It can be seen from Figure 17–5 that

$$\tan\lambda = L/(\pi D_p) \tag{17–3}$$

where πD_p = circumference of the pitch line of the screw.

Then if the rotation of the screw tends to raise the load (move it up the incline), the friction force opposes the motion and acts down the plane.

Conversely, if the rotation of the screw tends to lower the load, the friction force will act up the plane, as shown in Figure 17–5(b). The torque analysis changes, producing Equation (17–4):

➪ Torque Required to Lower a Load Down a Power Screw with a Square Thread

$$T_d = \frac{FD_p}{2}\left[\frac{\pi f D_p - L}{\pi D_p + fL}\right] \tag{17–4}$$

If the screw thread is steep (i.e., if it has a high lead angle), the friction force may not be able to overcome the tendency for the load to "slide" down the plane, and the load will fall due to gravity. In most cases for power screws with single threads, however, the lead angle is rather small, and the friction force is large enough to oppose the load and keep it from sliding down the plane. Such a screw is called *self-locking*, a desirable characteristic for jacks and similar devices. Quantitatively, the condition that must be met for self-locking is

$$f > \tan\lambda \tag{17–5}$$

The coefficient of friction must be greater than the tangent of the lead angle. For $f = 0.15$, the corresponding value of the lead angle is 8.5°. For $f = 0.1$, for very smooth, well-lubricated surfaces, the lead angle for self-locking is 5.7°. The lead angles for the screw designs listed in Table 17–1 range from 1.94° to 5.57°. Thus, it is expected that all would be self-locking. However, operation under vibration should be avoided, as this may still cause movement of the screw.

Efficiency of a Power Screw

Efficiency for the transmission of a force by a power screw can be expressed as the ratio of the torque required to move the load without friction to that with friction. Equation (17–2) gives the torque required with friction, T_u. Letting $f = 0$, the torque required without friction, T', is

$$T' = \frac{FD_p}{2}\frac{L}{\pi D_p} = \frac{FL}{2\pi} \tag{17–6}$$

Then the efficiency, e, is

➪ Efficiency of a Power Screw

$$e = \frac{T'}{T_u} = \frac{FL}{2\pi T_u} \tag{17–7}$$

Alternate Forms of the Torque Equations

Equations (17–2) and (17–4) can be expressed in terms of the lead angle, rather than the lead and the pitch diameter, by noting the relationship in Equation (17–3). With this substitution, the torque required to move the load would be

➪ Torque to Raise a Load with a Square Thread

$$T_u = \frac{FD_p}{2}\left[\frac{(\tan\lambda + f)}{(1 - f\tan\lambda)}\right] \tag{17–8}$$

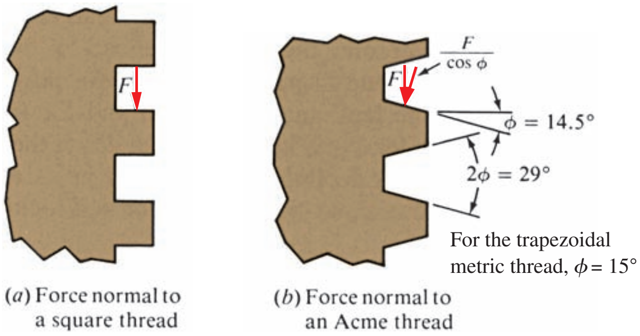
and the torque required to lower the load is

➪ Torque to Lower a Load with a Square Thread

$$T_d = \frac{FD_p}{2}\left[\frac{(f - \tan\lambda)}{(1 + f\tan\lambda)}\right] \tag{17–9}$$

Adjustment for Acme Threads and Trapezoidal Metric Threads

The difference between Acme threads and square threads is the presence of the thread angle, ϕ. Note from Figure 17–1 that $2\phi = 29°$, and therefore $\phi = 14.5°$. For the trapezoidal metric thread, $\phi = 15°$. This changes the direction of action of the forces on the thread from



(*a*) Force normal to a square thread (*b*) Force normal to an Acme thread

FIGURE 17–6 Force on an Acme thread

that depicted in Figure 17–5. Figure 17–6 shows that F would have to be replaced by $F/\cos\phi$. Carrying this through, the analysis for torque would give the following modified forms of Equations (17–8) and (17–9). The torque to move the load up the thread is

➪ Torque to Raise a Load with an Acme Thread

$$T_u = \frac{FD_p}{2}\left[\frac{(\cos\phi\tan\lambda + f)}{(\cos\phi - f\tan\lambda)}\right] \qquad (17\text{–}10)$$

and the torque to move the load down the thread is

➪ Torque to Lower a Load with an Acme Thread

$$T_d = \frac{FD_p}{2}\left[\frac{(f - \cos\phi\tan\lambda)}{(\cos\phi + f\tan\lambda)}\right] \qquad (17\text{–}11)$$

Power Required to Drive a Power Screw

If the torque required to rotate the screw is applied at a constant rotational speed, n, then the power in horsepower to drive the screw is

$$P = \frac{Tn}{63\,000}$$

Example Problem 17–1

Two Acme-threaded power screws are to be used to raise a heavy access hatch, as sketched in Figure 17–4. The total weight of the hatch is 25 000 lb, divided equally between the two screws. Select a satisfactory screw from Table 17–1 on the basis of tensile strength, limiting the tensile stress to 10 000 psi. Then determine the required thickness of the yoke that acts as the nut on the screw to limit the shear stress in the threads to 5000 psi. For the screw thus designed, compute the lead angle, the torque required to raise the load, the efficiency of the screw, and the torque required to lower the load. Use a coefficient of friction of 0.15.

Solution

The load to be lifted places each screw in direct tension. Therefore, the required tensile stress area is

$$A_t = \frac{F}{\sigma_d} = \frac{12\,500\text{ lb}}{10\,000\text{ lb/in}^2} = 1.25\text{ in}^2$$

From Table 17–1, a $1\frac{1}{2}$-in-diameter Acme thread screw with four threads per inch would provide a tensile stress area of 1.266 in^2.

For this screw, each inch of length of a nut would provide 2.341 in^2 of shear stress area in the threads. The required shear area is then

$$A_s = \frac{F}{\tau_d} = \frac{12\,500\text{ lb}}{5000\text{ lb/in}^2} = 2.50\text{ in}^2$$

Then the required length of the yoke would be

$$h = 2.5\text{ in}^2\left[\frac{1.0\text{ in}}{2.341\text{ in}^2}\right] = 1.07\text{ in}$$

For convenience, let's specify $h = 1.25$ in.

The lead angle is (remember that $L = p = 1/n = 1/4 = 0.250$ in)

$$\lambda = \tan^{-1}\frac{L}{\pi D_p} = \tan^{-1}\frac{0.250}{\pi(1.3429)} = 3.39°$$

The torque required to raise the load can be computed from Equation (17–10):

$$T_u = \frac{FD_p}{2}\left[\frac{(\cos\phi\tan\lambda + f)}{(\cos\phi - f\tan\lambda)}\right] \qquad (17\text{–}10)$$

Using $\cos\phi = \cos(14.5°) = 0.968$, and $\tan\lambda = \tan(3.39°) = 0.0592$, we have

$$T_u = \frac{(12\,500\text{ lb})(1.3429\text{ in})}{2}\,\frac{[(0.968)(0.0592) + 0.15]}{[0.968 - (0.15)(0.0592)]} = 1809\text{ lb}\cdot\text{in}$$

The efficiency can be computed from Equation (17–7):

$$e = \frac{FL}{2\pi T_u} = \frac{(12\,500\text{ lb})(0.250\text{ in})}{2(\pi)(1809\text{ lb}\cdot\text{in})} = 0.275\text{ or }27.5\%$$

The torque required to lower the load can be computed from Equation (17–11):

$$T_d = \frac{FD_p}{2}\left[\frac{(f - \cos\phi\tan\lambda)}{(\cos\phi + f\tan\lambda)}\right] \tag{17–11}$$

$$T_d = \frac{(12\,500\text{ lb})(1.3429\text{ in})}{2}\,\frac{[0.15 - (0.968)(0.0592)]}{[0.968 + (0.15)(0.0592)]} = 796\text{ lb}\cdot\text{in}$$

Example Problem 17–2 It is desired to raise the hatch in Figure 17–4 a total of 15.0 in in no more than 12.0 s. Compute the required rotational speed for the screws and the power required.

Solution The screw selected in the solution for Example Problem 17–1 was a $1\frac{1}{2}$-in Acme threaded screw with four threads per inch. Thus, the load would be moved 1/4 in with each revolution. The linear speed required is

$$V = \frac{15.0\text{ in}}{12.0\text{ s}} = 1.25\text{ in/s}$$

The required rotational speed would be

$$n = \frac{1.25\text{ in}}{\text{s}}\cdot\frac{1\text{ rev}}{0.25\text{ in}}\cdot\frac{60\text{ s}}{\text{min}} = 300\text{ rpm}$$

Then the power required to drive each screw would be

$$P = \frac{Tn}{63\,000} = \frac{(1809\text{ lb}\cdot\text{in})(300\text{ rpm})}{63\,000} = 8.61\text{ hp}$$

Multiple Start Thread Forms for Power Screws

The relatively low efficiency of standard single-thread Acme screws (approximately 30% or less) can be a strong disadvantage. Higher efficiencies can be achieved using high lead, multiple thread designs. The higher lead angle produces efficiencies in the 30% to 70% range. It should be understood that some mechanical advantage is lost so that higher torques are required to move a particular load as compared with single-thread screws. (See Internet sites 1, 6, 7, 10, 12, and 13.)

17–3 BALL SCREWS

The basic action of using screws to produce linear motion from rotation was described in Section 17–2 on power screws. A special adaptation of this action that minimizes the friction between the screw threads and the mating nut is the *ball screw*.

Figure 17–3 shows a cutaway view of a commercially available ball screw. It replaces the sliding friction of the conventional power screw with the rolling friction of bearing balls. The bearing balls circulate in hardened steel races formed by concave helical grooves in the screw and the nut. All reactive loads between the screw and the nut are carried by the bearing balls that provide the only physical contact between these members. As the screw and the nut rotate relative to each other, the bearing balls are diverted from one end and are carried by the ball-guide return tubes to the opposite end of the ball nut. This recirculation permits unrestricted travel of the nut in relation to the screw. (See Internet sites 1–10 and 14.)

Applications of ball screws occur in automotive steering systems, machine tool tables, linear actuators, jacking and positioning mechanisms, aircraft controls such as flap actuating devices, packaging equipment, and instruments. Figure 17–7 shows a linear actuator employing a ball screw driven by an electric motor

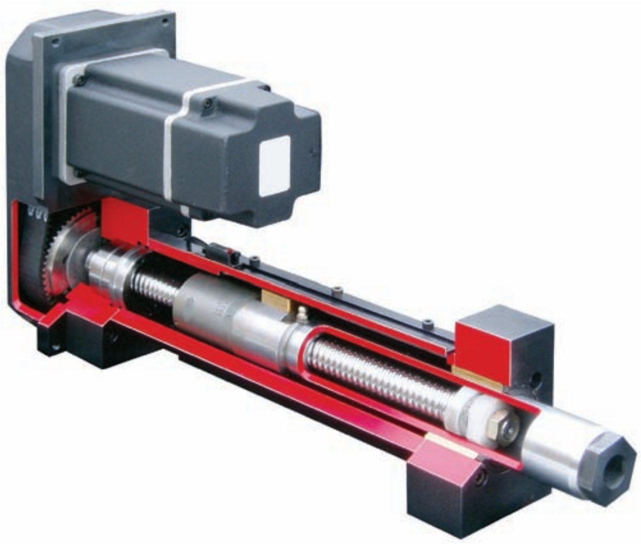


FIGURE 17–7 Application of a ball screw in a high-speed linear actuator (Joyce/Dayton Corp. and EDRIVE Actuators, Inc.); (see Internet sites 6 and 15)

through a synchronous belt drive. Rapid linear motion results with actuation forces up to 25 000 lb available. See Internet site 6.

The application parameters to be considered in selecting a ball screw include the following:

- Axial load to be exerted by the screw during rotation
- Rotational speed of the screw
- Maximum static load on the screw
- Direction of the load
- Manner of supporting the ends of the screw
- Length of the screw
- Expected life
- Environmental conditions

Load–Life Relationship

When transmitting a load, a ball screw experiences stresses similar to those on a ball bearing, as discussed in Chapter 14. The load is transferred from the screw to the balls, from the balls to the nut, and from the nut to the driven device. The contact stress between the balls and the races in which they roll eventually causes fatigue failure, indicated by pitting of the balls or the races.

Thus, the rating of ball screws gives the load capacity of the screw for a given life that 90% of the screws of a given design will survive. This is similar to the L_{10} life of ball bearings. Some manufacturers of ball screws use the term B_{10} life. Because ball screws are typically used as linear actuators, the most pertinent life parameter is the distance traveled by the nut relative to the screw.

Manufacturers usually report the rated load that a given screw can exert for 1 million in (25.4 km) of cumulative travel. The relationship between load, P, and life, L, is also similar to that for a ball bearing:

➪ Relationship between Bearing Load and Life

$$\frac{L_2}{L_1} = \left(\frac{P_1}{P_2}\right)^3 \qquad \textbf{(17–12)}$$

Thus, if the load on a ball screw is doubled, the life is reduced to one-eighth of the original life. If the load is cut

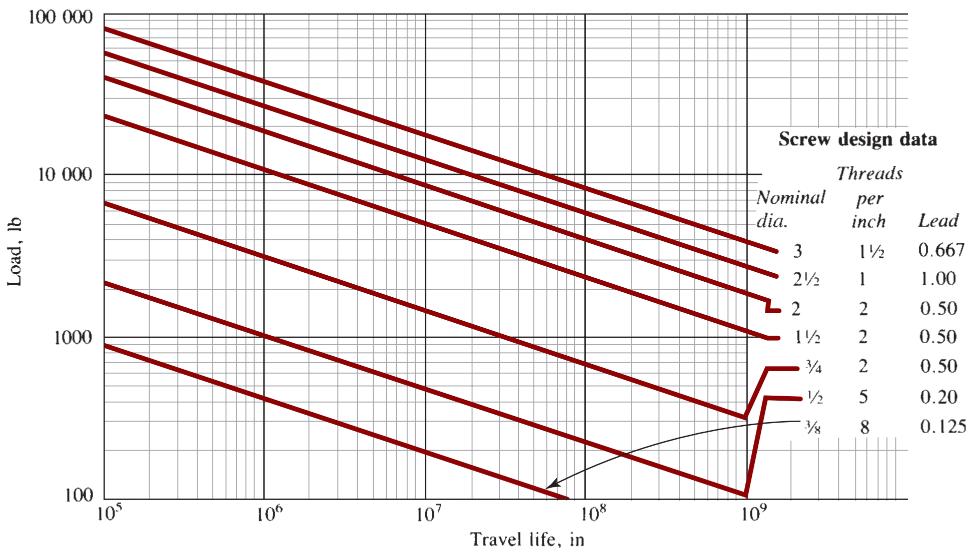


FIGURE 17–8 Ball screw performance

in half, the life is increased by eight times. Figure 17–8 shows the nominal load/life performance of ball screws of a small variety of sizes. Many more sizes and styles are available.

As an example of the use of Figure 17–8, consider the screw with a nominal diameter of 1/2 in and five threads per inch; the second line from the bottom of the chart. At 1.0 million in of rated travel life, the rated load is 1000 lb. If the actual applied load is 400 lb, the rated travel life would be approximately 2.0×10^7 in.

When a variety of load levels are experienced by a ball screw, methods for computing an equivalent load are available. Refer to the manufacturer's literature for the load/life data for the specific ball screw used. See Internet site 6 for an example.

Torque and Efficiency

The efficiency of a ball bearing screw is typically taken to be 90%. This far exceeds the efficiency for power screws without rolling contact that are typically in the range of 20% to 30%. Thus, far less torque is required to exert a given load with a given size of screw. Power is correspondingly reduced. The computation of the torque to turn a ball screw is adapted from Equation (17–7), relating efficiency to torque:

➩ Efficiency of a Ball Screw

$$e = \frac{FL}{2\pi T_u} \qquad \textbf{(17–7)}$$

Then, using $e = 0.90$,

➩ Torque to Drive a Ball Screw

$$T_u = \frac{FL}{2\pi e} = 0.177FL \qquad \textbf{(17–13)}$$

Because of the low friction, ball screws are virtually never self-locking. In fact, designers also use this property to advantage by purposely using the applied load on the nut to rotate the screw. This is called *backdriving*; backdriving torque can be computed from

➩ Backdriving Torque for a Ball Screw

$$T_b = \frac{FLe}{2\pi} = 0.143FL \qquad \textbf{(17–14)}$$

Example Problem 17–3

Select suitable ball screws for the application described in Example Problems 17–1 and 17–2 and illustrated in Figure 17–4. The hatch must be lifted 15.0 in to open it eight times per day, and then it must be closed. The design life is 10 years. The lifting or lowering is to be completed in no more than 12.0 s.

For the screw selected, compute the torque to turn the screw, the power required, and the actual expected life.

Solution

The data required to select a screw from Figure 17–8 are the load and the travel of the nut on the screw over the desired life. The load is 12 500 lb on each screw:

$$\text{Travel} = \frac{15.0\text{ in}}{\text{stroke}} \cdot \frac{2\text{ strokes}}{\text{cycle}} \cdot \frac{8\text{ cycles}}{\text{day}} \cdot \frac{365\text{ days}}{\text{year}} \cdot \frac{10\text{ years}}{} = 8.76 \times 10^5\text{ in}$$

From Figure 17–8, the 2-in screw with two threads per inch and a lead of 0.50 in is satisfactory.

The torque required to turn the screw is

$$T_u = 0.177FL = 0.177(12\,500\text{ lb})(0.50\text{ in}) = 1106\text{ lb}\cdot\text{in}$$

The rotational speed required is

$$n = \frac{1\text{ rev}}{0.50\text{ in}} \cdot \frac{15.0\text{ in}}{12.0\text{ s}} \cdot \frac{60\text{ s}}{\text{min}} = 150\text{ rpm}$$

The power required for each screw is

$$P = \frac{Tn}{63\,000} = \frac{(1106\text{ lb}\cdot\text{in})(150\text{ rpm})}{63\,000} = 2.63\text{ hp}$$

Compare this with the 8.61 hp required for the Acme screw in Example Problem 17–2.

The actual travel life expected for this screw at a load of 12 500 lb would be approximately 3.2×10^6 in, using Figure 17–8. This is 3.65 times longer than required.

17–4 APPLICATION CONSIDERATIONS FOR POWER SCREWS AND BALL SCREWS

This section discusses additional application considerations that apply to both power screws and ball screws. Details may change according to the specific geometry and manufacturing processes. Supplier data should be consulted.

Critical Speed

The proper application of ball screws must take into account their vibration tendencies, particularly when operating at relatively high speeds. Long slender screws may exhibit the phenomenon of *critical speed*, n_c, at which the screw would tend to vibrate or whirl about its axis, possibly reaching dangerous amplitudes. Therefore, it is recommended that the operating speed of the screw be below 0.80 times the critical speed. An estimate for the critical speed, offered by Roton Products, Inc. (Internet site 10), is

$$n_c = \frac{4.76 \times 10^6\, dK_s}{(SF)L^2} \qquad \textbf{(17–15)}$$

where,

d = minor diameter of the screw (in)
K_s = end fixity factor
L = length between supports (in)
SF = safety factor

The end fixity factor, K_s, depends on the manner of supporting the ends of the screw with the following possibilities:

1. Simply supported at each end by one bearing: $K_s = 1.00$
2. Fixed at each end by two bearings that prevent rotation at the support: $K_s = 2.24$
3. Fixed at one end and simply supported at the other: $K_s = 1.55$
4. Fixed at one end and free at the other: $K_s = 0.32$

The value of the safety factor is a design decision, often taken to be in the range from 1.25 to 3.0. Note that the screw length is squared in the denominator, indicating that a relatively long screw would have a low critical speed. The best designs would employ a short length, rigid fixed supports, and large diameter. See References 2 and 3 for additional information on critical speeds.

Column Buckling

Ball screws that carry axial compressive loads must be checked for column buckling. The parameters, similar to those discussed in Chapter 6, are the material from which the screw is made, the end fixity, the diameter, and the length. Long screws should be analyzed using the Euler formula Equation (6–4) or (6–6), while the J. B. Johnson formula Equation (6–9) is used for shorter screws. End fixity depends on the rigidity of supports similar to that described above for critical speed. However, the factors are different for column loading.

1. Simply supported at each end by one bearing: $K_s = 1.00$
2. Fixed at each end by two bearings that prevent rotation at the support: $K_s = 4.00$
3. Fixed at one end and simply supported at the other: $K_s = 2.00$
4. Fixed at one end and free at the other: $K_s = 0.25$

Suppliers of commercially available ball screws include data for allowable compressive load in their catalogs. (See Internet sites 1–10 and 14.)

Material for Screws

Ball screws are typically made from carbon or alloy steels using thread-rolling technology. After the threads are formed, induction heating improves the hardness and strength of the surfaces on which the circulating balls roll for wear resistance and long life. Many ball screw nuts are made from alloy steel that is case hardened by carburizing.

Power screws are typically produced from carbon or alloy steels such as SAE 1018, 1045, 1060, 4130, 4140, 4340, 4620, 6150, and 8620. For corrosive environments or where high temperatures are experienced, stainless steels are used, such as SAE 304, 305, 316, 384, 430, 431, or 440. Some are made from aluminum alloys 1100, 2014, or 3003.

Power screw nuts are made from steels for moderate loads and when operating at relatively low speeds. Grease lubrication is recommended. Higher speeds and loads call for lubricated bronze nuts that have superior wear performance. Applications requiring lighter loads can use plastic nuts that have inherently good lubricity without external lubrication. Examples of such applications are food processing equipment, medical devices, and clean manufacturing operations.

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2. Budynas, R. G., and K. J. Nisbett. *Shigley's Mechanical Engineering Design*. 10th ed. New York: McGraw-Hill, 2015.
3. Juvinall, R. C., and K. M. Marshek. *Fundamentals of Machine Component Design*. 5th ed. New York: John Wiley & Sons, 2011.
4. Oberg, E. et al. *Machinery's Handbook*. 30th ed. New York: Industrial Press, 2015.

INTERNET SITES FOR LINEAR MOTION ELEMENTS

1. **PowerTransmission.com.** Online listing of numerous manufacturers and suppliers of linear motion devices, including ball screws, lead (power) screws, linear actuators, screw jacks, and linear slides.
2. **Ball-screws.net.** Site lists numerous manufacturers of ball screws, some with online catalogs.
3. **Thomson Industries, Inc.** Manufacturer of ball screws, ball bushings, linear motion guides, clutches, brakes, actuators, and other motion control elements. Site contains selection software for U.S. and metric dimensions. Thomson is part of the Danaher Motion Group of Danaher Corporation.
4. **Danaher Motion Group.** Part of Danaher Corporation. Manufacturers of Thomson motion control devices, motors, drives, controls, air bearing rotary stages, and high-precision X-Y-Z stages.
5. **THK Linear Motion Systems.** Manufacturer of ball screws, ball splines, linear motion guides, linear bushings, linear actuators, and other motion control products.
6. **Joyce/Dayton Corp.** Manufacturer of a wide variety of jacks for commercial and industrial applications. Included are machine screw and ball screw types with integral gear drives and complete motorized actuator systems. High-speed linear actuators produced by EDRIVE Actuators, Inc. are also available through Joyce/Dayton's Internet site.
7. **SKF Linear Motion.** Manufacturer of high-efficiency metric screws, linear guiding systems, actuators, and controls.
8. **Bishop-Wisecarver Group.** Manufacturer and supplier of linear motion systems and components from miniature to large sizes. Ball screws, positioning stages, linear bearings and bushings, ball slides, rotary motion guides, rotary actuators, and others.
9. **Isel USA.** Manufacturer of a wide variety of linear motion products including slides, X-Y and X-Y-Z tables, rotary tables, ball screws, and controls.
10. **Roton Products, Inc.** Manufacturer of a large variety of power screws, ball screws, and worms in U.S. and metric dimensions.
11. **Ledex.** Manufacturer of small linear and rotary solenoid actuators and related products under the Ledex brand. A unit of Johnson Electric.
12. **Nook Industries.** Manufacturer of power screws and screw jacks in U.S. and metric dimensions and thread forms.
13. **Power Jacks Group.** Manufacturer of metric and U.S. screw jacks and actuators.
14. **Automation Direct.** Provider of a wide variety of automation products, including liner slides, stepper motors, proximity sensors, encoders, and pneumatic components.
15. **EDRIVE Actuators, Inc.** Producer of a wide range of heavy duty ball screw linear actuators for factory automation and machine tool industries.

PROBLEMS

1. Name four types of threads used for power screws.
2. Make a scale drawing of an Acme thread having a major diameter of $1\frac{1}{2}$in and four threads per inch. Draw a section 2.0 in long.
3. Repeat Problem 2 for a buttress thread.
4. Repeat Problem 2 for a square thread.
5. If an Acme-thread power screw is loaded in tension with a force of 30 000 lb, what size screw from Table 17–1 should be used to maintain a tensile stress below 10 000 psi?
6. For the screw chosen in Problem 5, what would be the required axial length of the nut on the screw that transfers the load to the frame of the machine if the shear stress in the threads must be less than 6000 psi?
7. Compute the torque required to raise the load of 30 000 lb with the Acme screw selected in Problem 5. Use a coefficient of friction of 0.15.
8. Compute the torque required to lower the load with the screw from Problem 5.
9. If a square-thread screw having a major diameter of 3/4 in and six threads per inch is used to lift a load of 4000 lb, compute the torque required to rotate the screw. Use a coefficient of friction of 0.15.
10. For the screw of Problem 9, compute the torque required to rotate the screw when lowering the load.
11. Compute the lead angle for the screw of Problem 9. Is it self-locking?
12. Compute the efficiency of the screw of Problem 9.
13. If the load of 4000 lb is lifted by the screw described in Problem 9 at the rate of 0.5 in/s, compute the required rotational speed of the screw and the power required to drive it.
14. A ball screw for a machine table drive is to be selected. The axial force to be transmitted by the screw is 600 lb. The table moves 24 in per cycle, and it is expected to cycle 10 times per hour for a design life of 10 years. Select an appropriate screw.
15. For the screw selected in Problem 14, compute the torque required to drive the screw.
16. For the screw selected in Problem 14, the normal travel speed of the table is 10.0 in/min. Compute the power required to drive the screw.
17. If the cycle time for the machine in Problem 14 were decreased to obtain 20 cycles/h instead of 10, what would be the expected life in years of the screw originally selected?
18. Specify a suitable size for a metric trapezoidal power screw that is subjected to a tensile load of 125 kN while keeping the tensile stress below 75 MPa.
19. Compute the torque required to raise the load vertically for the screw selected in Problem 18. Use a coefficient of friction of 0.15.
20. Compute the torque required to lower the load with the screw specified in Problem 18.
21. For the screw specified in Problem 18, compute the power required to raise the load a total of 4250 mm in 7.5 s.
22. Compute the lead angle for the screw specified in Problem 18. Will it be self-locking?

23. Compute the efficiency for the screw specified in Problem 18 when raising the load.
24. Specify a suitable size for a metric trapezoidal power screw that is subjected to a tensile load of 8500 N while keeping the tensile stress below 110 MPa.
25. Compute the torque required to raise the load vertically for the screw selected in Problem 24. Use a coefficient of friction of 0.15.
26. Compute the torque required to lower the load with the screw specified in Problem 24.
27. For the screw specified in Problem 24, compute the power required to raise the load a total of 240 mm in 3.5 s.
28. Compute the lead angle for the screw specified in Problem 24. Will it be self-locking?
29. Compute the efficiency for the screw specified in Problem 24 when raising the load.
30. Compute the critical speed for the ball screw specified in Problem 14. The total length of the screw between single bearings at each end is 28.0 in.

CHAPTER EIGHTEEN

SPRINGS

THE BIG PICTURE Springs

Discussion Map

- A *spring* is a flexible element used to exert a force or a torque and, at the same time, to store energy.
- There are numerous types of springs: compression, extension, torsion, and many more.

Discover

Look around you and see if you can find one or more springs. Describe them, giving their basic geometry, the type of force or torque produced, the way in which they are used, and other features.

Share your observations about springs with your colleagues, and learn from their observations.

Write a brief report about at least two different kinds of springs. Include sketches that show their basic size, geometry, and appearance. Describe their functions, including how they work and how they affect the operation of the device of which they are a part.

This chapter will help you develop skill in designing and analyzing springs of the helical compression, helical extension, and torsion types. Information on special treatments for spring wire and spring manufacture is also provided.

A *spring* is a flexible element used to exert a force or a torque and, at the same time, to store energy. The force can be a linear push or pull, or it can be radial, acting similarly to a rubber band around a roll of drawings. The torque can be used to cause a rotation, for example, to close a door on a cabinet or to provide a counterbalance force for a machine element pivoting on a hinge.

Springs inherently store energy when they are deflected and return the energy when the force that causes the deflection is removed. Consider the child's toy called a *Jack-in-the-Box*. When you push Jack into the box, you are exerting a force on a spring and delivering energy to it. Then when you close the lid of the box, the spring is captured and is held in the compressed state. What happens when you trip the latch on the lid? Jack leaps out of the box! More precisely, the spring force causes Jack to push the lid open, and then the stored energy in the spring is released, causing the spring to expand to its initial, free length when no load was applied. Some devices contain *power springs* or *motor springs* that are wound up, and then they deliver the energy at a measured pace to produce a long-term action. Examples are mechanical animated toys, toy race cars, some watches, timers, and clocks.

Look around you and see if you can find one or more springs. Or think about where you might have recently encountered a device that used springs.

Consider various appliances, an automobile, a truck, a bicycle, an office machine, a door latch, a toy, a machine in a production operation, or some other device that has moving parts.

Describe the springs. Did they exert a push or a pull? Or did they exert a torque, tending to cause rotation? What were the springs made from? How big were they? Was the force or torque very large, or was it light enough for you to actuate the spring easily? How were the springs mounted in the device of which they were a part? Was there a load on the spring at all times? Or did the spring relax completely at one point in its total possible operating cycle? Was the spring designed to be actuated often so that it experienced a very large number of cycles of load (and stress) during its expected life? What was the environment in which the spring operated? Hot or cold? Wet or dry? Exposed to corrosives? How did the environment affect the kind of material used for the spring or the kind of coating on it?

Share your observations with others in your group and with your instructor. Listen to the observations of others, and compare them with your examples of springs. Take at least two springs that are quite different from each other, and prepare a brief report about them, including sketches that show the basic size, geometry, and appearance. Describe their functions, including how they work and how they affect the operation of the device of which they are a part. Refer to the preceding paragraph for some factors that you might describe. Also include a listing with a brief description of each different kind of spring that you or your colleagues found.

This chapter will present basic information about a variety of spring types. Design procedures will be developed for helical compression springs, helical tension springs, and torsion springs. We will consider loads and stresses, deflection characteristics, material selection, life expectancy, attachment, and installation.

YOU ARE THE DESIGNER

One design for an automotive engine valve train is shown in Figure 18–1. As the cam rotates, it causes the push rod to move upward. The rocker arm then rotates and pushes the valve stem downward, opening the valve. Concurrently, the spring that surrounds the valve stem is compressed, and energy is stored. As the cam continues to rotate, it allows the movement of the train to return to its original position. The valve is aided in its upward motion and seating action by the spring exerting a force that closes the valve at the end of the cycle.

You are the designer of the spring for the valve train. What type of spring do you specify? What should its dimensions be, including the length, outside diameter, inside diameter, and diameter of the wire for the coils? How many coils should be used? What should the ends of the spring look like? How much force is exerted on the valve, and how does this force change as the valve train goes through a complete cycle? What material should be used? What stress levels are developed in the spring wire, and how can the spring be designed to be safe under the load, life, and environmental conditions in which it must operate? You must specify or calculate all of these factors to ensure a successful spring design. ■

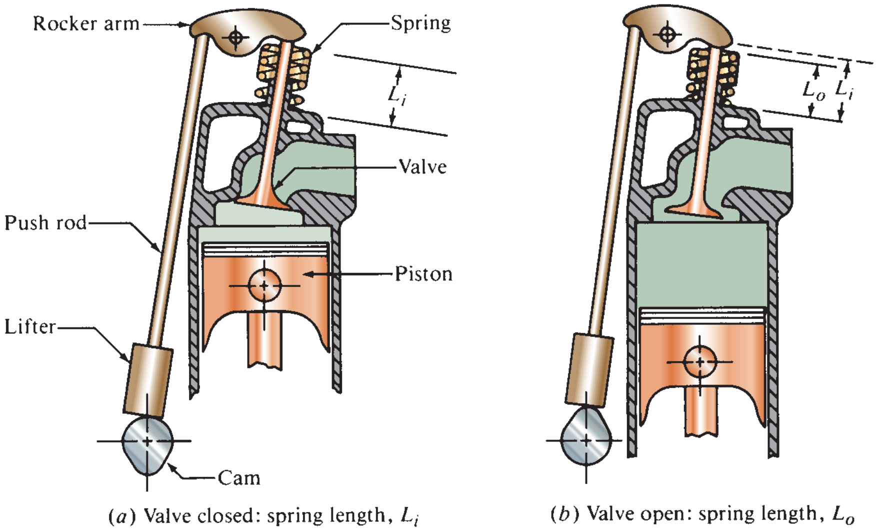


FIGURE 18–1 Engine valve train showing the application of a helical compression spring

18–1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Identify and describe several types of springs, including the helical compression spring, helical extension spring, torsion spring, Belleville spring, flat spring, drawbar spring, garter spring, constant-force spring, and power spring.
2. Design and analyze helical compression springs to conform to design requirements such as force/deflection characteristics, life, physical size, and environmental conditions.
3. Compute the dimensions of various geometric features of helical compression springs.
4. Specify suitable materials for springs based on strength, life, and deflection parameters.
5. Design and analyze helical extension springs.
6. Design and analyze torsion springs.
7. Use computer programs to assist in the design and analysis of springs.

18–2 KINDS OF SPRINGS

Springs can be classified according to the direction and the nature of the force exerted by the spring when it is deflected. Table 18–1 lists several kinds of springs classified as *push, pull, radial,* and *torsion*. Figure 18–2 shows several typical designs.

Helical compression springs are typically made from round wire, wrapped into a straight, cylindrical form with a constant pitch between adjacent coils. Square or rectangular wire may also be used. Four practical end configurations are shown in Figure 18–3. Without an applied load, the spring's length is called the *free length*. When a compression force is applied, the coils are pressed more closely together until they all touch, at which time the length is the minimum possible called the *solid length*.

TABLE 18–1 Types of Springs

Uses	Types of springs
Push	Helical compression spring
	Belleville spring
	Torsion spring: force acting at the end of the torque arm
	Flat spring, such as a cantilever or leaf spring
Pull	Helical extension spring
	Torsion spring: force acting at the end of the torque arm
	Flat spring, such as a cantilever or leaf spring
	Drawbar spring (special case of the compression spring)
	Constant-force spring
Radial	Garter spring, elastomeric band, and spring clamp
Torque	Torsion spring and power spring

A linearly increasing amount of force is required to compress the spring as its deflection is increased. Straight, cylindrical helical compression springs are among the most widely used types. Also shown in Figure 18–2 are the conical, barrel, hourglass, and variable-pitch types.

Helical extension springs appear to be similar to compression springs, having a series of coils wrapped into a cylindrical form. However, in extension springs, the coils either touch or are closely spaced under the no-load condition. Then as the external tensile load is applied, the coils separate. Figure 18–4 shows several end configurations for extension springs.

The *drawbar spring* [Figure 18–2(c)] incorporates a standard helical compression spring with two looped wire devices inserted through the inside of the spring. With such a design, a tensile force can be exerted by pulling on the loops while still placing the spring in compression. It also provides a definite stop as the compression spring is compressed to its solid height.

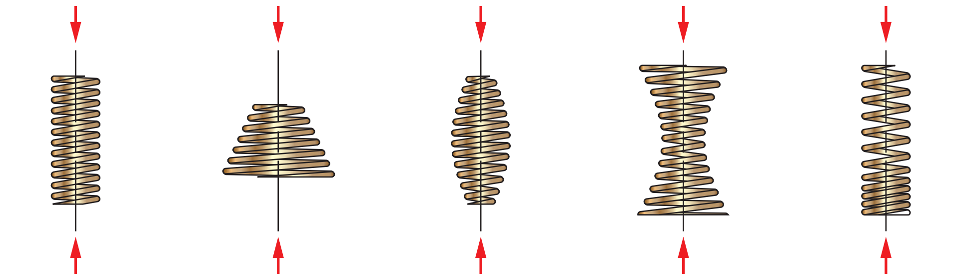
A *torsion spring,* as the name implies, is used to exert a torque as the spring is deflected by rotation about its axis. The common spring-action clothespin uses a torsion spring to provide the gripping action. Torsion springs are also used to rotate a door to its open or closed position or to counterbalance the lid of a container. Some timers and other controls use torsion springs to actuate switch contacts or to produce similar actions. Push or pull forces can be exerted by torsion springs if one end of the spring is attached to the member to be actuated.

Leaf springs are made from one or more flat strips of brass, bronze, steel, or other materials loaded as cantilevers or simple beams. They can provide a push or a pull force as they are deflected from their free condition. Large forces can be exerted within a small space by leaf springs. By tailoring the geometry of the leaves and by nesting leaves of different dimensions, the designer can achieve special force-deflection characteristics. The design of leaf springs uses the principles of stress and deflection analysis of beams as presented in courses in strength of materials and as reviewed in Chapter 3.

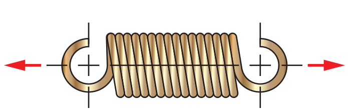
A *Belleville spring* has the shape of a shallow, conical disk with a central hole. It is sometimes called a *Belleville washer* because its appearance is similar to that of a flat washer. A very high spring rate or spring force can be developed in a small axial space with such springs. By varying the height of the cone relative to the thickness of the disk, the designer can obtain a variety of load-deflection characteristics. Also, nesting several springs face-to-face or back-to-back provides numerous spring rates.

Garter springs are coiled wires formed into a continuous ring shape so that they exert a radial force around the periphery of the object to which they are applied. Either inward or outward forces can be obtained with different designs. The action of a garter spring with an inward force is similar to that of a rubber band, and the spring action is similar to that of an extension spring.

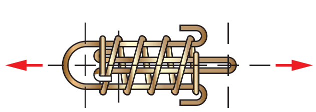
Constant-force springs take the form of a coiled strip. The force required to pull the strip off the coil is



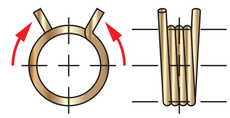
(*a*) Variations of helical compression springs



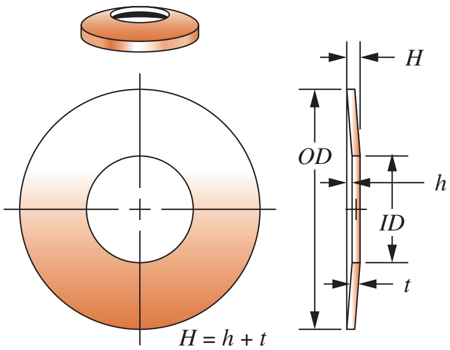
(*b*) Helical extension spring



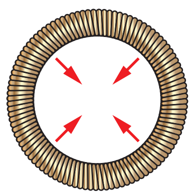
(*c*) Drawbar spring



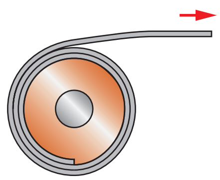
(*d*) Helical torsion spring



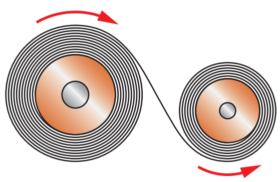
(*e*) Belleville spring



(*f*) Garter spring

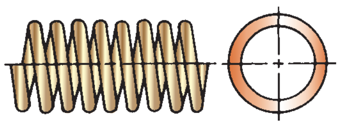


(*g*) Constant-force spring

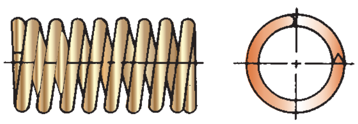


(*h*) Constant-force spring motor

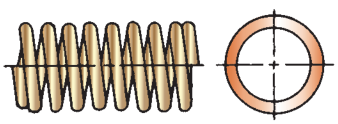
FIGURE 18–2 Several types of springs



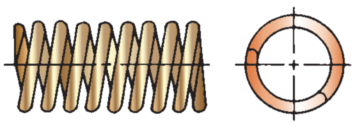
(*a*) Plain ends coiled right-hand



(*b*) Squared and ground ends coiled left-hand



(*c*) Squared or closed ends not ground, coiled right-hand



(*d*) Plain ends ground coiled left-hand

FIGURE 18–3 Appearance of helical compression springs showing end treatments

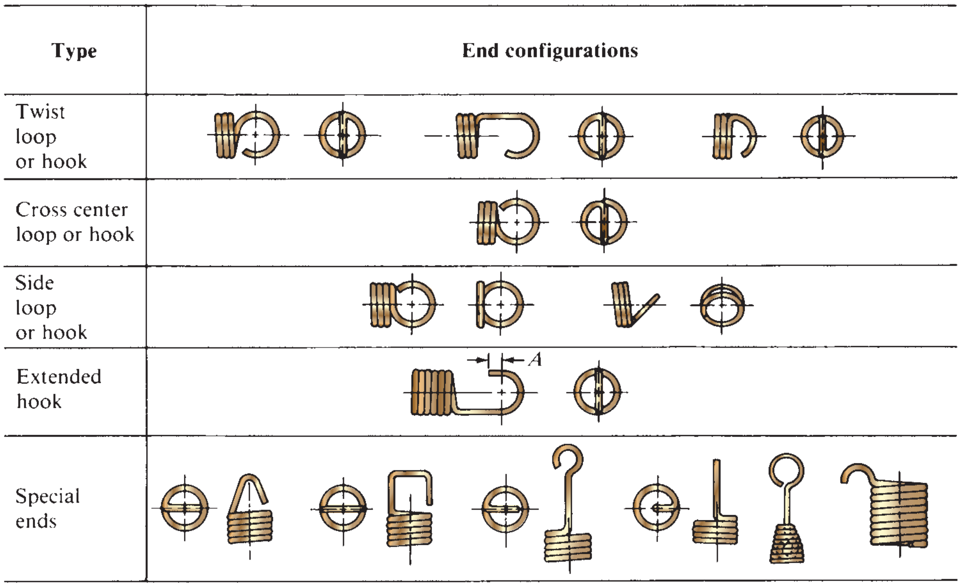


FIGURE 18–4 End configurations for extension springs

virtually constant over a long length of pull. The magnitude of the force is dependent on the width, thickness, and radius of curvature of the coil and on the elastic modulus of the spring material. Basically, the force is related to the deformation of the strip from its originally curved shape to the straight form.

Power springs, sometimes called *motor* or *clock springs,* are made from flat spring steel stock, wound into a spiral shape. A torque is exerted by the spring as it tends to unwrap the spiral. Figure 18–2 shows a spring motor made from a constant-force spring.

A *torsion bar,* as its name implies, is a bar loaded in torsion. When a round bar is used, the analyses for torsional stress and deflection are similar to the analysis presented for circular shafts in Chapters 3 and 12. Other cross-sectional shapes can be used, and special care must be exercised at the points of attachment.

Information about commercially available springs can be found in Internet sites 3–5 and 13. The methods of analyzing and designing springs used in this book have been developed by applying principles from References 1–10.

18–3 HELICAL COMPRESSION SPRINGS

In the most common form of helical compression spring, round wire is wrapped into a cylindrical form with a constant pitch between adjacent coils. This basic form is completed by a variety of end treatments, as shown in Figure 18–3.

For medium- to large-size springs used in machinery, the squared and ground-end treatment provides a flat surface on which to seat the spring. The end coil is collapsed against the adjacent coil (squared), and the surface is ground until at least 270° of the last coil is in contact with the bearing surface. Springs made from smaller wire (less than approximately 0.020 in, or 0.50 mm) are usually squared only, without grinding. In unusual cases the ends may be ground without squaring, or they may be left with plain ends, simply cut to length after coiling.

You are probably familiar with many uses of helical compression springs. The retractable ballpoint pen depends on the helical compression spring, usually installed around the ink supply barrel. Suspension systems for cars, trucks, and motorcycles frequently incorporate these springs. Other automotive applications include the valve springs in engines, hood linkage counterbalancing, and the clutch pressure-plate springs. In manufacturing, springs are used in dies to actuate stripper plates; in hydraulic control valves; as pneumatic cylinder return springs; and in the mounting of heavy equipment for shock isolation. Many small devices such as electrical switches and ball check valves incorporate helical compression springs. Desk chairs have stout springs to return the chair seat to its upright position. And don't forget the venerable pogo stick!

The following paragraphs define the many variables used to describe and analyze the performance of helical compression springs.

Diameters

Figure 18–5 shows the notation used in referring to the characteristic diameters of helical compression springs. The outside diameter (OD), the inside diameter (ID), and the wire diameter (D_w) are obvious and can be measured with standard measuring instruments. In calculating the stress and deflection of a spring, we use the mean diameter, D_m. Notice that

➩ Spring Diameters

$$OD = D_m + D_w$$
$$ID = D_m - D_w$$

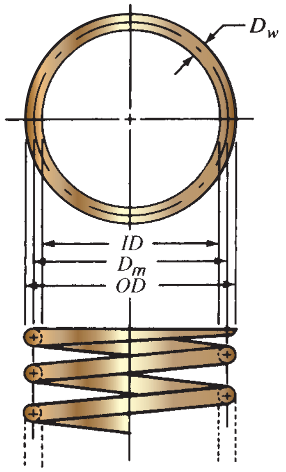


FIGURE 18–5 Notation for diameters

Standard Wire Diameters. The specification of the required wire diameter is one of the most important outcomes of the design of springs. Several types of materials are typically used for spring wire, and the wire is produced in sets of standard diameters covering a broad range. Table 18–2 lists the most common standard wire gages. Notice that except for music wire, the wire size gets smaller as the gage number gets larger. Also see the notes to the table. See Internet sites 8–11 and 14 for suppliers of spring wire.

Lengths

It is important to understand the relationship between the length of the spring and the force exerted by it (see Figure 18–6). The *free length*, L_f, is the length that the spring assumes when it is exerting no force, as if it were simply sitting on a table. The *solid length*, L_s, is found

TABLE 18–2 Wire Gages and Diameters for Springs

Gage no.	U.S. steel wire gage[1] (in)	Music wire gage[2] (in)	Brown & Sharpe Gage[3] (in)	Preferred metric diameters[4] (mm)
7/0	0.4900			13.0
6/0	0.4615	0.004	0.5800	12.0
5/0	0.4305	0.005	0.5165	11.0
4/0	0.3938	0.006	0.4600	10.0
3/0	0.3625	0.007	0.4096	9.0
2/0	0.3310	0.008	0.3648	8.5
0	0.3065	0.009	0.3249	8.0
1	0.2830	0.010	0.2893	7.0
2	0.2625	0.011	0.2576	6.5
3	0.2437	0.012	0.2294	6.0
4	0.2253	0.013	0.2043	5.5
5	0.2070	0.014	0.1819	5.0
6	0.1920	0.016	0.1620	4.8
7	0.1770	0.018	0.1443	4.5
8	0.1620	0.020	0.1285	4.0
9	0.1483	0.022	0.1144	3.8
10	0.1350	0.024	0.1019	3.5
11	0.1205	0.026	0.0907	3.0
12	0.1055	0.029	0.0808	2.8
13	0.0915	0.031	0.0720	2.5
14	0.0800	0.033	0.0641	2.0
15	0.0720	0.035	0.0571	1.8
16	0.0625	0.037	0.0508	1.6
17	0.0540	0.039	0.0453	1.4
18	0.0475	0.041	0.0403	1.2
19	0.0410	0.043	0.0359	1.0

TABLE 18–2 Wire Gages and Diameters for Springs (*continued*)

Gage no.	U.S. steel wire gage[1] (in)	Music wire gage[2] (in)	Brown & Sharpe Gage[3] (in)	Preferred metric diameters[4] (mm)
20	0.0348	0.045	0.0320	0.90
21	0.0317	0.047	0.0285	0.80
22	0.0286	0.049	0.0253	0.70
23	0.0258	0.051	0.0226	0.65
24	0.0230	0.055	0.0201	0.60 or 0.55
25	0.0204	0.059	0.0179	0.50 or 0.55
26	0.0181	0.063	0.0159	0.45
27	0.0173	0.067	0.0142	0.45
28	0.0162	0.071	0.0126	0.40
29	0.0150	0.075	0.0113	0.40
30	0.0140	0.080	0.0100	0.35
31	0.0132	0.085	0.00 893	0.35
32	0.0128	0.090	0.00 795	0.30 or 0.35
33	0.0118	0.095	0.00 708	0.30
34	0.0104	0.100	0.00 630	0.28
35	0.0095	0.106	0.00 501	0.25
36	0.0090	0.112	0.00 500	0.22
37	0.0085	0.118	0.00 445	0.22
38	0.0080	0.124	0.00 396	0.20
39	0.0075	0.130	0.00 353	0.20
40	0.0070	0.138	0.00 314	0.18

Sources: References 1, 4, and 5.

[1] Use the U.S. Steel Wire Gage for steel wire, except music wire. This gage has also been called the *Washburn and Moen Gage (W&M)*, the *American Steel Wire Co. Gage*, and the *Roebling Wire Gage*.

[2] Use the Music Wire Gage only for music wire (ASTM A228).

[3] Use the Brown & Sharpe Gage for nonferrous wires such as brass and phosphor bronze.

[4] The preferred metric sizes are from Reference 1 and are listed as the nearest preferred metric size to the U.S. Steel Wire Gage. The gage numbers do not apply.

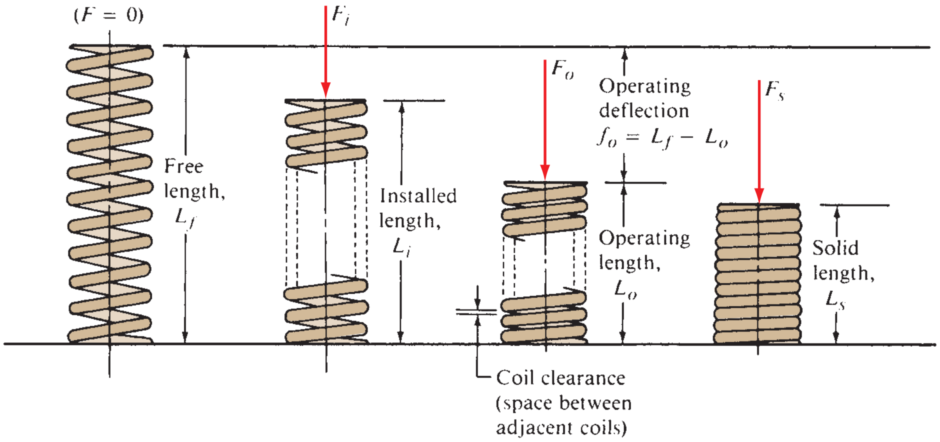


FIGURE 18–6 Notation for lengths and forces

when the spring is collapsed to the point where all coils are touching. This is obviously the shortest possible length that the spring can have. The spring is usually not compressed to the solid length during operation.

The shortest length for the spring during normal operation is the *operating length*, L_o. At times, a spring will be designed to operate between two limits of deflection. Consider the valve spring for an engine, for example, as shown in Figure 18–1. When the valve is open, the spring assumes its shortest length, which is L_o. Then when the valve is closed, the spring gets longer but still exerts a force to keep the valve securely on its seat. The length at this condition is called the *installed length*, L_i. So the valve spring length changes from L_o to L_i during normal operation as the valve itself reciprocates.

Forces

We will use the symbol F to indicate forces exerted by a spring, with various subscripts to specify which level of force is being considered. The subscripts correspond to those used for the lengths. Thus,

F_s = force at solid length, L_s: the maximum force that the spring ever sees

F_o = force at operating length, L_o: the maximum force the spring sees in *normal operation*

F_i = force at installed length, L_i: the force varies between F_o and F_i for a reciprocating spring

F_f = force at free length, L_f: this force is zero

Spring Rate

The relationship between the force exerted by a spring and its deflection is called its *spring rate*, k. Any change in force divided by the corresponding change in deflection can be used to compute the spring rate:

➪ Spring Rate

$$k = \Delta F/\Delta L \tag{18–1}$$

For example,

$$k = \frac{F_o - F_i}{L_i - L_o} \tag{18–1a}$$

or

$$k = \frac{F_o}{L_f - L_o} \tag{18–1b}$$

or

$$k = \frac{F_i}{L_f - L_i} \tag{18–1c}$$

In addition, if the spring rate is known, the force at any deflection can be computed. For example, if a spring had a rate of 42.0 lb/in, the force exerted at a deflection from free length of 2.25 in would be

$$F = k(L_f - L) = (42.0\ \text{lb/in})(2.25\ \text{in}) = 94.5\ \text{lb}$$

Spring Index

The ratio of the mean diameter of the spring to the wire diameter is called the *spring index*, C:

➪ Spring Index

$$C = D_m/D_w$$

It is recommended that C be greater than 5.0, with typical machinery springs having C values ranging from 5 to 12. For C less than 5, the forming of the spring will be very difficult, and the severe deformation required may create cracks in the wire. The stresses and the deflections in springs are dependent on C, and a larger C will help to eliminate the tendency for a spring to buckle.

Number of Coils

The total number of coils in a spring will be called N. But in the calculation of stress and deflections for a spring, some of the coils are inactive and are neglected. For example, in a spring with squared and ground ends or simply squared ends, each end coil is inactive, and the number of *active coils*, N_a, is $N - 2$. For plain ends, all coils are active: $N_a = N$. For plain coils with ground ends, $N_a = N - 1$.

Pitch

Pitch, p, refers to the axial distance from a point on one coil to the corresponding point on the next adjacent coil. The relationships among the pitch, free length, wire diameter, and number of active coils are as follows:

Squared and ground ends:	$L_f = pN_a + 2D_w$
Squared ends only:	$L_f = pN_a + 3D_w$
Plain and ground ends:	$L_f = p(N_a + 1)$
Plain ends:	$L_f = pN_a + D_w$

Pitch Angle

Figure 18–7 shows the pitch angle, λ; note that the larger the pitch angle is, the steeper the coils appear to be. Most practical spring designs produce a pitch angle less than about 12°. If the angle is greater than 12°, undesirable compressive stresses develop in the wire, and the formulas presented later are inaccurate. The pitch angle can be computed by the formula

$$\lambda = \tan^{-1}\left[\frac{p}{\pi D_m}\right] \tag{18–2}$$

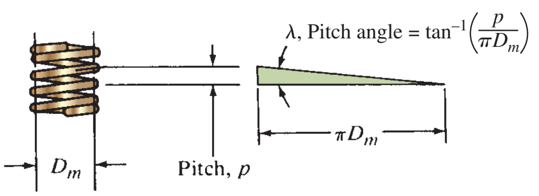


FIGURE 18–7 Pitch angle

You can see the logic of this formula by taking one coil of a spring and unwrapping it onto a flat surface, as illustrated in Figure 18–7. The horizontal line is the mean circumference of the spring, and the vertical line is the pitch, p.

Installation Considerations

Frequently, a spring is installed in a cylindrical hole or around a rod. When it is, adequate clearances must be provided. When a compression spring is compressed, its diameter gets larger. Thus, the inside diameter of a hole enclosing the spring must be greater than the outside diameter of the spring to eliminate rubbing. An initial diametral clearance of one-tenth of the wire diameter is recommended for springs having a diameter of 0.50 in (12 mm) or greater. If a more precise estimate of the actual outside diameter of the spring is required, the following formula can be used for the OD at the solid length condition:

$$OD_s = \sqrt{D_m^2 + \frac{p^2 - D_w^2}{\pi^2}} + D_w \qquad \textbf{(18–3)}$$

Even though the spring ID gets larger, it is also recommended that the clearance at the ID be approximately $0.1D_w$.

Springs with squared ends or squared and ground ends are frequently mounted on a button-type seat or in a socket with a depth equal to the height of just a few coils for the purpose of locating the spring.

Coil Clearance. The term *coil clearance* refers to the space between adjacent coils when the spring is compressed to its operating length, L_o. The actual coil clearance, cc, can be estimated from

➪ Coil Clearance

$$cc = (L_o - L_s)/N_a$$

One guideline is that the coil clearance should be greater than $D_w/10$, especially in springs loaded cyclically. Another recommendation relates to the overall deflection of the spring:

$$(L_o - L_s) > 0.15(L_f - L_s)$$

Materials Used for Springs

Virtually any elastic material can be used for a spring. However, most mechanical applications use metallic wire—high-carbon steel (most common), alloy steel, stainless steel, brass, bronze, beryllium copper, or nickel-base alloys. Most spring materials are made according to specifications of the ASTM. Table 18–3 lists some common types. (See Internet sites 8–11 and 14.)

Types of Loading and Allowable Stresses

The allowable stress to be used for a spring depends on the type of loading, the material, and the size of the wire. Loading is usually classified into three types:

- **Light service:** Static loads or up to 10 000 cycles of loading with a low rate of loading (nonimpact)
- **Average service:** Typical machine design situations; moderate rate of loading and up to 1 million cycles
- **Severe service:** Rapid cycling for above 1 million cycles; possibility of shock or impact loading; engine valve springs are a good example

The strength of a given material is greater for the smaller sizes. Figures 18–8 through 18–13 show the design stresses for six different materials. Note that some curves can be used for more than one material by the application of a factor. As a conservative approach to design, we will use the average service curve for most design examples, unless true high cycling conditions exist. We will use the light service curve as the upper limit on stress when the spring is compressed to its solid height. If the stress exceeds the light service value by a small amount, the spring will undergo permanent set because of yielding.

TABLE 18–3 Spring Materials

Material type	ASTM no.	Relative cost	Temperature limits, °F
A. High-carbon steels			
Hard-drawn	A227	1.0	0–250
General-purpose steel with 0.60%–0.70% carbon; low cost			
Music wire	A228	2.6	0–250
High-quality steel with 0.80%–0.95% carbon; very high strength; excellent surface finish; hard-drawn; good fatigue performance; used mostly in smaller sizes up to 0.125 in			
Oil-tempered	A229	1.3	0–350
General-purpose steel with 0.60%–0.70% carbon; used mostly in larger sizes above 0.125 in; not good for shock or impact			

(*continued*)

TABLE 18–3 Spring Materials (*continued*)

Material type	ASTM no.	Relative cost	Temperature limits, °F
B. Alloy steels			
Chromium-vanadium	A231	3.1	0–425
Good strength, fatigue resistance, impact strength, high-temperature performance; valve-spring quality			
Chromium-silicon	A401	4.0	0–475
Very high strength and good fatigue and shock resistance			
C. Stainless steels			
Type 302	A313(302)	7.6	<0–550
Very good corrosion resistance and high-temperature performance; nearly nonmagnetic; cold-drawn; types 304 and 316 also fall under this ASTM class and have improved workability but lower strength			
Type 17-7 PH	A313(631)	11.0	0–600
Good high-temperature performance			
D. Copper alloys: All have good corrosion resistance and electrical conductivity			
Spring brass	B134	High	0–150
Phosphor bronze	B159	8.0	<0–212
Beryllium copper	B197	27.0	0–300
E. Nickel-base alloys: All are corrosion-resistant, have good high- and low-temperature properties, and are nonmagnetic or nearly nonmagnetic (trade names of the International Nickel Company)			
Monel™			−100–425
K-Monel™			−100–450
Inconel™			Up to 700
Inconel-X™		44.0	Up to 850

Sources: References 1, 4, and 5.

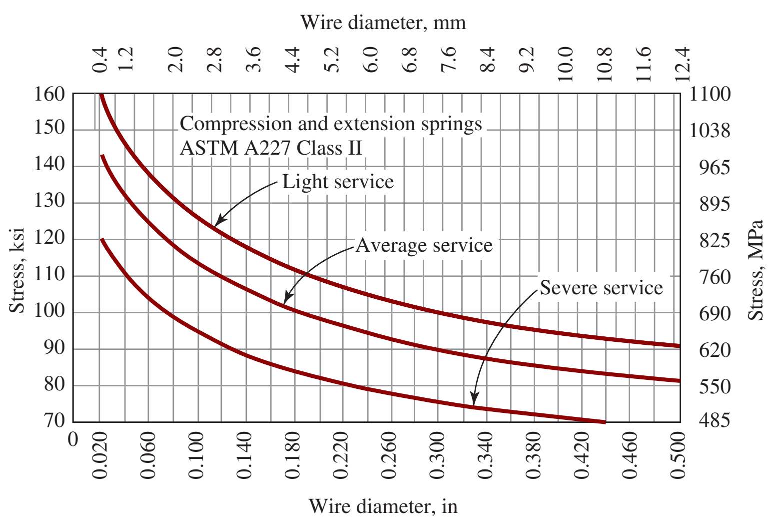


FIGURE 18–8 Design shear stresses for ASTM A227 steel wire, hard-drawn

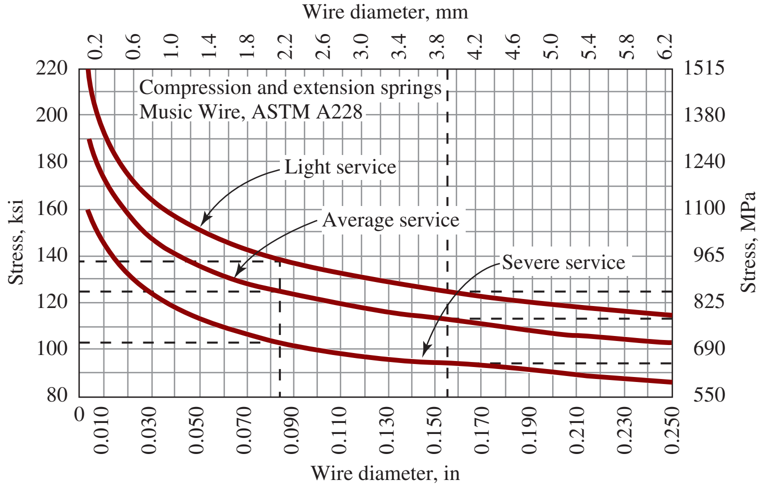


FIGURE 18–9 Design shear stresses for ASTM A228 steel wire (music wire)

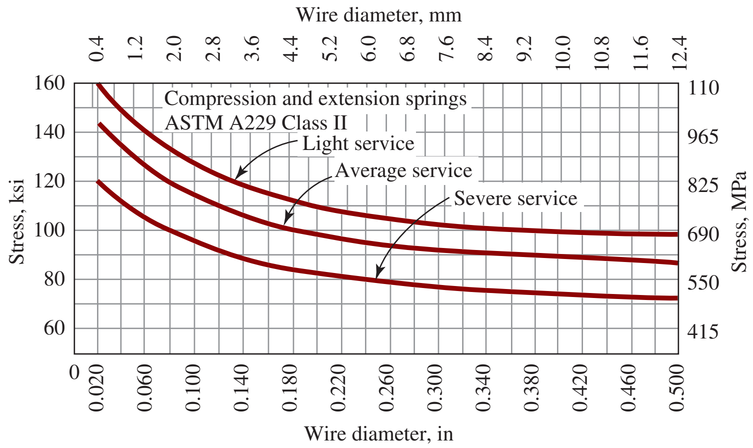


FIGURE 18–10 Design shear stresses for ASTM A229 steel wire, oil-tempered

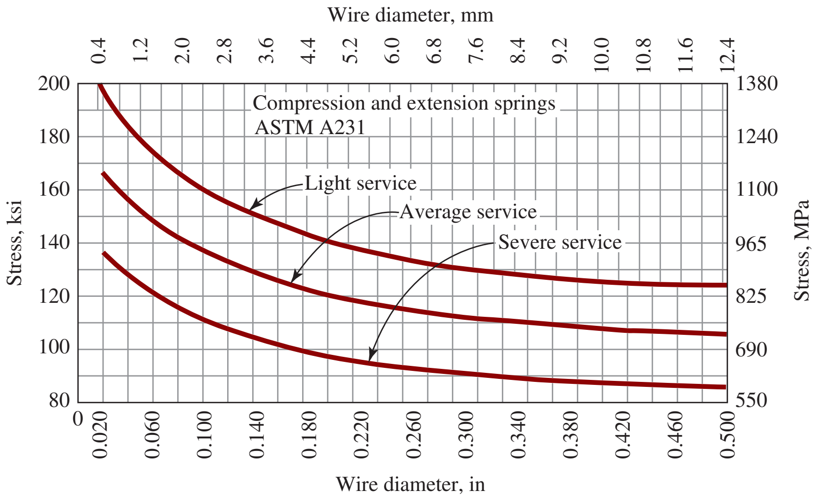


FIGURE 18–11 Design shear stresses for ASTM A231 steel wire, chromium-vanadium alloy, valve-spring quality

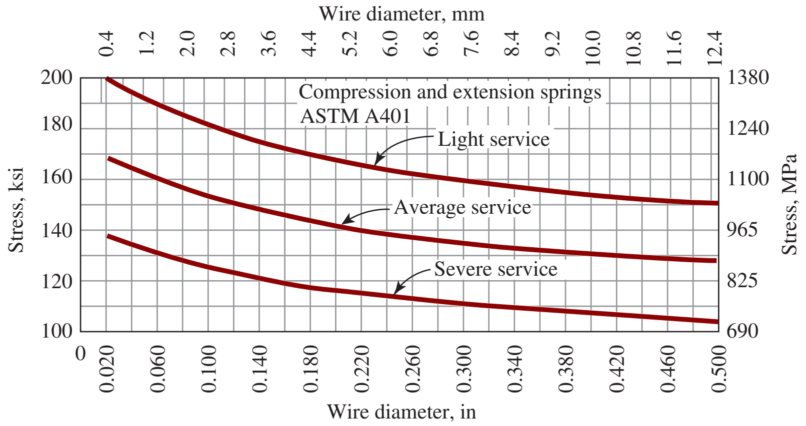


FIGURE 18–12 Design shear stresses for ASTM A401 steel wire, chromium-silicon alloy, oil-tempered

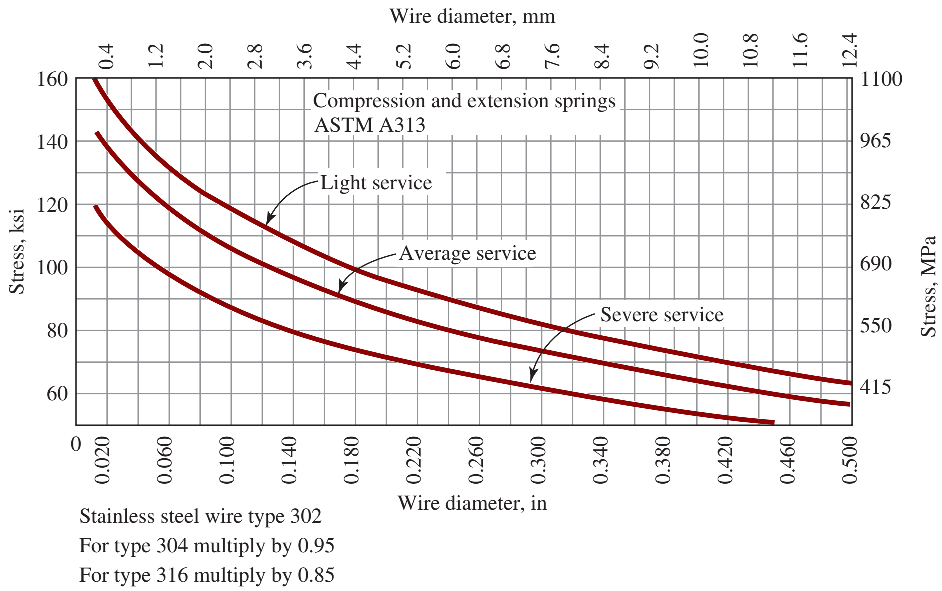


FIGURE 18–13 Design shear stresses for ASTM A313 corrosion-resistant stainless steel wire

18–4 STRESSES AND DEFLECTION FOR HELICAL COMPRESSION SPRINGS

As a compression spring is compressed under an axial load, the wire is twisted. Therefore, the stress developed in the wire is *torsional shear stress,* and it can be derived from the classical equation $\tau = Tc/J$.

When the equation is applied specifically to a helical compression spring, some modifying factors are needed to account for the curvature of the spring wire and for the direct shear stress created as the coils resist the vertical load. Also, it is convenient to express the shear stress in terms of the design variables encountered in springs. The resulting equation for stress is attributed to Wahl. (See Reference 10.) The maximum shear stress, which occurs at the inner surface of the wire, is

➩ Shear Stress in a Spring

$$\tau = \frac{8KFD_m}{\pi D_w^3} = \frac{8KFC}{\pi D_w^2} \qquad (18\text{–}4)$$

These are two forms of the same equation as the definition of $C = D_m/D_w$ demonstrates. The shear stress for any applied force, F, can be computed. Normally we will be concerned about the stress when the spring is compressed to solid length under the influence of F_s and when the spring is operating at its normal maximum load, F_o. Notice that the stress is inversely proportional to the *cube* of the wire diameter. This illustrates the great

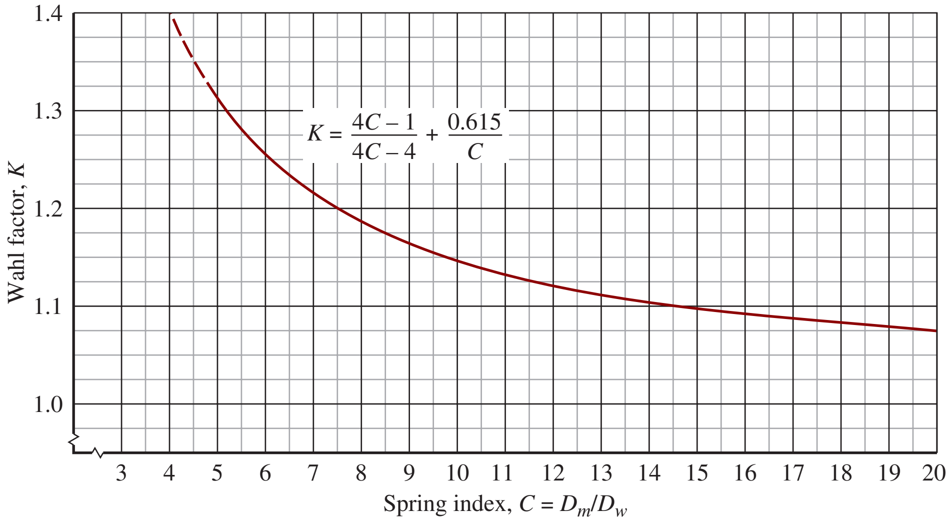


FIGURE 18–14 Wahl factor versus spring index for round wire

effect that variation in the wire size has on the performance of the spring.

The Wahl factor, K, in Equation (18–4) is the term that accounts for the curvature of the wire and the direct shear stress. Analytically, K is related to C:

➩ Wahl Factor

$$K = \frac{4C - 1}{4C - 4} + \frac{0.615}{C} \tag{18–5}$$

Figure 18–14 shows a plot of K versus C for round wire. Recall that $C = 5$ is the recommended minimum value of C. The value of K rises rapidly for $C < 5$.

Deflection

Because the primary manner of loading on the wire of a helical compression spring is torsion, the deflection is computed from the angle of twist formula:

$$\theta = TL/GJ$$

where θ = angle of twist in radians
T = applied torque
L = length of the wire
G = modulus of elasticity of the material in shear
J = polar moment of inertia of the wire

Again, for convenience, we will use a different form of the equation in order to calculate the linear deflection, f, of the spring from the typical design variables of the spring. The resulting equation is

➩ Deflection of a Spring

$$f = \frac{8FD_m^3 N_a}{GD_w^4} = \frac{8FC^3 N_a}{GD_w} \tag{18–6}$$

Recall that N_a is the number of *active* coils, as discussed in Section 18–3. Table 18–4 lists the values for G for typical spring materials. Note again, in Equation (18–6), that the wire diameter has a strong effect on the performance of the spring.

Buckling

The tendency for a spring to buckle increases as the spring becomes tall and slender, much as for a column. Figure 18–15 shows plots of the critical ratio of deflection to the free length versus the ratio of free length to the mean diameter for the spring. Three different end conditions are described in the figure. As an example of the use of this figure, consider a spring having squared and ground ends, a free length of 6.0 in, and a mean diameter of 0.75 in. We want to know what deflection would cause the spring to buckle. First compute

$$\frac{L_f}{D_m} = \frac{6.0}{0.75} = 8.0$$

Then, from Figure 18–15, the critical deflection ratio is 0.20. From this we can compute the critical deflection:

$$\frac{f_o}{L_f} = 0.20 \quad \text{or} \quad f_o = 0.20(L_f) = 0.20(6.0 \text{ in}) = 1.20 \text{ in}$$

That is, if the spring is deflected more than 1.20 in, the spring should buckle.

Additional development and discussion of formulas for stresses and deflection of helical compression springs can be found in References 4 and 6. Reference 3 provides valuable information on the analysis of spring failures.

18–5 ANALYSIS OF SPRING CHARACTERISTICS

This section demonstrates the use of the concepts developed in previous sections to analyze the geometry and the performance characteristics of a given spring. Assume that you have found a spring but there are no

TABLE 18–4 Spring Wire Modulus of Elasticity in Shear (G) and Tension (E)

Material and ASTM no.	Shear modulus, G (psi)	Shear modulus, G (GPa)	Tension modulus, E (psi)	Tension modulus, E (GPa)
Hard-drawn steel: A227	11.5×10^6	79.3	28.6×10^6	197
Music wire: A228	11.85×10^6	81.7	29.0×10^6	200
Oil-tempered: A229	11.2×10^6	77.2	28.5×10^6	196
Chromium-vanadium: A231	11.2×10^6	77.2	28.5×10^6	196
Chromium-silicon: A401	11.2×10^6	77.2	29.5×10^6	203
Stainless steels: A313				
Types 302, 304, 316	10.0×10^6	69.0	28.0×10^6	193
Type 17-7 PH	10.5×10^6	72.4	29.5×10^6	203
Spring brass: B134	5.0×10^6	34.5	15.0×10^6	103
Phosphor bronze: B159	6.0×10^6	41.4	15.0×10^6	103
Beryllium copper: B197	7.0×10^6	48.3	17.0×10^6	117
Monel and K-Monel	9.5×10^6	65.5	26.0×10^6	179
Inconel and Inconel-X	10.5×10^6	72.4	31.0×10^6	214

Note: Data are average values. Slight variations with wire size and treatment may occur.

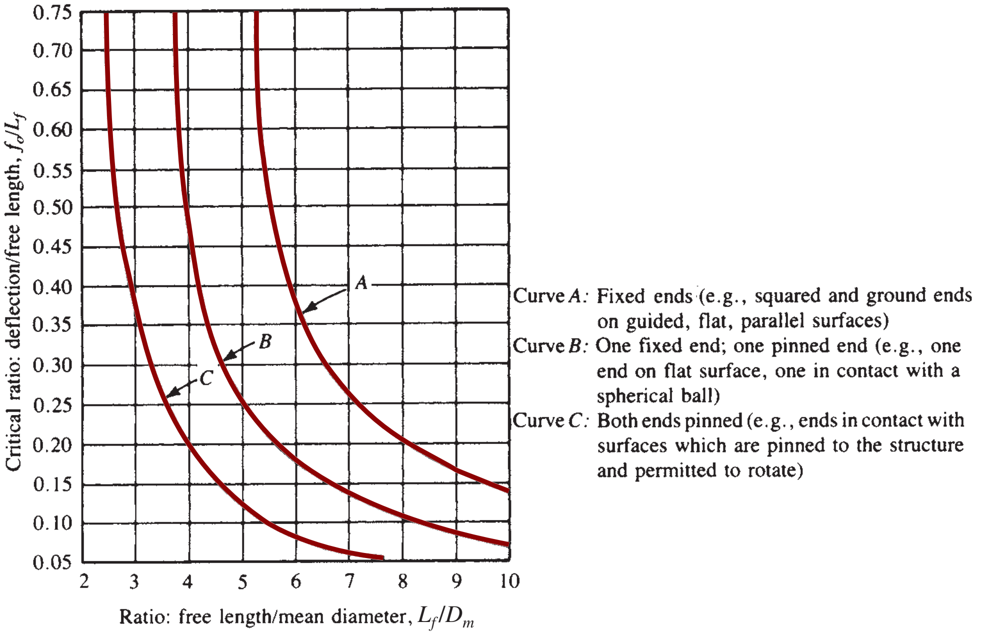


FIGURE 18–15 Spring buckling criteria. If the actual ratio of f_o/L_f is greater than the critical ratio, the spring will buckle at operating deflection.

performance data available for it. By making a few measurements and computations, you should be able to determine those characteristics. One piece of information you *would* have to know is the material from which the spring is made so that you could evaluate the acceptability of calculated stress levels.

The method of analysis is presented in Example Problem 18–1.

Example Problem 18–1 A spring is known to be made from music wire, ASTM A228 steel, but no other data are known. You are able to measure the following features using simple measurement tools:

Free length $= L_f = 1.75$ in
Outside diameter $= OD = 0.561$ in
Wire diameter $= D_w = 0.055$ in
The ends are squared and ground.
The total number of coils is 10.0.

This spring will be used in an application where the normal operating load is to be 14.0 lb. Approximately 300 000 cycles of loading are expected.

For this spring, compute and/or do the following:

1. The music wire gage number, mean diameter, inside diameter, spring index, and Wahl factor
2. The expected stress at the operating load of 14.0 lb
3. The deflection of the spring under the 14.0-lb load
4. The operating length, solid length, and spring rate
5. The force on the spring when it is at its solid length and the corresponding stress at solid length
6. The design stress for the material; then compare it with the actual operating stress
7. The maximum permissible stress; then compare it with the stress at solid length
8. Check the spring for buckling and coil clearance
9. Specify a suitable diameter for a hole in which to install the spring

Solution The solution is presented in the same order as the requested items just listed. The formulas used are found in the preceding sections of this chapter.

Step 1. The wire is 24-gage music wire (Table 18–2). Thus,

$$D_m = OD - D_w = 0.561 - 0.055 = 0.506 \text{ in}$$
$$ID = D_m - D_w = 0.506 - 0.055 = 0.451 \text{ in}$$
$$\text{Spring index} = C = D_m/D_w = 0.506/0.055 = 9.20$$
$$\text{Wahl factor} = K = (4C - 1)/(4C - 4) + 0.615/C$$
$$K = [4(9.20) - 1]/[4(9.20) - 4] + 0.615/9.20 = 1.158$$
$$K = 1.158$$

Step 2. Stress in spring at $F = F_o = 14.0$ lb [Equation (18–4)]:

$$\tau_o = \frac{8KF_oC}{\pi D_w^2} = \frac{8(1.158)(14.0)(9.20)}{\pi(0.055)^2} = 125\,560 \text{ psi}$$

Step 3. Deflection at operating force [Equation (18–6)]:

$$f_o = \frac{8F_oC^3N_a}{GD_w} = \frac{8(14.0)(9.20)^3(8.0)}{(11.85 \times 10^6)(0.055)} = 1.071 \text{ in}$$

Note that the number of active coils for a spring with squared and ground ends is $N_a = N - 2 = 10.0 - 2 = 8.0$. Also, the spring wire modulus, G, was found in Table 18–4. The value of f_o is the deflection *from free length* to the operating length.

Step 4. Operating length: We compute operating length as

$$L_o = L_f - f_o = 1.75 - 1.071 = 0.679 \text{ in}$$
$$\text{Solid length} = L_s = D_w(N) = 0.055(10.0) = 0.550 \text{ in}$$

Spring Index: We use Equation (18–1).

$$k = \frac{\Delta F}{\Delta L} = \frac{F_o}{L_f - L_o} = \frac{F_o}{f_o} = \frac{14.0 \text{ lb}}{1.071 \text{ in}} = 13.07 \text{ lb/in}$$

Step 5. We can find the force at solid length by multiplying the spring rate times the deflection from the free length to the solid length. Then

$$F_s = k(L_f - L_s) = (13.07\ \text{lb/in})(1.75\ \text{in} - 0.550\ \text{in}) = 15.69\ \text{lb}$$

The stress at solid length, τ_s, could be found from Equation (18–4), using $F = F_s$. However, an easier method is to recognize that the stress is directly proportional to the force on the spring and that all of the other data in the formula are the same as those used to compute the stress under the operating force, F_o. We can then use the simple proportion

$$\tau_s = \tau_o(F_s/F_o) = (125\,560\ \text{psi})(15.69/14.0) = 140\,700\ \text{psi}$$

Step 6. Design stress, τ_d: From Figure 18–9, in the graph of design stress versus spring wire diameter for ASTM A228 steel, we can use the *average service* curve based on the expected number of cycles of loading. We read $\tau_d = 135\,000$ psi for the 0.055-in wire. Because the actual operating stress, τ_o, is less than this value, it is satisfactory.

Step 7. Maximum allowable stress, τ_{max}: It is recommended that the *light service* curve be used to determine this value. For $D_w = 0.055$, $\tau_{max} = 150\,000$ psi. The actual expected maximum stress that occurs at solid length ($\tau_s = 140\,700$ psi) is less than this value, and therefore the design is satisfactory with regard to stresses.

Step 8. Buckling: To evaluate buckling, we must compute

$$L_f/D_m = (1.75\ \text{in})/(0.506\ \text{in}) = 3.46$$

Referring to Figure 18–15 and using curve *A* for squared and ground ends, we see that the critical deflection ratio is very high and that buckling should not occur. In fact, for any value of $L_f/D_m < 5.2$, we can conclude that buckling will not occur.

Coil clearance, *cc*: We evaluate *cc* as follows:

$$cc = (L_o - L_s)/N_a = (0.679 - 0.550)/(8.0) = 0.016\ \text{in}$$

Comparing this to the recommended minimum clearance of

$$D_w/10 = (0.055\ \text{in})/10 = 0.0055\ \text{in}$$

we can judge this clearance to be acceptable.

***Step* 9.** Hole diameter: It is recommended that the hole into which the spring is to be installed should be greater in diameter than the *OD* of the spring by the amount of $D_w/10$. Then

$$D_{hole} > OD + D_w/10 = 0.561\ \text{in} + (0.055\ \text{in})/10 = 0.567\ \text{in}$$

A diameter of 5/8 in (0.625 in) would be a satisfactory standard size.

This completes the example problem.

18–6 DESIGN OF HELICAL COMPRESSION SPRINGS

The objective of the design of helical compression springs is to specify the geometry for the spring to operate under specified limits of load and deflection, possibly with space limitations, also. We will specify the material and the type of service by considering the environment and the application.

A typical problem statement follows. Then two solution procedures are shown, and each is implemented with the aid of a spreadsheet.

Example Problem 18–2

A helical compression spring is to exert a force of 8.0 lb when compressed to a length of 1.75 in. At a length of 1.25 in, the force must be 12.0 lb. The spring will be installed in a machine that cycles slowly, and approximately 200 000 cycles total are expected. The temperature will not exceed 200°F. The spring will be installed in a hole having a diameter of 0.75 in.

For this application, specify a suitable material, wire diameter, mean diameter, *OD, ID,* free length, solid length, number of coils, and type of end condition. Check the stress at the maximum operating load and at the solid length condition.

The first of two solution procedures will be shown. The numbered steps can be used as a guide for future problems and as a kind of algorithm for the spreadsheet approach that follows the manual solution.

Solution Method 1 The procedure works directly toward the overall geometry of the spring by specifying the mean diameter to meet the space limitations. The process requires that the designer have tables of data available for wire diameters (such as Table 18–2) and graphs of design stresses for the material from which the spring will be made (such as Figures 18–8 through 18–13). We must make an initial estimate for the design stress for the material by consulting the charts of design stress versus wire diameter to make a reasonable choice. In general, more than one trial must be made, but the results of early trials will help you decide the values to use for later trials.

Step 1. Specify a material and its shear modulus of elasticity, G.

For this problem, several standard spring materials can be used. Let's select ASTM A231 chromium-vanadium steel wire, having a value of $G = 11\ 200\ 000$ psi (see Table 18–4).

Step 2. From the problem statement, identify the operating force, F_o; the operating length at which that force must be exerted, L_o; the force at some other length, called the *installed force,* F_i; and the installed length, L_i.

Remember, F_o is the maximum force that the spring experiences under normal operating conditions. Many times, the second force level is not specified. In that case, let $F_i = 0$, and specify a design value for the free length, L_f, in place of L_i.

For this problem, $F_o = 12.0$ lb; $L_o = 1.25$ in; $F_i = 8.0$ lb; and $L_i = 1.75$ in.

Step 3. Compute the spring rate, k, using Equation (18–1a):

$$k = \frac{F_o - F_i}{L_i - L_o} = \frac{12.0 - 8.0}{1.75 - 1.25} = 8.00 \text{ lb/in}$$

Step 4. Compute the free length, L_f:

$$L_f = L_i + F_i/k = 1.75 \text{ in} + [(8.00 \text{ lb})/(8.00 \text{ lb/in})] = 2.75 \text{ in}$$

The second term in the preceding equation is the amount of deflection from free length to the installed length in order to develop the installed force, F_i. Of course, this step becomes unnecessary if the free length is specified in the original data.

Step 5. Specify an initial estimate for the mean diameter, D_m.

Keep in mind that the mean diameter will be smaller than the *OD* and larger than the *ID*. Judgment is necessary to get started. For this problem, let's specify $D_m = 0.60$ in. This should permit the installation into the 0.75-in-diameter hole and it will be checked later in the problem solution.

Step 6. Specify an initial design stress.

The charts for the design stresses for the selected material can be consulted, considering also the service. In this problem, we should use average service. Then for the ASTM A231 steel, as shown in Figure 18–11, a nominal design stress would be 130 000 psi. This is strictly an estimate based on the strength of the material. The process includes a check on stress later.

Step 7. Compute the trial wire diameter by solving Equation (18–4) for D_w. Notice that everything else in the equation is known except the Wahl factor, K, because it depends on the wire diameter itself. But K varies only little over the normal range of spring indexes, C. From Figure 18–14, note that $K = 1.2$ is a nominal value. This, too, will be checked later. With the assumed value of K, some simplification can be done:

$$D_w = \left[\frac{8KF_oD_m}{\pi\tau_d}\right]^{1/3} = \left[\frac{(8)(1.2)(F_o)(D_m)}{(\pi)(\tau_d)}\right]^{1/3}$$

Combining constants gives

➪ Trial Wire Diameter

$$D_w = \left[\frac{8KF_oD_m}{\pi\tau_d}\right]^{1/3} = \left[\frac{(3.06)(F_o)(D_m)}{(\tau_d)}\right]^{1/3} \tag{18–7}$$

For this problem,

$$D_w = \left[\frac{(3.06)(F_o)(D_m)}{\tau_d}\right]^{1/3} = \left[\frac{(3.06)(12)(0.6)}{130\ 000}\right]^{0.333}$$

$$D_w = 0.0553 \text{ in}$$

Step 8. Select a standard wire diameter from the tables, and then determine the design stress and the maximum allowable stress for the material at that diameter. The design stress will normally be for average service, unless high cycling rates or shock indicate that severe service is warranted. The light service curve should be used with care because it is very near to the yield strength. In fact, we will use the light service curve as an estimate of the maximum allowable stress.

For this problem, the next larger standard wire size is 0.0625 in, no. 16 on the U.S. Steel Wire Gage chart. For this size, the curves in Figure 18–11 for ASTM A231 steel wire show the design stress to be approximately 145 000 psi for average service, and the maximum allowable stress to be 170 000 psi from the light service curve.

***Step* 9.** Compute the actual values of C and K, the spring index and the Wahl factor:

$$C = \frac{D_m}{D_w} = \frac{0.60}{0.0625} = 9.60$$

$$K = \frac{4C - 1}{4C - 4} + \frac{0.615}{C} = \frac{4(9.60) - 1}{4(9.60) - 4} + \frac{0.615}{9.60} = 1.15$$

Step 10. Compute the actual expected stress due to the operating force, F_o, from Equation (18–4):

$$\tau_o = \frac{8KF_oD_m}{\pi D_w^3} = \frac{(8)(1.15)(12.0)(0.60)}{(\pi)(0.0625)^3} = 86\ 450 \text{ psi}$$

Comparing this with the design stress of 145 000 psi, we see that it is safe.

Step 11. Compute the number of active coils required to give the proper deflection characteristics for the spring. Using Equation (18–6) and solving for N_a, we have

$$f = \frac{8FC^3N_a}{GD_w}$$

➪ **Number of Active Coils**

$$N_a = \frac{fGD_w}{8FC^3} = \frac{GD_w}{8kC^3} \quad (\textit{Note}: F/f = k, \text{ the spring rate}) \tag{18–8}$$

Then, for this problem,

$$N_a = \frac{GD_w}{8kC^3} = \frac{(11\ 200\ 000)(0.0625)}{(8)(8.0)(9.60)^3} = 12.36 \text{ coils}$$

Notice that $k = 8.0$ lb/in is the spring rate. Do not confuse this with K, the Wahl factor.

Step 12. Compute the solid length, L_s; the force on the spring at solid length, F_s; and the stress in the spring at solid length, τ_s. This computation will give the maximum stress that the spring will receive.

The solid length occurs when all of the coils are touching, but recall that there are two inactive coils for springs with squared and ground ends. Thus,

$$L_s = D_w(N_a + 2) = 0.0625(14.36) = 0.898 \text{ in}$$

The force at solid length is the product of the spring rate times the deflection to solid length ($L_f - L_s$):

$$F_s = k(L_f - L_s) = (8.0 \text{ lb/in})(2.75 - 0.898) \text{ in} = 14.8 \text{ lb}$$

Because the stress in the spring is directly proportional to the force, a simple method of computing the solid length stress is

$$\tau_s = (\tau_o)(F_s/F_o) = (86\ 450 \text{ psi})(14.8/12.0) = 106\ 750 \text{ psi}$$

When this value is compared with the maximum allowable stress of 170 000 psi, we see that it is safe, and the spring will not yield when compressed to solid length.

Step 13. Complete the computations of geometric features, and compare them with space and operational limitations:

$$OD = D_m + D_w = 0.60 + 0.0625 = 0.663 \text{ in}$$

$$ID = D_m - D_w = 0.60 - 0.0625 = 0.538 \text{ in}$$

These dimensions are satisfactory for installation in a hole having a diameter of 0.75 in.

Step 14. The tendency to buckle is checked, along with the coil clearance.

Buckling check: $L_f/D_m = 2.75/0.60 = 4.58$
Because this ratio is less than 5.2, buckling will not occur.

Coil clearance: $cc = (L_o - L_s)/N_a = (1.25 - 0.898)/12.36 = 0.029$ in
Check: $D_w/10 = 0.0625/10 = 0.00625$ in < 0.029 in
Coil clearance is acceptable.

$D_{hole} > OD + D_w/10 = 0.663 + 0.00625 = 0.669$ in
The specified $D_{hole} = 0.750$ in is acceptable.

This procedure completes the design of one satisfactory spring for this application. It may be desirable to make other trials to attempt to find a more nearly optimum spring.

Spreadsheet for Spring Design Method 1

The 14 steps required to complete one trial design using Method 1 which was demonstrated in Example Problem 18–2 are fairly involved, tedious, and time consuming. Furthermore, it is highly likely that several iterations are required to produce an optimum solution that meets application considerations of the physical size of the spring, acceptable levels of stress under all loads, cost, and other factors. You might want to investigate the use of different materials for the same basic design goals for the forces, lengths, and spring rate.

For these and many other reasons, it is recommended that computer-assisted design approaches be developed to perform most of the calculations and to guide you through the solution procedure. This could be done with a computer program, a spreadsheet, mathematical analysis software, or a programmable calculator. Once written, the program or spreadsheet can be used for any similar design problem in the future by yourself or others.

Figure 18–16 shows one approach using a spreadsheet with the data from Example Problem 18–2 used for illustration. This approach is attractive because the entire solution is presented on one page, and the user is guided through the solution procedure. Let's summarize the use of this spreadsheet. As you read this, you should compare the spreadsheet entries with the details of the solution of Example Problem 18–2. The various formulas used there are programmed into the appropriate cells of the spreadsheet.

1. The process begins, as with virtually any spring design procedure, with the specification of the relationship between force and length for two separate conditions, typically called the *operating force-length* and the *installed force-length*. From these data, the spring rate is effectively specified as well. Sometimes the free length is known, which then equals the installed length, and the installed force equals zero. Also, the designer would know roughly the space into which the spring is to be installed.
2. The heading for the spreadsheet gives a brief overview of the design approach built into Method 1. The designer specifies a target mean diameter for the spring to fit a particular application. The spring material is specified, and the graph of the strength of that material is used as a guide for estimating the design stress. This requires the user to make a rough estimate also of the wire diameter, but a specific size is not yet chosen. The service (light, average, or severe) must also be specified at this time.
3. The spreadsheet then computes the resulting spring rate, the free length, and a trial wire diameter to produce an acceptable stress. Equation (18–7) is used to compute the wire size.
4. The designer then enters a standard wire size, typically greater than the computed value. Table 18–2 is a listing of standard wire sizes. At this time, also, the designer must look again at the design stress graph for the chosen material and must determine a revised value for the design stress corresponding with the new wire diameter. The maximum allowable stress is also entered, found from the light service curve for the material at the specified wire size.
5. With the data entered, the spreadsheet completes the entire set of remaining calculations. The spring must be manufactured with exactly the specified diameters and number of active coils. The end condition for the spring, from the possibilities shown in Figure 18–3, should also be specified.
6. The designer's task is to evaluate the suitability of the results for basic geometry, stresses, potential for buckling, coil clearance, and installation of the spring into a hole. The **Comments** section to the right includes several prompts. But this is the designer's responsibility: to make judgments and design decisions.

Notice the advantage of using a spreadsheet: The designer does the thinking, and the spreadsheet does the computing. For subsequent design iterations, only the values that change need to be entered. For example, if the designer wants to try a different wire diameter for the same spring material, only the three data values in the section called **Secondary Input Data** need to be changed, and a new result is produced immediately. Many design iterations can be completed in a short time with this approach.

You might see ways to enhance the utility of the spreadsheet, and you are encouraged to do that.

Spreadsheet for Spring Design Method 2

An alternative spring design method is shown in Figure 18–17. This method allows the designer more

Design of helical compression spring—Method 1 **Specify mean diameter and design stresses. Compute wire diameter and number of coils.**			
1. Enter data for forces and lengths. **2. Specify material; shear modulus, G; and an estimate for design stress.** **3. Enter trial mean diameter for spring, considering space available.** **4. Check computed values for spring rate, free length, and new trial wire diameter.** **5. Enter your choice for wire diameter of a standard size.** **6. Enter design stress and maximum allowable stress from Figures 18–8 through 18–13 for new D_w.**			
Numerical values in italics in shaded areas must be inserted for each problem.	Problem ID:		Example Problem 18–2
Initial Input Data:			**Comments**
Maximum operating force =	F_o =	*12.0 lb*	
Operating length =	L_o =	*1.25 in*	
Installed force =	F_i =	*8.0 lb*	*Note:* $F_i = 0$, if
Installed length =	L_i =	*1.75 in*	L_i = free length.
Trial mean diameter =	D_m =	*0.60 in*	
Type of spring wire:	*ASTM A231 steel*		See Figures 18–8 to 18–13.
Shear modulus of elasticity of spring wire =	G =	*1.12E + 07 psi*	From Table 18–4
Initial estimate of design stress =	τ_{di} =	*130 000 psi*	From design stress graph
Computed Values:			
Computed spring rate =	k =	8.00 lb/in	
Computed free length =	L_f =	2.75 in	
Computed trial wire diameter =	D_{wt} =	0.055 in	
Secondary Input Data:			
Standard wire diameter =	D_w =	*0.0625 in*	From Table 18–2
Design stress =	τ_d =	*145 000 psi*	From design stress graph
Maximum allowable stress =	τ_{max} =	*170 000 psi*	Use light service curve.
Computed Values:			
Outside diameter =	D_o =	0.663 in	
Inside diameter =	D_i =	0.538 in	
Number of active coils =	N_a =	12.36	
Spring index =	C =	9.60	**Should not be < 5.0**
Wahl factor =	K =	1.15	
Stress at operating force =	τ_o =	86 459 psi	**Cannot be > 145 000**
Solid length =	L_s =	0.898 in	**Cannot be > 1.25**
Force at solid length =	F_s =	14.82 lb	
Stress at solid length =	τ_s =	106 768 psi	**Cannot be > 170 000**
Check Buckling, Coil Clearance, and Hole Size:			
Buckling: Ratio =	L_f/D_m =	4.58	Check Figure 18–15 if >5.2.
Coil clearance =	cc =	0.029 in	Should be > 0.00 625
If installed in hole, minimum hole diameter =	D_{hole} >	0.669 in	For side clearance

FIGURE 18–16 Spreadsheet for spring design Method 1 of Example Problem 18–2

freedom in manipulating the parameters. The data used are basically the same as those of Example Problem 18–2 except that there is no requirement for installing the spring in a certain size hole. We refer to this modified problem statement as Example Problem 18–3.

Example Problem 18–3

The process is similar to that of Method 1, so only the major differences are discussed below. Follow along with the spreadsheet in Figure 18–17 as we describe Method 2, Trials 1 and 2.

Solution Method 2, Trial 1

1. The general procedure is outlined at the top of the spreadsheet. The designer selects a material, estimates the wire diameter as an initial trial, and inputs a corresponding estimate of the design stress.
2. The spreadsheet then computes a new trial wire diameter using a formula derived from the fundamental equation for shear stress in a helical compression spring, Equation (18–4). The development is described here.

$$\tau = \frac{8KFC}{\pi D_w^2} \tag{18–4}$$

Let $F = F_o$ and $\tau = \tau_d$ (the design stress). Solving for the wire diameter gives

$$D_w = \sqrt{\frac{8KF_oC}{\pi\tau_d}} \tag{18–9}$$

The values of K and C are not yet known, but a good estimate for the wire diameter can be computed if the spring index is assumed to be approximately 7.0, a reasonable value. The corresponding value for the Wahl factor is $K = 1.2$, from Equation (18–5). Combining these assumed values with the other constants in the preceding equation gives

$$D_w = \sqrt{21.4(F_o)/(\tau_d)} \tag{18–10}$$

This formula is programmed into the spreadsheet in the cell to the right of the one labeled D_{wt}, the computed trial wire diameter.

3. The designer then enters a standard wire size and determines revised values for the design stress and the maximum allowable stress from the graphs of material properties, Figures 18–8 through 18–13.
4. The spreadsheet then computes the maximum permissible number of active coils for the spring. The logic here is that *the solid length must be less than the operating length*. The solid length is the product of the wire diameter and the total number of coils. For squared and ground ends, this is

$$L_s = D_w(N_a + 2)$$

Note that different relationships are used for the total number of coils for springs with other end conditions. See the discussion "Number of Coils" in Section 18–3. Now letting $L_s = L_o$ as a limit and solving for the number of coils gives

$$(N_a)_{\max} = (L_o - 2D_w)/D_w \tag{18–11}$$

This is the formula programmed into the spreadsheet cell to the right of the one labeled $N_{\max}$.

5. The designer now has the freedom to choose any number of active coils less than the computed maximum value. Note the effects of that decision. Choosing a small number of coils will provide more clearance between adjacent coils and will use less wire per spring. However, the stresses produced for a given load will be higher, so there is a practical limit. One approach is to try progressively fewer coils until the stress approaches the design stress. Whereas any number of coils can be produced, even fractional numbers, we suggest trying integer values for the convenience of the manufacturer.
6. After the selected number of coils has been entered, the spreadsheet can complete the remaining calculations. One additional new formula is used in this spreadsheet to compute the value of the spring index, C. It is developed from the second form of Equation (18–6) relating the deflection

Design of helical compression spring—Method 2 Specify wire diameter, design stresses, and number of coils. Compute mean diameter.				
1. Enter data for forces and lengths. 2. Specify material; shear modulus, G; and an estimate for design stress. 3. Enter trial wire diameter. 4. Check computed values for spring rate, free length, and new trial wire diameter. 5. Enter your choice for wire diameter of a standard size. 6. Enter design stress and maximum allowable stress from Figures 18–8 through 18–13 for new D_w. 7. Check computed maximum number of coils. Enter selected actual number of coils.				
Numerical values in italics in shaded areas must be inserted for each problem.	Problem ID:			Example Problem 18–3, Trial 1:
Initial Input Data:				**Comments**
Maximum operating force =	F_o =		*12.0 lb*	
Operating length =	L_o =		*1.25 in*	
Installed force =	F_i =		*8.0 lb*	*Note:* $F_i = 0$, if
Installed length =	L_i =		*1.75 in*	L_i = free length.
Type of spring wire:	*ASTM A231 steel*			See Figures 18–8 to 18–13.
Shear modulus of elasticity of spring wire =	G =		*1.12E + 07 psi*	From Table 18–4
Initial estimate of design stress =	τ_{di} =		*144 000 psi*	From design stress graph
Initial trial wire diameter =	D_w =		*0.06 in*	
Computed Values:				
Computed spring rate =	k =		8.00 lb/in	
Computed free length =	L_f =		2.75 in	
Computed trial wire diameter =	D_{wt} =		0.042 in	
Secondary Input Data:				
Standard wire diameter =	D_w =		*0.0475 in*	From Table 18–2
Design stress =	τ_d =		*149 000 psi*	From design stress graph
Maximum allowable stress =	τ_{max} =		*174 000 psi*	Use light service curve.
Computed: Maximum number of coils =	N_{max} =		24.32	Suggest using an integer.
Input: Number of active coils =	N_a =		*22*	
Computed Values:				
Spring index =	C =		7.23	**Should not be < 5.0**
Wahl factor =	K =		1.21	
Mean diameter =	D_m =		0.343 in	
Outside diameter =	D_o =		0.391 in	
Inside diameter =	D_i =		0.296 in	
Solid length =	L_s =		1.140 in	**Cannot be > 1.25**
Stress at operating force =	τ_o =		118 030 psi	**Cannot be > 149 000**
Force at solid length =	F_s =		12.88 lb	
Strength at solid length =	τ_s =		126 685 psi	**Cannot be > 149 000**
Check Buckling, Coil Clearance, and Hole Size:				
Buckling: Ratio =	Lf/Dm =		8.01	Check Figure 18–15 if > 5.2.
Coil clearance =	cc =		0.0050 in	Should be > 0.00475
If installed in hole, minimum hole diameter =	D_{hole} >		0.396 in	For side clearance

FIGURE 18–17 Spreadsheet for spring design Method 2 of Example Problem 18–3, Trial 1

for the spring, f, to a corresponding applied force, F, the value of C, and other parameters that are already known. First we solve for C^3:

➪ **Deflection of a Spring**

$$f = \frac{8FD_m^3N_a}{GD_w^4} = \frac{8FC^3N_a}{GD_w}$$

$$C^3 = \frac{fGD_w}{8FN_a}$$

Now notice that we have the force F in the denominator and the corresponding deflection f in the numerator. But the spring rate k is defined as the ratio of F/f. Then we can substitute k into the denominator and solve for C:

$$C = \left[\frac{GD_w}{8kN_a}\right]^{1/3} \tag{18–12}$$

This formula is programmed into the spreadsheet cell to the right of the one labeled $C =$.

7. Recall that C is defined as the ratio, D_m/D_w. We can now solve for the mean diameter:

$$D_m = CD_w$$

This is used to compute the mean diameter in the cell to the right of $D_m =$.

8. The remaining calculations use equations already developed and used earlier. Again, the designer is responsible for evaluating the suitability of the results and for performing any additional iterations to search for an optimum result.

Solution Method 2, Trial 2

Now notice that the solution obtained in Trial 1 and shown in Figure 18–17 is far from optimum. Its free length of 2.75 in is quite long compared with the mean diameter of 0.343 in. The buckling ratio of $L_f/D_m = 8.01$ indicates that the spring is long and slender. Checking Figure 18–15, we can see that buckling is predicted.

One way to work toward a more suitable geometry is to increase the wire diameter and reduce the number of coils. The net result will be a larger mean diameter, improving the buckling ratio.

Figure 18–18 shows the result of several iterations, finally using $D_w = 0.0625$ in (larger than the former value of 0.0475 in) and 16 active coils (down from 22 in the first trial). The buckling ratio is down to 4.99, indicating that buckling is unlikely. The stress at operating force is comfortably lower than the design stress. The other geometrical features also appear to be satisfactory.

This example should demonstrate the value of using spreadsheets or other computer-based computational aids. You should also have a better feel for the kinds of design decisions that can work toward a more optimum design. (See Internet sites 1, 2, and 15 for commercially available spring design software.)

18–7 EXTENSION SPRINGS

Extension springs are designed to exert a pulling force and to store energy. They are made from closely coiled helical coils similar in appearance to helical compression springs. Most extension springs are made with adjacent coils touching in such a manner that an initial force must be applied to separate the coils. Once the coils are separated, the force is linearly proportional to the deflection, as it is for helical compression springs. Figure 18–19 shows a typical extension spring, and Figure 18–20 shows the characteristic type of load-deflection curve. By convention, the initial force is found by projecting the straight line portion of the curve back to zero deflection.

The stresses and deflections for an extension spring can be computed by using the formulas used for compression springs. Equation (18–4) is used for the torsional shear stress, Equation (18–5) for the Wahl factor to account for the curvature of the wire and the direct shear stress, and Equation (18–6) for the deflection characteristics. All coils in an extension spring are active. In addition, since the end loops or hooks deflect, their deflection may affect the actual spring rate.

The initial tension in an extension spring is typically 10% to 25% of the maximum design force. Figure 18–21 shows one manufacturer's recommendation of the preferred torsional stress due to initial tension as a function of the spring index.

End Configurations for Extension Springs

A wide variety of end configurations may be obtained for attaching the spring to mating machine elements, some of which are shown in Figure 18–4. The cost of the spring can be greatly affected by its end type, and

Design of helical compression spring—Method 2 Specify wire diameter, design stresses, and number of coils. Compute mean diameter.				
1. Enter data for forces and lengths. 2. Specify material; shear modulus, G; and an estimate for design stress. 3. Enter trial wire diameter. 4. Check computed values for spring rate, free length, and new trial wire diameter. 5. Enter your choice for wire diameter of a standard size. 6. Enter design stress and maximum allowable stress from Figures 18–8 through 18–13 for new D_w. 7. Check computed maximum number of coils. Enter selected actual number of coils.				
Numerical values in italics in shaded areas must be inserted for each problem.		Problem ID:		*Example Problem 18–3,* Trial 2: D_w = 0.0625
Initial Input Data:				**Comments**
Maximum operating force	=	F_o =	*12.0 lb*	
Operating length	=	L_o =	*1.25 in*	
Installed force	=	F_i =	*8.0 lb*	*Note:* $F_i = 0$ if:
Installed length	=	L_i =	*1.75 in*	L_i = free length
Type of spring wire:		*ASTM A231 steel*		See Figures 18–8 to 18–13.
Shear modulus of elasticity of spring wire	=	G =	*1.12E + 07 psi*	From Table 18–4
Initial estimate of design stress	=	τ_{di} =	*144 000 psi*	From design stress graph
Initial trial wire diameter	=	D_w =	*0.06 in*	
Computed Values:				
Computed spring rate	=	k =	8.00 lb/in	
Computed free length	=	L_f =	2.75 in	
Computed trial wire diameter	=	D_{wt} =	0.042 in	
Secondary Input Data:				
Standard wire diameter	=	D_w =	*0.0625 in*	From Table 18–2
Design stress	=	τ_d =	*142 600 psi*	From design stress graph
Maximum allowable stress	=	τ_{max} =	*167 400 psi*	Use light service curve.
Computed: Maximum number of active coils	=	$(N_a)_{max}$ =	18.00	
Input: Number of active coils	=	N_a =	*16*	Suggest using an integer.
Computed Values:				
Spring index	=	C =	8.81	**Should not be < 5.0**
Wahl factor	=	K =	1.17	
Mean diameter	=	D_m =	0.551 in	
Outside diameter	=	D_o =	0.613 in	
Inside diameter	=	D_i =	0.488 in	
Solid length	=	L_s =	1.125 in	**Cannot be > 1.25**
Stress at operating force	=	τ_o =	80 341 psi	**Cannot be > 142 600**
Force at solid length	=	F_s =	13.00 lb	
Stress at solid length	=	τ_s =	87 036 psi	**Cannot be > 167 400**
Check Buckling, Coil Clearance, and Hole Size:				
Buckling: Ratio	=	L_f/D_m =	4.99	Check Figure 18–15 if > 5.2.
Coil clearance	=	cc =	0.0078 in	Should be > 0.00625
If installed in hole, minimum hole diameter	=	D_{hole} >	0.619 in	For side clearance

FIGURE 18–18 Spreadsheet for spring design Method 2 of Example Problem 18–3, Trial 2

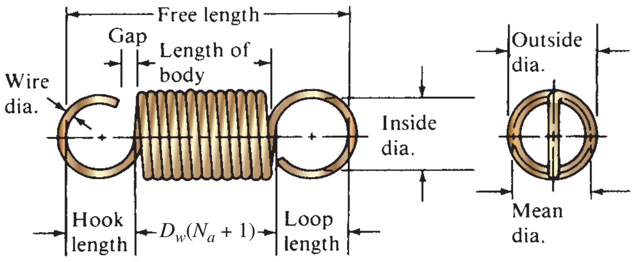


FIGURE 18–19 Extension spring

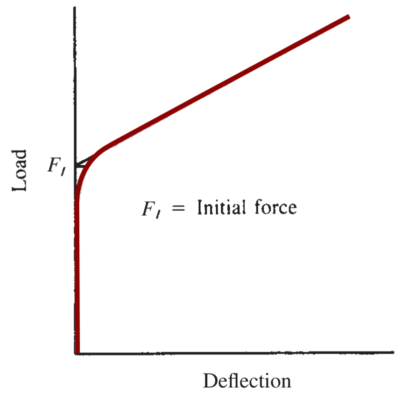


FIGURE 18–20 Load-deflection curve for an extension spring

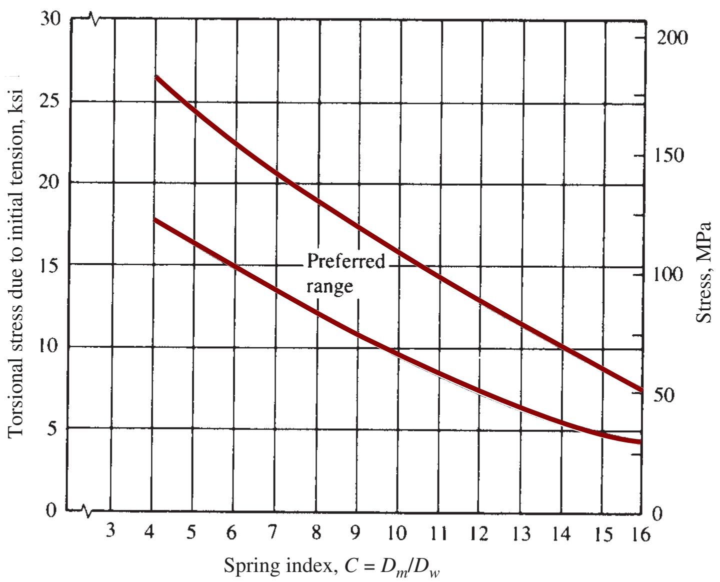


FIGURE 18–21 Recommended torsional shear stress in an extension spring due to initial tension (Data from Associated Spring, Barnes Group, Inc.)

it is recommended that the manufacturer be consulted before the ends are specified.

Frequently, the weakest part of an extension spring is its end, especially in fatigue loading cases. The loop end shown in Figure 18–22, for example, has a high bending stress at point A and a torsional shear stress at point B. Approximations for the stresses at these points can be computed as follows:

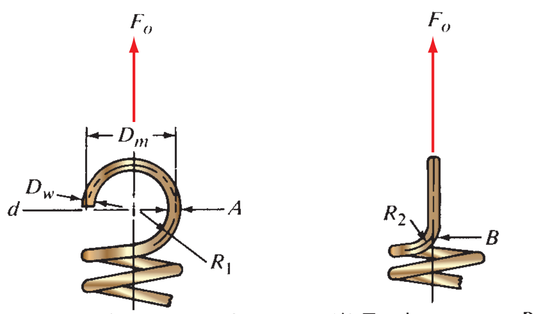


FIGURE 18–22 Stresses in ends of extension spring

Bending Stress at *A*

$$\sigma_A = \frac{16 D_m F_o K_1}{\pi D_w^3} + \frac{4F_o}{\pi D_w^2} \tag{18–13}$$

$$K_1 = \frac{4C_1^2 - C_1 - 1}{4C_1(C_1 - 1)} \tag{18–14}$$

$$C_1 = 2R_1/D_w$$

Torsional Stress at *B*

$$\tau_B = \frac{8 D_m F_o K_2}{\pi D_w^3} \tag{18–15}$$

$$K_2 = \frac{4C_2 - 1}{4C_2 - 4} \tag{18–16}$$

$$C_2 = 2R_2/D_w$$

The ratios C_1 and C_2 relate to the curvature of the wire and should be large, typically greater than 4, to avoid high stresses.

Allowable Stresses for Extension Springs

The torsional shear stress in the coils of the spring and in the end loops can be compared to the curves in Figures 18–8 through 18–13. Some designers reduce these allowable stresses by about 10%. The bending stress in the end loops, such as that from Equation (18–13), should be compared to the allowable bending stresses for torsion springs, as presented in the next section.

Example Problem 18–4

A helical extension spring is to be installed in a latch for a large, commercial laundry machine. When the latch is closed, the spring must exert a force of 16.25 lb at a length between attachment points of 3.50 in. As the latch is opened, the spring is pulled to a length of 4.25 in with a maximum force of 26.75 lb. An outside diameter of 5/8 in (0.625 in) is desired. The latch will be cycled only about 10 times per day, and thus the design stress will be based on average service. Use ASTM A227 steel wire. Design the spring.

Solution

As before, the suggested design procedure will be given as numbered steps, followed by the calculations for this set of data.

Step 1. Assume a trial mean diameter and a trial design stress for the spring.

Let the mean diameter be 0.500 in. For ASTM A227 wire under average service, a design stress of 110 000 psi is reasonable (from Figure 18–8).

Step 2. Compute a trial wire diameter from Equation (18–4) for the maximum operating force, the assumed mean diameter and design stress, and an assumed value for K of about 1.20.

$$D_w = \left[\frac{8KF_oD_m}{\pi\tau_d}\right]^{1/3} = \left[\frac{(8)(1.20)(26.75)(0.50)}{(\pi)(110\,000)}\right]^{1/3} = 0.072 \text{ in}$$

A standard wire size in the U.S. Steel Wire Gage system is available in this size. Use 15-gage wire.

Step 3. Determine the actual design stress for the selected wire size.

From Figure 18–8, at a wire size of 0.072 in, the design stress is 120 000 psi.

Step 4. Compute the actual values for outside diameter, mean diameter, inside diameter, spring index, and the Wahl factor, K. These factors are the same as those defined for helical compression springs.

Let the outside diameter be as specified, 0.625 in. Then

$$D_m = OD - D_w = 0.625 - 0.072 = 0.553 \text{ in}$$

$$ID = OD - 2D_w = 0.625 - 2(0.072) = 0.481 \text{ in}$$

$$C = D_m/D_w = 0.553/0.072 = 7.68$$

$$K = \frac{4C - 1}{4C - 4} + \frac{0.615}{C} = \frac{4(7.68) - 1}{4(7.68) - 4} + \frac{0.615}{7.68} = 1.19$$

Step 5. Compute the actual expected stress in the spring wire under the operating load from Equation (18–4):

$$\tau_o = \frac{8KF_oD_m}{\pi D_w^3} = \frac{8(1.19)(26.75)(0.553)}{\pi(0.072)^3} = 120\,000 \text{ psi} \qquad \text{(okay)}$$

Step 6. Compute the required number of coils to produce the desired deflection characteristics. Solve Equation (18–6) for the number of coils, and substitute k = force/deflection = F/f:

$$k = \frac{26.75 - 16.25}{4.25 - 3.50} = 14.0 \text{ lb/in}$$

$$N_a = \frac{GD_w}{8C^3k} = \frac{(11.5 \times 10^6)(0.072)}{8(7.68)^3(14.0)} = 16.3 \text{ coils}$$

Step 7. Compute the body length for the spring, and propose a trial design for the ends.

$$\text{Body length} = D_w(N_a + 1) = (0.072)(16.3 + 1) = 1.25 \text{ in}$$

Let's propose to use a full loop at each end of the spring, adding a length equal to the *ID* of the spring at each end. Then the total free length is

$$L_f = \text{body length} + 2(ID) = 1.25 + 2(0.481) = 2.21 \text{ in}$$

***Step* 8.** Compute the deflection from free length to operating length:

$$f_o = L_o - L_f = 4.25 - 2.21 = 2.04 \text{ in}$$

***Step* 9.** Compute the initial force in the spring at which the coils just begin to separate. This is done by subtracting the amount of force due to the deflection, f_o:

$$F_I = F_o - kf_o = 26.75 - (14.0)(2.04) = -1.81 \text{ lb}$$

The negative force resulting from a free length that is too small for the specified conditions is clearly impossible.

Let's try $L_f = 2.50$ in, which will require a redesign of the end loops. Then

$$f_o = 4.25 - 2.50 = 1.75 \text{ in}$$

$$F_I = 26.75 - (14.0)(1.75) = 2.25 \text{ lb} \quad \text{(reasonable)}$$

***Step* 10.** Compute the stress in the spring under the initial tension, and compare it with the recommended levels in Figure 18–21.

Because the stress is proportional to the load,

$$\tau_I = \tau_o(F_I/F_o) = (120\,000)(2.25/26.75) = 10\,100 \text{ psi}$$

For $C = 7.68$, this stress is slightly below the preferred range from Figure 18–21. Some small adjustments could be made to bring the torsional shear stress due to initial tension within the desired range, producing a more optimal design. Making and testing a prototype may also be done.

The final configuration of the end loops must also be completed and analyzed for stress.

18–8 HELICAL TORSION SPRINGS

Many machine elements require a spring that exerts a rotational moment, or torque, instead of a push or a pull force. The helical torsion spring is designed to satisfy this requirement. The spring has the same general appearance as either the helical compression or the extension spring, with round wire wrapped into a cylindrical shape. Usually the coils are close together but with a small clearance, allowing no initial tension in the spring as there is for extension springs. Figure 18–23 shows a few examples of torsion springs with a variety of end treatments.

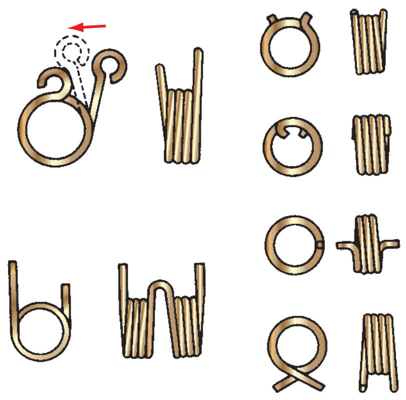


FIGURE 18–23 Torsion springs showing a variety of end types

The familiar clip-type clothespin uses a torsion spring to provide the gripping force. Many cabinet doors are designed to close automatically under the influence of a torsion spring. Some timers and switches use torsion springs to actuate mechanisms or close contacts. Torsion springs often provide counterbalancing of machine elements that are mounted to a hinged plate.

The following are some of the special features and design guides for torsion springs:

1. The moment applied to a torsion spring should always act in a direction that causes the coils to wind more tightly, rather than to open the spring up. This takes advantage of favorable residual stresses in the wire after forming.
2. In the free (no-load) condition, the definition of mean diameter, outside diameter, inside diameter, wire diameter, and spring index are the same as those used for compression springs.
3. As the load on a torsion spring increases, its mean diameter, D_m, decreases, and its length, L, increases, according to the following relations:

$$D_m = D_{mI}N_a/(N_a + \theta) \quad \textbf{(18–17)}$$

where D_{mI} = initial mean diameter at the free condition
N_a = number of active coils in the spring (to be defined later)
θ = angular deflection of the spring from the free condition, expressed in revolutions or a fraction of a revolution

$$L = D_w(N_a + 1 + \theta) \quad \textbf{(18–18)}$$

This equation assumes that all coils are touching. If any clearance is provided, as is often desirable to reduce friction, a length of N_a times the clearance must be added.

4. Torsion springs must be supported at three or more points. They are usually installed around a rod to provide location and to transfer reaction forces to the structure. The rod diameter should be approximately 90% of the *ID* of the spring at the maximum load.

Stress Calculations

The stress in the coils of a helical torsion spring is *bending stress* created because the applied moment tends to bend each coil into a smaller diameter. Thus, the stress is computed from a form of the flexure formula, $\sigma = Mc/I$, modified to account for the curved wire. Also, because most torsion springs are made from round wire, the section modulus I/c is $S = \pi D_w^3/32$. Then

$$\sigma = \frac{McK_b}{I} = \frac{MK_b}{S} = \frac{MK_b}{\pi D_w^3/32} = \frac{32MK_b}{\pi D_w^3} \quad \textbf{(18–19)}$$

K_b is the curvature correction factor and is reported by Wahl (see Reference 10) to be

$$K_b = \frac{4C^2 - C - 1}{4C(C - 1)} \quad \textbf{(18–20)}$$

where C is the spring index

Deflection, Spring Rate, and Number of Coils

The basic equation governing deflection is

$$\theta' = ML_w/EI$$

where θ' = angular deformation of the spring in radians (rad)
M = applied moment, or torque
L_w = length of wire in the spring
E = tensile modulus of elasticity
I = moment of inertia of the spring wire

We can substitute equations for L_w and I and convert θ' (radians) to θ (revolutions) to produce a more convenient form for application to torsion springs:

$$\theta = \frac{ML_w}{EI} = \frac{M(\pi D_m N_a)}{E(\pi D_w^4/64)}\,\frac{1\text{ rev}}{2\pi\text{ rad}} = \frac{10.2MD_mN_a}{ED_w^4} \quad \textbf{(18–21)}$$

To compute the spring rate, k_θ (moment per revolution), solve for M/θ:

$$k_\theta = \frac{M}{\theta} = \frac{ED_w^4}{10.2D_mN_a} \quad \textbf{(18–22)}$$

Friction between coils and between the *ID* of the spring and the guide rod may decrease the rate slightly from this value.

The number of coils, N_a, is composed of a combination of the number of coils in the body of the spring, called N_b, and the contribution of the ends as they are subjected to bending, also. Calling N_e the contribution of the ends having lengths L_1 and L_2, we have

$$N_e = (L_1 + L_2)/(3\pi D_m) \quad \textbf{(18–23)}$$

Then compute $N_a = N_b + N_e$.

Design Stresses

Because the stress in a torsion spring is bending and not torsional shear, the design stresses are different from those used for compression and extension springs. Figures 18–24 through 18–29 include six graphs of the design stress versus wire diameter for the same alloys used earlier for compression springs.

Design Procedure for Torsion Springs

A general procedure for designing torsion springs is presented and illustrated with Example Problem 18–5.

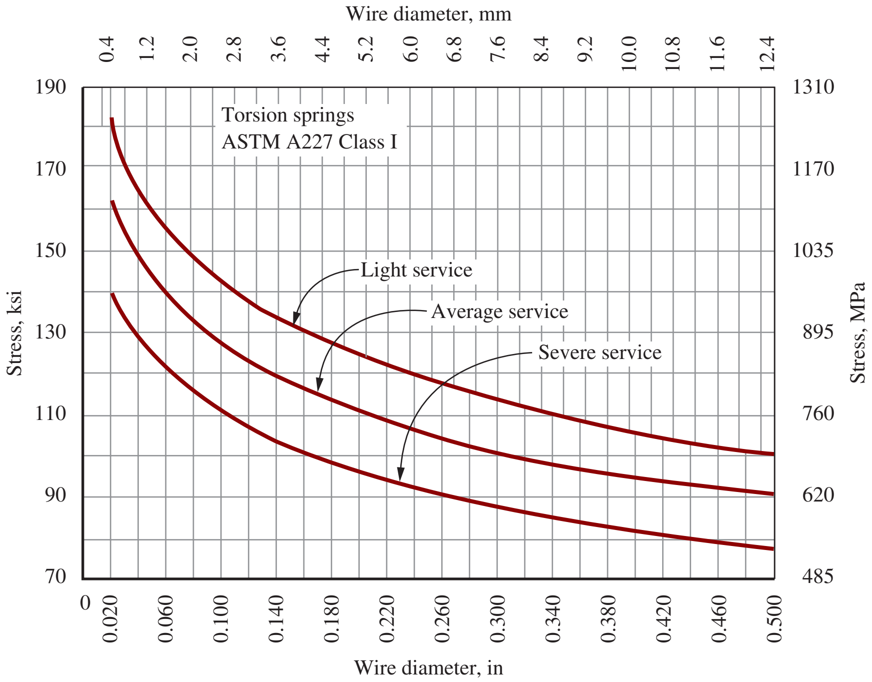


FIGURE 18–24 Design bending stresses for torsion springs ASTM A227 steel wire, hard-drawn

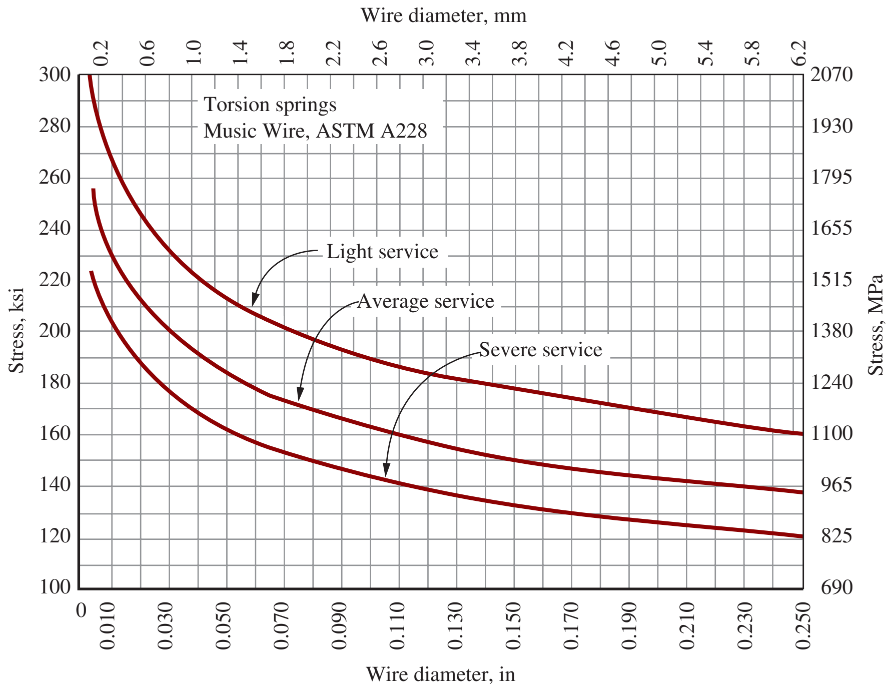


FIGURE 18–25 Design bending stresses for torsion springs ASTM A228 music wire

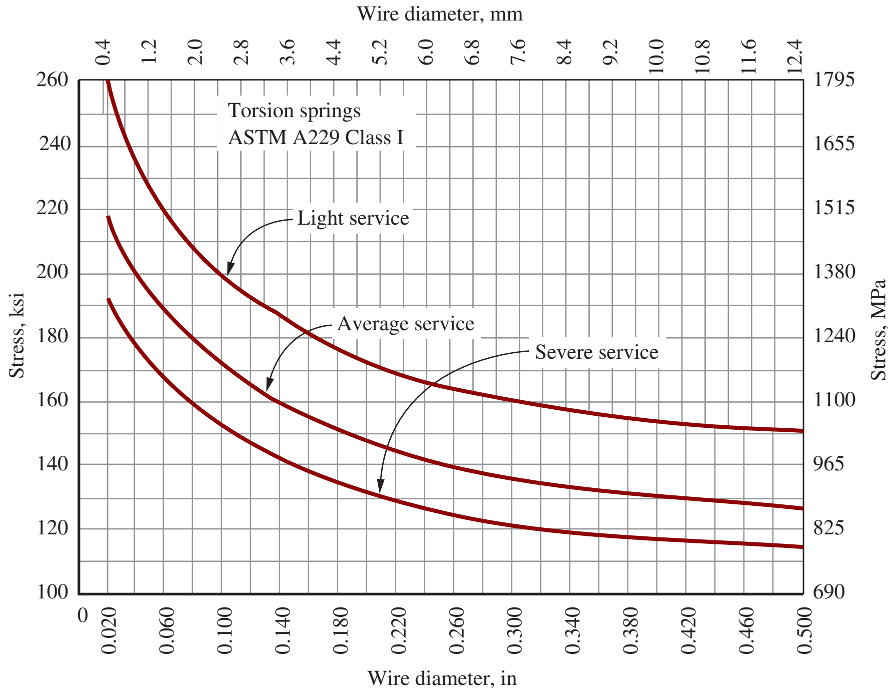


FIGURE 18–26 Design bending stresses for torsion springs ASTM A229 steel wire, oil-tempered, MB grade

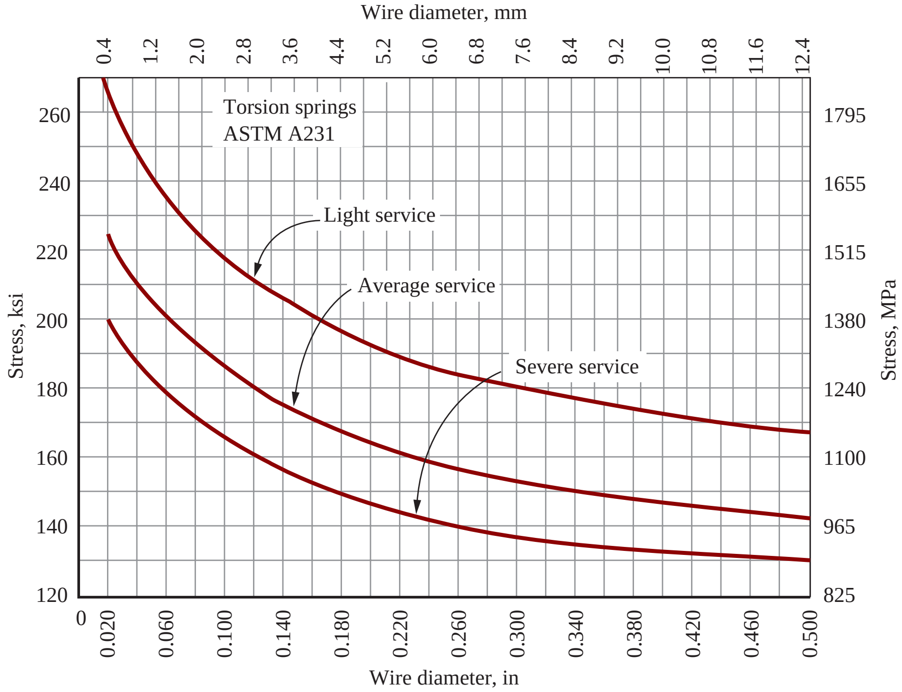


FIGURE 18–27 Design bending stresses for torsion springs ASTM A231 steel wire, chromium-vanadium alloy, valve-spring quality

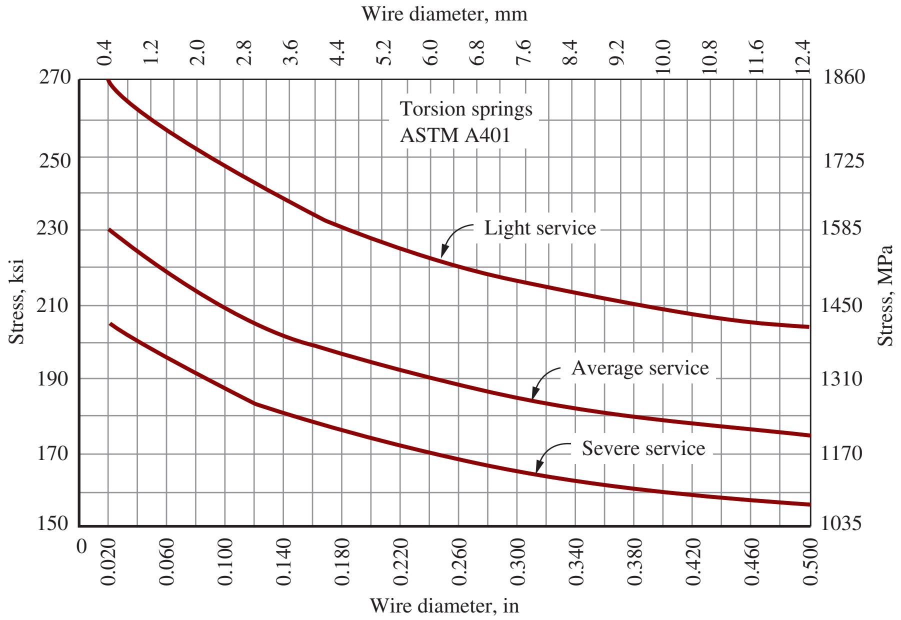


FIGURE 18–28 Design bending stresses for torsion springs ASTM A401 steel wire, chromium-silicon alloy, oil-tempered

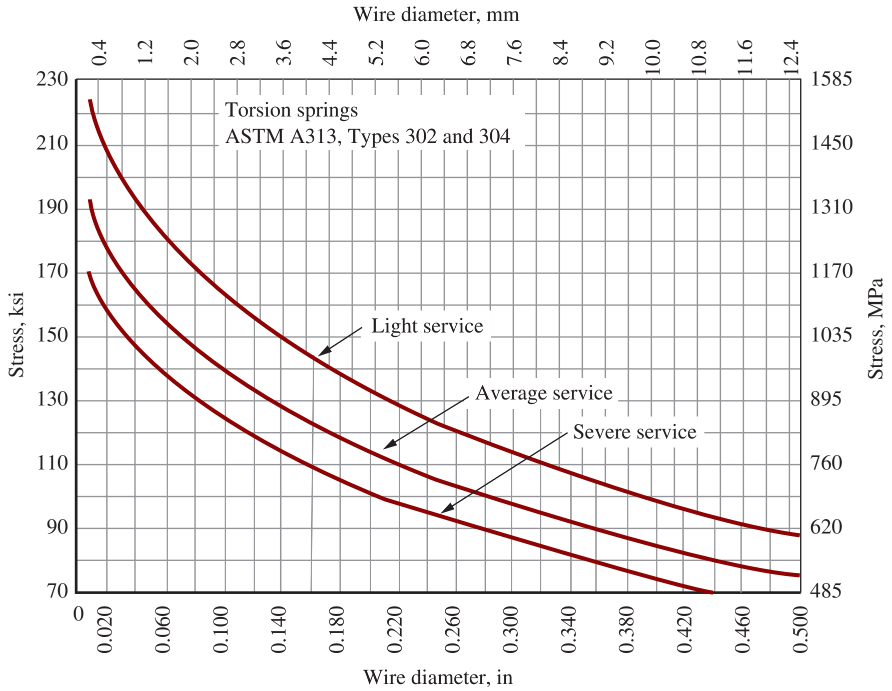


FIGURE 18–29 Design bending stresses for torsion springs ASTM A313 stainless steel wire, types 302 and 304, corrosion-resistant

Example Problem 18–5

A timer incorporates a mechanism to close a switch after the timer rotates one complete revolution. The switch contacts are actuated by a torsion spring that is to be designed. A cam on the timer shaft slowly moves a lever attached to one end of the spring to a point where the maximum torque on the spring is 3.00 lb·in. At the end of the revolution, the cam permits the lever to rotate 60° suddenly with the motion produced by the energy stored in the spring. At this new position, the torque on the spring is 1.60 lb·in. Because of space limitations, the *OD* of the spring should not be greater than 0.50 in, and the length should not be greater than 0.75 in. Use music wire, ASTM A228 steel wire. The number of cycles of the spring will be moderate, so use design stresses for average service.

Solution

Step 1. Assume a trial value for the mean diameter and an estimate for the design stress.

Let's use a mean diameter of 0.400 in and estimate the design stress for A228 music wire, average service, to be 180 000 psi (Figure 18–25).

Step 2. Solve Equation (18–19) for the wire diameter, compute a trial size, and select a standard wire size. Let $K_b = 1.15$ be an estimate. Also use the largest applied torque:

$$D_w = \left[\frac{32MK_b}{\pi\sigma_d}\right]^{1/3} = \left[\frac{32(3.0)(1.15)}{\pi(180\,000)}\right]^{1/3} = 0.058 \text{ in}$$

From Table 18–2, we can choose 25-gage music wire with a diameter of 0.059 in. For this size wire, the actual design stress for average service is 178 000 psi.

Step 3. Compute the *OD*, *ID*, spring index, and the new K_b:

$$OD = D_m + D_w = 0.400 + 0.059 = 0.459 \text{ in} \quad \text{(okay)}$$

$$ID = D_m - D_w = 0.400 - 0.059 = 0.341 \text{ in}$$

$$C = D_m/D_w = 0.400/0.059 = 6.78$$

$$K_b = \frac{4C^2 - C - 1}{4C(C-1)} = \frac{4(6.78)^2 - 6.78 - 1}{4(6.78)(6.78 - 1)} = 1.123$$

Step 4. Compute the actual expected stress from Equation (18–19):

$$\sigma = \frac{32MK_b}{\pi D_w^3} = \frac{32(3.0)(1.123)}{(\pi)(0.059)^3} = 167\,000 \text{ psi} \quad \text{(okay)}$$

Step 5. Compute the spring rate from the given data.

The torque exerted by the spring decreases from 3.00 to 1.60 lb·in as the spring rotates 60°. Convert 60° to a fraction of a revolution (rev):

$$\theta = \frac{60}{360} = 0.167 \text{ rev}$$

$$k_\theta = \frac{M}{\theta} = \frac{3.00 - 1.60}{0.167} = 8.40 \text{ lb·in/rev}$$

Step 6. Compute the required number of coils by solving for N_a from Equation (18–22):

$$N_a = \frac{ED_w^4}{10.2D_m k_\theta} = \frac{(29 \times 10^6)(0.059)^4}{(10.2)(0.400)(8.40)} = 10.3 \text{ coils}$$

Step 7. Compute the equivalent number of coils due to the ends of the spring from Equation (18–23).

This requires some design decisions. Let's use straight ends, 2.0 in long on one side and 1.0 in long on the other. These ends will be attached to the structure of the timer during operation. Then

$$N_e = (L_1 + L_2)/(3\pi D_m) = (2.0 + 1.0)/[3\pi(0.400)] = 0.80 \text{ coil}$$

Step 8. Compute the required number of coils in the body of the spring:

$$N_b = N_a - N_e = 10.3 - 0.8 = 9.5 \text{ coils}$$

Step 9. Complete the geometric design of the spring, including the size of the rod on which it will be mounted.

We first need the total angular deflection for the spring from the free condition to the maximum load. In this case, we know that the spring rotates 60° during operation. To this we must add the rotation from the free condition to the initial torque, 1.60 lb·in. Thus,

$$\theta_I = M_I/k_\theta = 1.60 \text{ lb} \cdot \text{in}/(8.4 \text{ lb} \cdot \text{in/rev}) = 0.19 \text{ rev}$$

Then the total rotation is

$$\theta_t = \theta_I + \theta = 0.19 + 0.167 = 0.357 \text{ rev}$$

From Equation (18–17), the mean diameter at the maximum operating torque is

$$D_m = D_{mI}N_a/(N_a + \theta_t) = [(0.400)(10.3)]/[(10.3 + 0.357)] = 0.387 \text{ in}$$

The minimum inside diameter is

$$ID_{\min} = 0.387 - D_w = 0.387 - 0.059 = 0.328 \text{ in}$$

The rod diameter on which the spring is mounted should be approximately 0.90 times this value. Then

$$D_r = 0.9(0.328) = 0.295 \text{ in} \quad \text{(say, 0.30 in)}$$

The length of the spring, assuming that all coils are originally touching, is computed from Equation (18–18):

$$L_{\max} = D_w(N_a + 1 + \theta_t) = (0.059)(10.3 + 1 + 0.356) = 0.688 \text{ in} \quad \text{(okay)}$$

This value for length is the maximum space required in the direction along the axis of the coil when the spring is fully actuated. The specifications allow an axial length of 0.75 in, so this design is acceptable.

18–9 IMPROVING SPRING PERFORMANCE BY SHOT PEENING AND LASER PEENING

The data presented in this chapter for spring wire materials are for commercially available spring wire with good surface finish. Care should be exercised to avoid nicks and scratches on the spring wire that may present sites for initiating fatigue cracks. When forming ends of extension springs or when creating other special geometry, bend radii should be as large as practical to avoid regions of high residual stress after the bending process.

Critical applications may call for the use of *shot peening* or *laser peening* to enhance the fatigue performance of springs and other machine elements. Examples are valve springs for engines, rapid cycling springs in automation equipment, clutch springs, aerospace applications, medical equipment, gears, pump impellers, and military systems. Shot peening is a process in which small hard pellets, called shot, are directed with high velocity at the surface to be treated. The shot causes small local plastic deformation to occur near the surface. The material below the deformed area is subsequently placed in compression as it tries to return the surface to its original shape. High residual compressive stresses are produced at the surface, a favorable situation. Fatigue cracks typically initiate at points of high tensile stress. Therefore, the residual compressive stress tends to preclude such cracks from starting, and the fatigue strength of the material is significantly increased.

Laser peening provides a similar improvement in the fatigue behavior of steels used for springs. A high-energy laser is fired at the surface of the metal spring wire which sends shock waves through the wire. Deep levels of compressive stress remain after laser peening treatment, enhancing the fatigue resistance of the spring, while also reducing corrosion failures.

(See Internet site 12 for more information about both shot peening and laser peening.)

18–10 SPRING MANUFACTURING

Spring forming machines are fascinating examples of high-speed, flexible, programmable, multiaxis, multifunction devices. Internet sites 6 and 7 show images of several types that include wire feeding systems, coiling wheels, and multiple forming heads. The systems use CNC (computer numerical control) to permit rapid setup and adjustment to produce automatically the varied geometries for the body and ends of springs such as those shown in Figures 18–2 and 18–4 . Furthermore, other complex shapes can be made for wire forms used for special clips, clamps, and brackets. Several YouTube videos can be found that show spring-making machines in operation.

Accessories to the spring forming machines are as follows:

- Systems to handle the large coils of raw wire, uncoiling it smoothly as needed.
- Straighteners that remove the curved shape of the coiled wire prior to forming the spring.

- Grinders to form the squared and ground ends of compression springs.
- Heat treating ovens that perform stress relieving on springs immediately after forming so they will maintain the specified geometry.
- Shot peening and laser peening equipment as described in Section 18–9.

General information on the spring-making industry can be found at Internet sites 1, 2, and 14. Manufacturers of spring-making equipment are listed at Internet sites 6 and 7.

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8. Spring Manufacturers Institute. *Handbook of Spring Design*. Oak Brook, IL: Spring Manufacturers Institute, 2002.
9. Spring Manufacturers Institute. *Encyclopedia of Spring Design*. Oak Brook, IL: Spring Manufacturers Institute. A set of four volumes are listed below:
   - Fundamentals of Spring Design
   - Compression, Extension, Torsion, and Garter Springs
   - Other Types of Springs
   - Testing and Tolerancing
10. Wahl, A. M. *Mechanical Springs*. 2nd ed. New York: McGraw-Hill, 1963. (Reprinted by Spring Manufacturers Institute in 1991 with permission of McGraw-Hill Book Co.)

INTERNET SITES RELATED TO SPRING DESIGN

1. **Spring Manufacturers Institute.** Industrial association serving spring manufacturers and providers of technical publications, spring design software, education, and a variety of services. The software called *Advanced Spring Design Software (ASD)* facilitates the design of numerous types of springs, including compression, extension, and torsion.
2. **Institute of Spring Technology.** Provides a wide range of research, development, problem solving, failure analysis, training, Spring Calculation software, materials and mechanical testing for the spring industry.
3. **Associated Spring–Barnes Group, Inc.** A large manufacturer of precision springs for industries such as transportation, telecommunications, electronics, home appliances, and farm equipment. Products include compression springs, extension springs, Belleville washers, retaining rings, and many others.
4. **Associated Spring Raymond.** Producer of a wide variety of springs and related components. Online catalog of stock compression, extension, and torsion springs, spring washers, gas springs, constant-force springs, retaining springs, and urethane composite springs.
5. **Century Spring Corporation.** Producer of a wide variety of springs. Online catalog of stock compression, extension, torsion, and die springs, urethane springs, and drawbar springs.
6. **Unidex Machinery Company.** Manufacturer of CNC spring forming machinery, coiling machines, and related equipment.
7. **Oriimec Corporation of America.** Manufacturer of multiaxis CNC spring forming machinery, CNC coiling machines, and tension spring machines.
8. **American Spring Wire Corporation.** Manufacturer of valve grade and commercial quality spring wire in carbon and alloy steels.
9. **Mapes Piano String Company.** Manufacturer of spring wire, piano wire, and specialty wire for guitar strings. Spring wires include music wire (plain and coated), high tensile missile wire, and stainless steel.
10. **Little Falls Alloys, Inc.** Manufacturer of nonferrous wire such as beryllium copper, brass, phosphor bronze, nickel, and cupro nickel. Product listings include mechanical and physical properties data.
11. **Alloy Wire International.** Manufacturer of round and shaped wire in superalloys such as Inconel®, Incoloy®, Monel® (tradenames of Special Metals Group of Companies), hastelloy (trade name of Haynes International), stainless steel, and titanium.
12. **Curtiss-Wright Surface Technologies.** Provider of shot peening, laser peening, and engineered coating services to aerospace, automotive, chemical, marine, agriculture, mining, and medical industries. Their processes improve the fatigue performance of springs, gears, and many other products. The website includes discussions of the technical aspects of their processes. From the home page, click on *Surface Technologies Division*.
13. **Lee Spring Company.** Manufacturer of springs and provider of a large selection of stock springs including compression, extension, torsion, plastic compression, Belleville washers, die springs, and custom springs.
14. **Wire Links.** An online directory for the wire and cable industry. Provides information on suppliers to the wire, cable, and spring-making industry. From the home page, search for *spring wire* under the "Keyword" feature.

15. **Spring Pro.** Professional spring design software for the design of compression, extension, torsion, conical, simple beam, and cantilever springs. Integrated use of U.S. and metric units. Generates CAD drawings of the designs.

PROBLEMS

Compression Springs

1. A spring has an overall length of 2.75 in when it is not loaded and a length of 1.85 in when carrying a load of 12.0 lb. Compute its spring rate.
2. A spring is loaded initially with a load of 4.65 lb and has a length of 1.25 in. The spring rate is given to be 18.8 lb/in. What is the free length of the spring?
3. A spring has a spring rate of 76.7 lb/in. At a load of 32.2 lb, it has a length of 0.830 in. Its solid length is 0.626 in. Compute the force required to compress the spring to solid height. Also compute the free length of the spring.
4. A spring has an overall length of 63.5 mm when it is not loaded and a length of 37.1 mm when carrying a load of 99.2 N. Compute its spring rate.
5. A spring is loaded initially with a load of 54.05 N, and it has a length of 39.47 mm. The spring rate is given to be 1.47 N/mm. What is the free length of the spring?
6. A spring has a spring rate of 8.95 N/mm. At a load of 134 N, it has a length of 29.4 mm. Its solid length is 21.4 mm. Compute the force required to compress the spring to solid height. Also compute the free length of the spring.
7. A helical compression spring with squared and ground ends has an outside diameter of 1.100 in, a wire diameter of 0.085 in, and a solid height of 0.563 in. Compute the *ID*, the mean diameter, the spring index, and the approximate number of coils.
8. The following data are known for a spring:
   Total number of coils = 19
   Squared and ground ends
   Outside diameter = 0.560 in
   Wire diameter = 0.059 in (25-gage music wire)
   Free length = 4.22 in
   For this spring, compute the spring index, the pitch, the pitch angle, and the solid length.
9. For the spring of Problem 8, compute the force required to reduce its length to 3.00 in. At that force, compute the stress in the spring. Would that stress be satisfactory for average service?
10. Would the spring of Problems 8 and 9 tend to buckle when compressed to a length of 3.00 in?
11. For the spring of Problem 8, compute the estimate for the outside diameter when it is compressed to solid length.
12. The spring of Problem 8 must be compressed to solid length to install it. What force is required to do this? Compute the stress at solid height. Is that stress satisfactory?
13. A support bar for a machine component is suspended to soften applied loads. During operation, the load on each spring varies from 180 to 220 lb. The position of the rod is to move no more than 0.500 in as the load varies. Design a compression spring for this application. Several million cycles of load application are expected. Use ASTM A229 steel wire.
14. Design a helical compression spring to exert a force of 22.0 lb when compressed to a length of 1.75 in. When its length is 3.00 in, it must exert a force of 5.0 lb. The spring will be cycled rapidly, with severe service required. Use ASTM A401 steel wire.
15. Design a helical compression spring for a pressure relief valve. When the valve is closed, the spring length is 2.0 in, and the spring force is to be 1.50 lb. As the pressure on the valve increases, a force of 14.0 lb causes the valve to open and compress the spring to a length of 1.25 in. Use corrosion-resistant ASTM A313, type 302 steel wire, and design for average service.
16. Design a helical compression spring to be used to return a pneumatic cylinder to its original position after being actuated. At a length of 10.50 in, the spring must exert a force of 60 lb. At a length of 4.00 in, it must exert a force of 250 lb. Severe service is expected. Use ASTM A231 steel wire.
17. Design a helical compression spring using music wire that will exert a force of 14.0 lb when its length is 0.68 in. The free length is to be 1.75 in. Use average service.
18. Design a helical compression spring using stainless steel wire, ASTM A313, type 316, for average service, which will exert a force of 8.00 lb after deflecting 1.75 in from a free length of 2.75 in.
19. Repeat Problem 18 with the additional requirement that the spring must operate around a rod having a diameter of 0.625 in.
20. Repeat Problem 17 with the additional requirement that the spring is to be installed in a hole having a diameter of 0.750 in.
21. Design a helical compression spring using ASTM A231 steel wire for severe service. The spring will exert a force of 45.0 lb at a length of 3.05 in and a force of 22.0 lb at a length of 3.50 in.
22. Design a helical compression spring using round steel wire, ASTM A227. The spring will actuate a clutch and must withstand several million cycles of operation. When the clutch discs are in contact, the spring will have a length of 2.50 in and must exert a force of 20 lb. When the clutch is disengaged, the spring will be 2.10 in long and must exert a force of 35 lb. The spring will be installed around a round shaft having a diameter of 1.50 in.
23. Design a helical compression spring using round steel wire, ASTM A227. The spring will actuate a clutch and must withstand several million cycles of operation. When the clutch discs are in contact, the spring will have a length of 60 mm and must exert a force of 90 N. When the clutch is disengaged, the spring will be 50 mm long and must exert a force of 155 N. The spring will be installed around a round shaft having a diameter of 38 mm.
24. Evaluate the performance of a helical compression spring made from 17-gage ASTM A229 steel wire that has an outside diameter of 0.531 in. It has a free length of 1.25 in, squared and ground ends, and a total of 7.0 coils. Compute the spring rate and the deflection and stress when carrying 10.0 lb. At this stress level, for what service (light, medium, or severe) would the spring be suitable?

Extension Springs

For Problems 25–31, be sure that the stress in the spring under initial tension is within the range suggested in Figure 18–21.

25. Design a helical extension spring using music wire to exert a force of 7.75 lb when the length between attachment points is 2.75 in, and a force of 5.25 lb at a length of 2.25 in. The outside diameter must be less than 0.300 in. Use severe service.
26. Design a helical extension spring for average service using music wire to exert a force of 15.0 lb when the length between attachment points is 5.00 in, and a force of 5.20 lb at a length of 3.75 in. The outside diameter must be less than 0.75 in.
27. Design a helical extension spring for severe service using music wire to exert a maximum force of 10.0 lb at a length of 3.00 in. The spring rate should be 6.80 lb/in. The outside diameter must be less than 0.75 in.
28. Design a helical extension spring for severe service using music wire to exert a maximum force of 10.0 lb at a length of 6.00 in. The spring rate should be 2.60 lb/in. The outside diameter must be less than 0.75 in.
29. Design a helical extension spring for average service using music wire to exert a maximum force of 10.0 lb at a length of 9.61 in. The spring rate should be 1.50 lb/in. The outside diameter must be less than 0.75 in.
30. Design a helical extension spring for average service using stainless steel wire, ASTM A313, type 302, to exert a maximum force of 162 lb at a length of 10.80 in. The spring rate should be 38.0 lb/in. The outside diameter should be approximately 1.75 in.
31. An extension spring has an end similar to that shown in Figure 18–22. The pertinent data are as follows: U.S. Wire Gage no. 19; mean diameter = 0.28 in; $R_1 = 0.25$ in; $R_2 = 0.094$ in. Compute the expected stresses at points *A* and *B* in the figure for a force of 5.0 lb. Would those stresses be satisfactory for ASTM A227 steel wire for average service?

Torsion Springs

32. Design a helical torsion spring for average service using stainless steel wire, ASTM A313, type 302, to exert a torque of 1.00 lb·in after a deflection of 180° from the free condition. The outside diameter of the coil should be no more than 0.500 in. Specify the diameter of a rod on which to mount the spring.
33. Design a helical torsion spring for severe service using stainless steel wire, ASTM A313, type 302, to exert a torque of 12.0 lb·in after a deflection of 270° from the free condition. The outside diameter of the coil should be no more than 1.250 in. Specify the diameter of a rod on which to mount the spring.
34. Design a helical torsion spring for severe service using music wire to exert a maximum torque of 2.50 lb·in after a deflection of 360° from the free condition. The outside diameter of the coil should be no more than 0.750 in. Specify the diameter of a rod on which to mount the spring.
35. A helical torsion spring has a wire diameter of 0.038 in; an outside diameter of 0.368 in; 9.5 coils in the body; one end 0.50 in long; the other end 1.125 in long; and a material of ASTM A401 steel. What torque would cause the spring to rotate 180°? What would the stress be then? Would it be safe?

CHAPTER NINETEEN

FASTENERS

THE BIG PICTURE Fasteners

Discussion Map

- *Fasteners* connect or join two or more components. Common types are *bolts* and *screws* such as those illustrated in Figure 19–1 through 19–4.

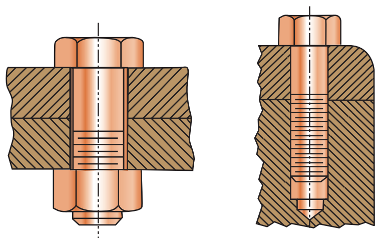
Discover

Look for examples of bolts and screws. List how many types you have found. For what functions were they being used? What kinds of forces are the fasteners subjected to? What materials are used for the fasteners?

In this chapter, you will learn to analyze the performance of fasteners and to select suitable types and sizes.

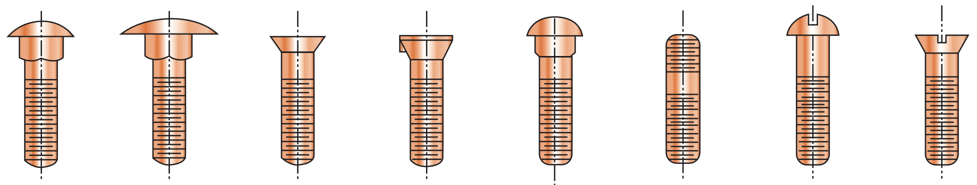
A *fastener* is any device used to connect or join two or more components. Literally hundreds of fastener types and variations are available. The most common are threaded fasteners referred to by many names, among them bolts, screws, nuts, studs, lag screws, and set screws.

A *bolt* is a threaded fastener designed to pass through holes in the mating members and to be secured by tightening a nut from the end opposite the head of the bolt. See Figure 19–1(a), called a *hex head bolt*. Several other types of bolts are shown in Figure 19–2.



(*a*) Hex head bolt (*b*) Hex head cap screw

FIGURE 19–1 Comparison of a bolt with a screw



(*a*) Carriage bolt (*b*) Elevator bolt (*c*) Coundtersunk bolt (*d*) Plow bolt (*e*) Track bolt (*f*) Stud bolt (*g*) Stove bolt (*h*) Stove bolt

FIGURE 19–2 Bolt styles. See also the hex head bolt in Figure 19–1

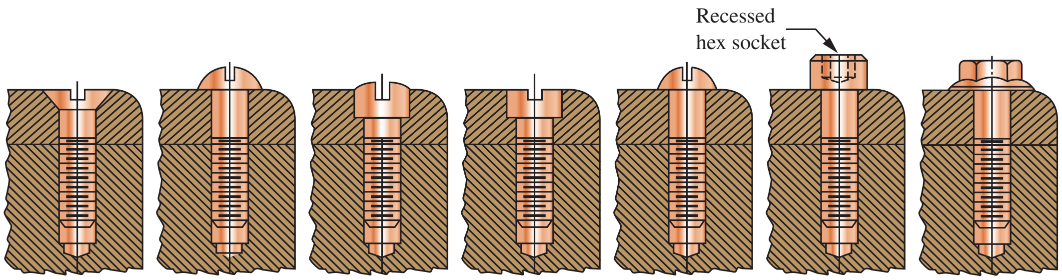


FIGURE 19–3 Cap screws or machine screws. See also the hex head cap screw in Figure 19–1

A *screw* is a threaded fastener designed to be inserted through a hole in one member to be joined and into a threaded hole in the mating member. See Figure 19–1(b). The threaded hole may have been preformed, for example, by tapping, or it may be formed by the screw itself as it is forced into the material. *Machine screws*, also called *cap screws*, are precision fasteners with straight-threaded bodies that are turned into tapped holes (see Figure 19–3). A popular type of machine screw is the socket head cap screw. The usual configuration, shown in Figure 19–3(f), has a cylindrical head with a recessed hex socket. Also readily available are flat head styles for countersinking to produce a flush surface, button head styles for a low-profile appearance, and shoulder screws providing a precision bearing surface for location or pivoting. See Internet sites 9 and 11. *Sheet-metal screws, lag screws, self-tapping screws*, and *wood screws* usually form their own threads. Figure 19–4 shows a few styles.

Search for examples where the kinds of fasteners illustrated in Figure 19–1 through 19–4 are used. How many can you find? Make a list using the names for the fasteners in the figures. Describe the application. What function is the fastener performing? What kinds of forces are exerted on each fastener during service? How large is the fastener? Measure as many dimensions as you can. What material is each fastener made from?

Look in your car, particularly under the hood in the engine compartment. If you can, also look under the chassis to see where fasteners are used to hold different components onto the frame or some other structural member.

Look also at bicycles, lawn and garden equipment, grocery carts, display units in a store, hand tools, kitchen appliances, toys, exercise equipment, and furniture. If you have access to a factory, you should be able to identify hundreds or thousands of examples. Try to get some insight about where certain types of fasteners are used and for what purposes.

In this chapter, you will learn about many of the types of fasteners that you will encounter, including how to analyze their performance.

References 1–7 and Internet sites 1, 2, 14, 15, and 18 provide extensive coverage of the science and technology of fasteners and are recommended as sources to supplement the treatment in this book.

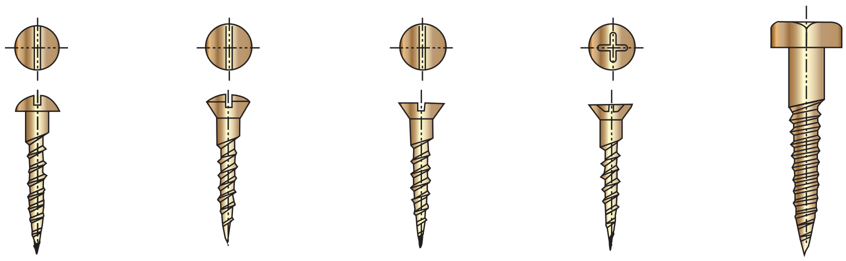


FIGURE 19–4 Sheet-metal and lag screws

YOU ARE THE DESIGNER

Review Figure 15–8 which shows the assembly of the gear-type power transmission that was designed in that chapter. Fasteners are called for in several places on the housing for the transmission, but they were not specified in that chapter. The four *bearing caps* are to be fastened to the housing and the cover by threaded fasteners. The cover itself is to be attached to the housing by fasteners. Finally, the mounting base has provisions for using fasteners to hold the entire transmission to a support structure.

You are the designer. What kinds of fasteners would you consider for these applications? What material should be used to make them? What strength should the material have? If threaded fasteners are used, what size should the threads be, and how long must they be? What head style would you specify? How much torque should be applied to the fastener to ensure that there is sufficient clamping force between the joined members? How does the design of the gasket between the cover and the housing affect the choice of the fasteners and the specification of the tightening torque for them? What alternatives are there to the use of threaded fasteners to hold the components together and to still allow disassembly?

This chapter presents information that you can use to make such design decisions. The references at the end of this chapter give other valuable sources of information from the large body of knowledge about fasteners. ■

19–1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Describe a bolt in comparison with a machine screw.
2. Name and describe nine styles of heads for bolts.
3. Name and describe six styles of heads for machine screws.
4. Describe sheet-metal screws and lag screws.
5. Describe six styles of set screws and their application.
6. Describe nine types of locking devices that restrain a nut from becoming loose on a bolt.
7. Use tables of data for various grades of steel materials used for bolts as published by the SAE International (SAE) and the ASTM International (ASTM), and for standard metric grades.
8. List at least 10 materials other than steel that are used for fasteners.
9. Use tables of data for standard screw threads in the American Standard and metric systems for dimensions and stress analysis.
10. Define *proof load, clamping load*, and *tightening torque* as applied to bolts and screws, and compute design values.
11. Compute the effect of adding an externally applied force on a bolted joint, including the final force on the bolts and the clamped members.
12. List and describe 16 different coating and finishing techniques that are used for metal fasteners.
13. Describe rivets, quick-operating fasteners, welding, brazing, and adhesives, and contrast them with bolts and screws for fastening applications.

19–2 BOLT MATERIALS AND STRENGTH

In machine design, most fasteners are made from steel because of its high strength, high stiffness, good ductility, and good machinability and formability. But varying compositions and conditions of steel are used. The strength of steels used for bolts and screws is used to determine its *grade*, according to one of several standards. Three strength ratings are frequently available: the familiar tensile strength and yield strength plus the proof strength. The *proof strength*, similar to the elastic limit, is defined as the stress at which the bolt or the screw would undergo permanent deformation. It usually ranges between 0.90 and 0.95 times the yield strength.

Three major standards-setting organizations relevant to fasteners in the United States and internationally are listed in References 1, 2, 11, and 17–28. See Internet sites 1, 2, 14, 15, and 18. Note that some of the listed standards are for metric fasteners whereas most others focus on the U.S. inch-pound system and a few address both units. The fastener-producing industry is represented by the Industrial Fasteners Institute that publishes a comprehensive document listed in Reference 10. There is much cross-referencing among these organizations. It is essential that designers are familiar with the entire content of the latest versions of applicable standards that present far more information than is mentioned in this book.

You should become familiar with a variety of vendors of commercially available screws, bolts, and other types of fasteners. As a sample, check out Internet sites 3–13.

Table 19–1 lists selected basic data for four SAE grades of fasteners from Reference 18 with the tensile strength generally increasing with higher grade numbers. Take note that strengths are related to specified ranges of sizes of fasteners. Some grades are made with special marking on the heads to assist users in the field to ensure that the proper grade is being used.

Table 19–2 lists data extracted from six different ASTM standards, each focused on a particular grade of steel or a type of fastener. The full designations for these grades are listed in References 20–24 and 27. Reference 25 is the specification for nuts most applicable to the listed bolt grades. Numerous grades are described. Reference 10 contains tables showing the commercially available grades of nuts for each style of bolt.

TABLE 19–1 SAE Grades of Steels for Fasteners

Grade number	Bolt size (in)	Tensile strength (ksi)	Yield strength (ksi)	Proof strength (ksi)	Head marking
1	1/4–1$\frac{1}{2}$	60	36	33	None
2	1/4–3/4	74	57	55	None
	>3/4–1$\frac{1}{2}$	60	36	33	
5	1/4–1	120	92	85	
	>1–1$\frac{1}{2}$	105	81	74	
8	1/4–1$\frac{1}{2}$	150	130	120	

TABLE 19–2 ASTM Grades for Bolt Steels

ASTM grade	Bolt size (in)	Tensile strength (ksi)	Yield strength (ksi)	Proof strength (ksi)	Head marking
A307	1/4–4	60	(Not reported)		None
A325	1/2–1	120	92	85	A 325
	>1–$1\frac{1}{2}$	105	81	74	
A354-BC	$1/4$–$2\frac{1}{2}$	125	109	105	BC
A354-BD	$1/4$–$2\frac{1}{2}$	150	130	120	
A449	1/4–1	120	92	85	
	>1–$1\frac{1}{2}$	105	81	74	
	$>1\frac{1}{2}$–3	95	58	55	
A490	1/2–1	150	130	120	A 490
A574	0.060–1/2	180		140	(Socket head cap screws)
	5/8–4	170		135	

TABLE 19–3 Metric Grades of Steels for Bolts

Grade	Bolt size	Tensile strength (MPa)	Yield strength (MPa)	Proof strength (MPa)
4.6	M5–M36	400	240	225
4.8	M1.6–M16	420	340[a]	310
5.8	M5–M24	520	415[a]	380
8.8	M17–M36	830	660	600
9.8	MI.6–M16	900	720[a]	650
10.9	M6–M36	1040	940	830
12.9	M1.6–M36	1220	1100	970

[a] Yield strengths are approximate and are not included in the standard.

Examples of metric grades of steels used for bolts are shown in Table 19–3, with data extracted from a variety of sources including References 1, 2, 19, and 28. Note the significantly different system for denoting the grades.

Metric bolts and screws use a numerical code system ranging from 4.6 to 12.9, with higher numbers indicating higher strengths. The numbers before the decimal point are approximately 0.01 times the tensile strength of the material in MPa. The last digit with the decimal point is the approximate ratio of the yield strength of the material to the tensile strength.

Approximate Equivalencies among SAE, ASTM, and Metric Grades of Bolt Steels. The following list shows approximate equivalents that may be useful when comparing designs for which specifications include combinations of SAE, ASTM, and metric grades of bolt steels. The individual standards should be consulted for specific strength data.

SAE grade	ASTM grade	Metric grade
J429 Grade 1	A307 Grade A	Grade 4.6
J429 Grade 2	______	Grade 5.8
J429 Grade 5	A449, A325	Grade 8.8
J429 Grade 8	A354 Grade BD, A490	Grade 10.9
	A574	Grade 12.9

Listed below are some of the more commonly used bolt steels according to Reference 10.

U.S. units	SI units
SAE J429, grades 5 and higher	SAE J1199 and ISO, grades 8.8, 10.9, and 12.9
ASTM A325, A384, A449, A490	ASTM A325M, A490M, A574M

The company listed in Internet site 9 provides high-strength heat-treated alloy steel socket head cap screws having the following strengths:

Size range	Tensile strength (ksi)	Tensile strength (MPa)	Yield strength (ksi)	Yield strength (MPa)
0–5/8	190	1310	170	1172
3/4–3	180	1241	155	1069

Roughly equivalent performance is obtained from metric socket head cap screws made to the metric strength grade 12.9. The same geometry is available in corrosion-resistant stainless steel, typically 19–8, at somewhat lower strength levels. See Internet site 11.

Aluminum is used for its corrosion resistance, light weight, and fair strength level. Its good thermal and electrical conductivity may also be desirable. The most widely used alloys are 2024-T4, 6061-T6, and 7075-T73. Properties of these materials are listed in Appendix 9.

Brass, copper, and *bronze* are also used for their corrosion resistance. Ease of machining and an attractive appearance are also advantages. Certain alloys are particularly good for resistance to corrosion in marine applications. See Appendix 12.

Nickel and its alloys, such as *Monel* and *Inconel* (from the International Nickel Company), provide good performance at elevated temperatures while also having good corrosion resistance, toughness at low temperatures, and an attractive appearance. See Appendix 11–1.

Stainless steels are used primarily for their corrosion resistance. Alloys used for fasteners include 19–8, 410, 416, 430, and 431. In addition, stainless steels in the 300 series are nonmagnetic. See Appendix 6 for properties.

A high strength-to-weight ratio is the chief advantage of *titanium* alloys used for fasteners in aerospace applications. Appendix 11–2 gives a list of properties of several alloys.

Plastics are used widely because of their light weight, corrosion resistance, insulating ability, and ease of manufacture. Nylon 6/6 is the most frequently used material, but others include ABS, acetal, TFE fluorocarbons, polycarbonate, polyethylene, polypropylene, and polyvinylchloride. Appendix 13 lists several plastics and their properties. In addition to being used in screws and bolts, plastics are used extensively where the fastener is designed specially for the particular application. See also Reference 13.

Coatings and *finishes* are provided for metallic fasteners to improve appearance or corrosion resistance. Some also lower the coefficient of friction for more consistent results relating tightening torque to clamping force. Steel fasteners can be finished with black oxide, bluing, bright nickel, phosphate, and hot-dip zinc. Plating can be used to deposit cadmium, copper, chromium, nickel, silver, tin, and zinc. Various paints, lacquers, and chromate finishes are also used. Aluminum is usually anodized. Check environmental hazards for coatings and finishes.

19–3 THREAD DESIGNATIONS AND STRESS AREA

Table 19–4 shows pertinent dimensions for threads in the American Standard styles, and Table 19–5 gives SI metric styles. For consideration of strength and size, the

TABLE 19–4 American Standard Screw Threads

A. American Standard thread dimensions, numbered sizes

Basic major diameter, D (in)	Coarse threads: UNC		Fine threads: UNF	
	Size—Threads per inch, n	Tensile stress area (in^2)	Size—Threads per inch, n	Tensile stress area (in^2)
0.0600	——	——	0–80	0.001 80
0.0730	1–64	0.002 63	1–72	0.002 78
0.0860	2–56	0.003 70	2–64	0.003 94
0.0990	3–48	0.004 87	3–56	0.005 23
0.1120	4–40	0.006 04	4–48	0.006 61
0.1250	5–40	0.007 96	5–44	0.008 30
0.1380	6–32	0.009 09	6–40	0.010 15
0.1640	8–32	0.0140	8–36	0.014 74
0.1900	10–24	0.0175	10–32	0.0200
0.2160	12–24	0.0242	12–28	0.0258

B. American Standard thread dimensions, fractional sizes

Basic major diameter, D (in)	Coarse threads: UNC		Fine threads: UNF	
	Size—Threads per inch, n	Tensile stress area (in^2)	Size—Threads per inch, n	Tensile stress area (in^2)
0.2500	1/4–20	0.0318	1/4–28	0.0364
0.3125	5/16–18	0.0524	5/16–24	0.0580
0.3750	3/8–16	0.0775	3/8–24	0.0878
0.4375	7/16–14	0.1063	7/16–20	0.1187
0.5000	1/2–13	0.1419	1/2–20	0.1599
0.5625	9/16–12	0.182	9/16–18	0.203
0.6250	5/8–11	0.226	5/8–18	0.256
0.7500	3/4–10	0.334	3/4–16	0.373
0.8750	7/8–9	0.462	7/8–14	0.509
1.000	1–8	0.606	1–12	0.663
1.125	$1\frac{1}{8}$–7	0.763	$1\frac{1}{8}$–12	0.856
1.250	$1\frac{1}{4}$–7	0.969	$1\frac{1}{4}$–12	1.073
1.375	$1\frac{3}{8}$–6	1.155	$1\frac{3}{8}$–12	1.315
1.500	$1\frac{1}{2}$–6	1.405	$1\frac{1}{2}$–12	1.581
1.750	$1\frac{3}{4}$–5	1.90		
2.000	2–$4\frac{1}{2}$	2.50		

TABLE 19–5 Metric Sizes of Screw Threads

	Coarse threads		Fine threads	
Basic major diameter, D (mm)	Basic thread designation MD × Pitch (mm) (mm)	Tensile stress area (mm^2)	MD × Pitch (mm) (mm)	Tensile stress area (mm^2)
1	M1 × 0.25	0.460	—	—
1.6	M1.6 × 0.35	1.27	M16 × 0.20	1.57
2	M2 × 0.4	2.07	M2 × 0.25	2.45
2.5	M2.5 × 0.45	3.39	M25 × 0.35	3.70
3	M3 × 0.5	5.03	M3 × 0.35	5.61
4	M4 × 0.7	8.78	M4 × 0.5	9.79
5	M5 × 0.8	14.2	M5 × 0.5	16.1
6	M6 × 1	20.1	M6 × 0.75	22.0
8	M8 × 1.25	36.6	M8 × 1	39.2
10	M10 × 1.5	58.0	M10 × 1.25	61.2
12	M12 × 1.75	84.3	M12 × 1.25	92.1
16	M16 × 2	157	M16 × 1.5	167
20	M20 × 2.5	245	M20 × 1.5	272
24	M24 × 3	353	M24 × 2	384
30	M30 × 3.5	561	M30 × 2	621
36	M36 × 4	817	M36 × 3	865
42	M42 × 4.25	1121		
48	M48 × 5	1473		

designer must know the basic major diameter, the pitch of the threads, and the area available to resist tensile loads. Note that the pitch is equal to $1/n$, where n is the number of threads per inch in the American Standard system. In the SI, the pitch in millimeters is designated directly. The tensile stress area listed in Tables 19–4 and 19–5 takes into account the actual area cut by a transverse plane. Because of the helical path of the thread on the screw, such a plane will cut near the root on one side of the screw but will cut near the major diameter on the other. The equation for the tensile stress area for American Standard threads is

➪ Tensile Stress Area for UNC or UNF Threads

$$A_t = (0.7854)[D - (0.9743)p]^2 \qquad \textbf{(19–1)}$$

where D = major diameter

p = pitch of the thread

For metric threads, the tensile stress area is

➪ Tensile Stress Area for Metric Threads

$$A_t = (0.7854)[D - (0.9382)p]^2 \qquad \textbf{(19–2)}$$

For most standard screw thread sizes, at least two pitches are available: the *coarse* series and the *fine thread* series. Both are included in Tables 19–4 and 19–5.

The smaller American Standard threads use a number designation from 0 to 12. The corresponding major diameter is listed in Table 19–4(A). The larger sizes use fractional-inch designations. The decimal equivalent for the major diameter is shown in Table 19–4(B). Metric threads list the major diameter and the pitch in millimeters, as shown in Table 19–5. Samples of the standard designations for a thread are given next.

American Standard: Basic size followed by number of threads per inch and the thread series designation.

10–24 UNC	10–32 UNF
1/2–13 UNC	1/2–20 UNF
$1\frac{1}{2}$–6 UNC	$1\frac{1}{2}$–12 UNF

Metric: M (for "metric"), followed by the basic major diameter and then the pitch in millimeters.

M3 × 0.5 M3 × 0.35 M10 × 1.5

19–4 CLAMPING LOAD AND TIGHTENING OF BOLTED JOINTS

Clamping Load

When a bolt or a screw is used to clamp two parts, the force exerted between the parts is the *clamping load*. The designer is responsible for specifying the clamping load and for ensuring that the fastener is capable of withstanding the load.

The maximum clamping load is often taken to be 0.75 times the proof load, where the proof load is the product of the proof stress times the tensile stress area of the bolt or screw.

Tightening Torque

The clamping load is created in the bolt or the screw by exerting a tightening torque on the nut or on the head of the screw. An approximate relationship between the torque and the axial tensile force in the bolt or screw (the clamping force) is

➪ Tightening Torque

$$T = KDP \tag{19–3}$$

where T = torque, lb·in
D = nominal outside diameter of threads, in
P = clamping load, lb
K = constant dependent on the lubrication present

For average commercial conditions, use $K = 0.15$ if any lubrication at all is present. Even cutting fluids or other residual deposits on the threads will produce conditions consistent with $K = 0.15$. If the threads are well cleaned and dried, $K = 0.20$ is better. Of course, these values are approximate, and variations among seemingly identical assemblies should be expected. Testing and statistical analysis of the results are recommended.

Example Problem 19–1 A set of three bolts is to be used to provide a clamping force of 12 000 lb between two components of a machine. The load is shared equally among the three bolts. Specify suitable bolts, including the grade of the material, if each is to be stressed to 75% of its proof strength. Then compute the required tightening torque.

Solution The load on each screw is to be 4000 lb. Let's specify a bolt made from SAE Grade 5 steel, having a proof strength of 85 000 psi. Then the allowable stress is

$$\sigma_a = 0.75(85\ 000\ \text{psi}) = 63\ 750\ \text{psi}$$

The required tensile stress area for the bolt is then

$$A_t = \frac{\text{load}}{\sigma_a} = \frac{4000\ \text{lb}}{63\ 750\ \text{lb/in}^2} = 0.0627\ \text{in}^2$$

From Table 19–4(B), we find that the 3/8–16 UNC thread has the required tensile stress area. The required tightening torque will be

$$T = KDP = 0.15(0.375\ \text{in})(4000\ \text{lb}) = 225\ \text{lb}\cdot\text{in}$$

Equation (19–3) is adequate for general mechanical design. A more complete analysis of the torque to create a given clamping force requires more information about the joint design. There are three contributors to the torque. One, which we will refer to as T_1, is the torque required to develop the tensile load in the bolt, P_t, using the inclined plane nature of the thread.

$$T_1 = \frac{P_t\, l}{2\pi} = \frac{P_t}{2\pi n} \tag{19–4}$$

where l is the lead of the bolt thread and $l = p = 1/n$.

The second component of the torque, T_2, is that required to overcome friction between the mating threads, computed from,

$$T_2 = \frac{d_p\, \mu_1\, P_t}{2 \cos \alpha} \tag{19–5}$$

where

d_p = pitch diameter of the thread
μ_1 = coefficient of friction between the thread surfaces
α = 1/2 of the thread angle, typically 30°

The third component of the torque, T_3, is the friction between the underside of the head of the bolt or nut and the clamped surface. This friction force is assumed to act at the middle of the friction surface and is computed from

$$T_3 = \frac{(d + b)\, \mu_2\, P_t}{4} \tag{19–6}$$

where

d = major diameter of the bolt
b = outside diameter of the friction surface on the underside of the bolt
μ_2 = coefficient of friction between the bolt head and the clamped surface

The total torque is then,

$$T_{\text{tot}} = T_1 + T_2 + T_3 \tag{19–7}$$

See References 4–8 and 12–16 for additional discussion on torque for bolts. It is important to note that many variables are involved in the contributors to the relationship between the applied torque and the tensile preload given to the bolt. Accurate prediction of the coefficients of friction is difficult. The accuracy with which the specified torque is applied is affected by the precision of the measurement device used, such as a torque wrench, pneumatic nut runner, or hydraulic nut driver, as well as the operator's skill. Reference 4 has an extensive discussion of the large variety of torque wrenches available.

References 4–7, 12, 15, and 16 provide extended methods of calculating tightening torque, performance under fatigue loading, analyzing gasketed joints, joint stiffness, and joint behavior under tensile and shear loading conditions.

Methods similar to those shown in Chapter 5 of this book are used for fatigue analysis using the Goodman approach. The design of the bolted joint should be done using approaches that tend to minimize fatigue problems. Examples are minimizing overall and fluctuating stress levels, increasing the root radius of the thread, using rolled threads instead of cut threads, providing generous fillets between the head and the shank of the bolt, ensuring that the underside of the bolt head is accurately perpendicular to the axis of the threads, providing gradual run-out where threads begin or end on the bolt shaft, reducing bending action, and protecting the bolted joint from corrosion.

Other Methods of Bolt Tightening

Measuring the torque applied to the bolt, screw, or nut during installation is convenient. However, because of the many variables involved, the actual clamping force created may vary significantly. Fastening of critical connections often uses other methods of bolt tightening that more directly relate to the clamping force. Situations where these methods may be used are structural steel connections, flanges for high-pressure systems, nuclear power plant components, cylinder head and connecting rod bolts for engines, aerospace structures, turbine engine components, propulsion systems, and military equipment.

Turn of the Nut Method. The bolt is first tightened to a snug fit to bring all of the parts of the joint into intimate contact. Then the nut is given an additional turn with a wrench of between one-third and one full turn, depending on the size of the bolt. One full turn would produce a stretch in the bolt equal to the lead of the thread, where $l = p = 1/n$. The elastic behavior of the bolt determines the amount of the resulting clamping force. References 4, 8, and 17 give more details.

Tension Control Bolting Products. Special bolts are available that include a carefully sized neck on one end connected to a splined section. The spline is held fixed as the nut is turned. When a predetermined torque is applied to the nut, the neck section breaks and the tightening stops. Consistent connection performance results.

Another form of tension control bolt employs a tool that exerts direct axial tension on the bolt, swages a collar into annular grooves or the threads of the fastener, and then breaks a small-diameter part of the bolt at a predetermined force. The result is a predictable amount of clamping force in the joint. See Internet site 13.

Wavy Flanged Bolt. The underside of the head of this bolt is formed in a wavy pattern during manufacture. When torque is applied to the joint, the wavy surface is deformed to become flat against the clamped surface when the proper amount of tension has been created in the body of the bolt. See Internet site 9.

Direct Tension Indicator (DTI) Washers. The DTI washer has several raised areas on its upper surface. A regular washer is then placed over the DTI washer and a nut tightens the assembly until the raised areas are flattened to a specified degree, creating a predictable tension in the bolt. See Internet site 10.

Ultrasonic Tension Measurement and Control. Recent developments have resulted in the availability of equipment that imparts ultrasonic acoustic waves to bolts as they are tightened with the timing of the reflected waves being correlated to the amount of stretch and tension in the bolt. See Reference 4.

Tighten to Yield Method. Most fasteners are provided with guaranteed yield strength; therefore, it takes a predictable amount of tensile force to cause the bolt to yield. Some automatic systems use this principle by sensing the relationship between applied torque and the rotation of the nut and stopping the process when the bolt begins to yield. During the elastic part of the stress–strain curve for the bolt, a linear change in torque versus rotation occurs. At yield, a dramatic increase in rotation with little or no increase in torque signifies yielding. A variation on this method, called the *logarithmic rate method (LRM)*, determines the peak of the curve of the logarithm of the rate of torque versus turn data and then applies a preset amount of additional turn to the nut. See Internet site 16.

Internet site 12 describes a testing system that aids in analyzing the effectiveness of bolt-tightened joints. A pressure sensitive thin film is placed at the interface between the surfaces to be clamped, somewhat like a gasket. Then the bolted connection is made with bolts tightened to specified values. Subsequent removal of the bolts reveals quantitative and qualitative data about the distribution of clamping pressure on the critical surfaces. The company also offers consulting services on joint design and production methods.

19–5 EXTERNALLY APPLIED FORCE ON A BOLTED JOINT

The analysis shown in Example Problem 19–1 considers the stress in the bolt only under static conditions and only for the clamping load. It was recommended that the tension on the bolt be very high, approximately 75% of the proof load for the bolt. Such a load will use the available strength of the bolt efficiently and will prevent the separation of the connected members.

When a load is applied to a bolted joint over and above the clamping load, special consideration must be given to the behavior of the joint. Initially, the force on the bolt (in tension) is equal to the force on the clamped members (in compression). Then some of the additional load will act to stretch the bolt beyond its length assumed after the clamping load was applied. Another portion will result in a *decrease* in the compressive force in the clamped member. Thus, only part of the applied force is carried by the bolt. The amount is dependent on the relative stiffness of the bolt and the clamped members.

If a stiff bolt is clamping a flexible member, such as a resilient gasket, most of the added force will be taken by the bolt because it takes little force to change the compression in the gasket. In this case, the bolt design must take into account not only the initial clamping force but also the added force.

Conversely, if the bolt is relatively flexible compared with the clamped members, virtually all of the externally applied load will initially go to decreasing the clamping force until the members actually separate, a condition usually interpreted as failure of the joint. Then the bolt will carry the full amount of the external load.

In practical joint design, a situation between the extremes previously described would normally occur. In typical "hard" joints (without a soft gasket), the stiffness of the clamped members is approximately three times that of the bolt. The externally applied load is then shared by the bolt and the clamped members according to their relative stiffnesses as follows:

$$F_b = P + \frac{k_b}{k_b + k_c}F_e \tag{19–8}$$

$$F_c = P - \frac{k_c}{k_b + k_c}F_e \tag{19–9}$$

where F_e = externally applied load
P = initial clamping load [as used in Equation (19–3)]
F_b = final force in bolt
F_c = final force on clamped members
k_b = stiffness of bolt
k_c = stiffness of clamped members

Example Problem 19–2

Assume that the joint described in Example Problem 19–1 was subjected to an additional external load of 3000 lb after the initial clamping load of 4000 lb was applied. Also assume that the stiffness of the clamped members is three times that of the bolt. Compute the force in the bolt, the force in the clamped members, and the final stress in the bolt after the external load is applied.

Solution

We will first use Equations (19–8) and (19–9) with $P = 4000$ lb, $F_e = 3000$ lb, and $k_c = 3k_b$:

$$F_b = P + \frac{k_b}{k_b + k_c}F_e = P + \frac{k_b}{k_b + 3k_b}F_e = P + \frac{k_b}{4k_b}F_e$$

$$F_b = P + F_e/4 = 4000 + 3000/4 = 4750 \text{ lb}$$

$$F_c = P - \frac{k_c}{k_b + k_c}F_e = P - \frac{3k_b}{k_b + 3k_b}F_e = P - \frac{3k_b}{4k_b}F_e$$

$$F_c = P - 3F_e/4 = 4000 - 3(3000)/4 = 1750 \text{ lb}$$

Because F_c is still greater than zero, the joint is still tight. Now the stress in the bolt can be found. For the 3/8–16 bolt, the tensile stress area is 0.0775 in^2. Thus,

$$\sigma = \frac{P}{A_t} = \frac{4750 \text{ lb}}{0.0775 \text{ in}^2} = 61\,300 \text{ psi}$$

The proof strength of the Grade 5 material is 85 000 psi, and this stress is approximately 72% of the proof strength. Therefore, the selected bolt is still safe. But consider what would happen with a relatively "soft" joint, discussed in Example Problem 19–3.

Example Problem 19–3

Solve Example Problem 19–2 again, but assume that the joint has a flexible elastomeric gasket separating the clamping members and that the stiffness of the bolt is then 10 times that of the joint.

Solution

The procedure will be the same as that used previously, but now $k_b = 10k_c$. Thus,

$$F_b = P + \frac{k_b}{k_b + k_c}F_e = P + \frac{10k_c}{10k_c + k_c}F_e = P + \frac{10k_c}{11k_c}F_e$$

$$F_b = P + 10F_e/11 = 4000 + 10(3000)/11 = 6727 \text{ lb}$$

$$F_c = P - \frac{k_c}{k_b + k_c}F_e = P - \frac{k_c}{10k_c + k_c}F_e = P - \frac{k_c}{11k_c}F_e$$

$$F_c = P - F_e/11 = 4000 - 3000/11 = 3727 \text{ lb}$$

The stress in the bolt would be

$$\sigma = \frac{6727 \text{ lb}}{0.0775 \text{ in}^2} = 86\,800 \text{ psi}$$

This exceeds the proof strength of the Grade 5 material and is dangerously close to the yield strength.

19–6 THREAD STRIPPING STRENGTH

In addition to sizing a bolt on the basis of axial tensile stress, the threads must be checked to ensure that they will not be stripped off by shearing. The variables involved in the shear strength of the threads are the materials of the bolt, the nut, or the internal threads of a tapped hole, the length of engagement, L_e, and the size of the threads. The details of analysis depend on the relative strength of the materials.

Internal Thread Material Stronger than Bolt Material. For this case, the strength of the threads of the bolt will control the design. Here, we present an equation for the required length of engagement, L_e, of the bolt threads that will have at least the same strength in shear as the bolt itself does in tension.

$$L_e = \frac{2A_{tB}}{\pi\,(ID_{\text{Nmax}})[0.5 + 0.57735\,n(PD_{\text{Bmin}} - ID_{\text{Nmax}})]} \tag{19–10}$$

where A_{tB} = tensile stress area of bolt
ID_{Nmax} = maximum inside (root) diameter of nut threads
n = number of threads per inch
PD_{Bmin} = minimum pitch diameter of bolt threads

The subscripts B and N refer to the bolt and nut, respectively. The subscripts min and max refer to the minimum and maximum values, respectively, considering the tolerances on thread dimensions. Reference 14 gives data for tolerances as a function of the class of thread specified.

For a given length of engagement, the resulting shear area for the bolt threads is

$$A_{sB} = \pi\, L_e\, ID_{\text{Nmax}}[0.5 + 0.57735\,n\,(PD_{\text{Bmin}} - ID_{\text{Nmax}})] \tag{19–11}$$

Nut Material Weaker than Bolt Material. This is particularly applicable when the bolt is inserted into a tapped hole in cast iron, aluminum, or some other material with relatively low strength. The required length of engagement to develop at least the full strength of the bolt is

$$L_e = \frac{s_{utB}\,(2\,A_{tB})}{s_{utN}\,\pi\,OD_{\text{Bmin}}\,[0.5 + 0.57735\,n\,(OD_{\text{Bmin}} - PD_{\text{Nmax}})]} \tag{19–12}$$

where

s_{utB} = ultimate tensile strength of the bolt material
s_{utN} = ultimate tensile strength of the nut material
OD_{Bmin} = minimum outside diameter of the bolt threads
PD_{Nmax} = maximum pitch diameter of the nut threads

The shear area of the root of the threads of the nut is

$$A_{sN} = \pi L_e OD_{\text{Bmin}}[0.5 + 0.57735\,n\,(OD_{\text{Bmin}} - PD_{\text{Nmax}})] \tag{19–13}$$

Equal Strength for Nut and Bolt Material. For this case failure is predicted as shear of either part at the nominal pitch diameter, PD_{nom}. The required length of engagement to develop at least the full strength of the bolt is

$$L_e = \frac{4\,A_{tB}}{\pi\,PD_{\text{nom}}} \tag{19–14}$$

The shear stress area for the nut or bolt threads is

$$A_s = \pi\,PD_{\text{nom}}\,L_e/2 \tag{19–15}$$

19–7 OTHER TYPES OF FASTENERS AND ACCESSORIES

Most bolts and screws have enlarged heads that bear down on the part to be clamped and thus exert the clamping force. *Set screws* are headless, are inserted into tapped holes, and are designed to bear directly on the mating part, locking it into place. Figure 19–5 shows several styles of points and drive means for set screws. Caution must be used with set screws, as with any threaded fastener, so that vibration does not loosen the screw.

A *washer* may be used under either or both the bolt head and the nut to distribute the clamping load over a wide area and to provide a bearing surface for the relative rotation of the nut. The basic type of washer is the plain flat washer, a flat disc with a hole in it through which the bolt or screw passes. Other styles, called *lockwashers*, have axial deformations or projections that produce axial forces on the fastener when compressed. These forces keep the threads of the mating parts in intimate contact and decrease the probability that the fastener will loosen in service.

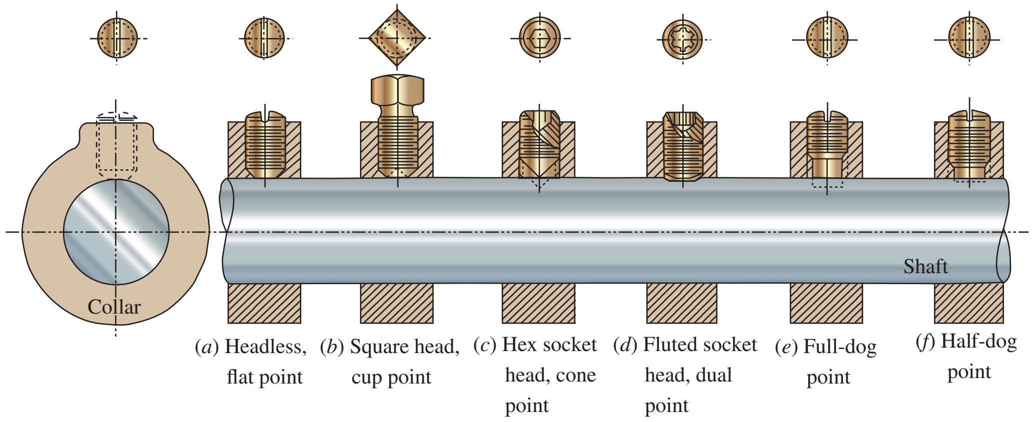


FIGURE 19–5 Set screws with different head and point styles applied to hold a collar on a shaft

Figure 19–6 shows several means of using washers and other types of locking devices. Part (a) is a jam nut tightened against the regular nut. Part (b) is the standard lockwasher. Part (c) is a locking tab that keeps the nut from turning. Part (d) is a cotter inserted through a hole drilled through the bolt. Part (e) uses a cotter, but it also passes through slots in the nut. Part (f) is one of several types of thread-deformation techniques used. Part (g) is an *elastic stop nut* that uses a plastic insert to keep the threads of the nut in tight contact with the bolt. This may be used on machine screws as well. In part (h), the elastic stop nut is riveted to a thin plate, allowing a mating part to be bolted from the opposite side. The thin metal device in (i) bears against the top of the nut and grips the threads, preventing axial motion of the nut.

A *stud* is like a stationary bolt attached permanently to a part of one member to be joined. The mating member is then placed over the stud, and a nut is tightened to clamp the parts together.

Additional variations occur when these types of fasteners are combined with different head styles. Several of these are shown in the figures already discussed. Others are listed next:

Square	Hex	Heavy hex	Hex jam
Hex castle	Hex flat	Hex slotted	12-point
High crown	Low crown	Round	T-head
Pan	Truss	Hex washer	Flat countersunk
Plow	Cross recess	Fillister	Oval countersunk
Hex socket	Spline socket	Button	Binding

Additional combinations are created by consideration of the American National Standards or British Standard (metric); material grades; finishes; thread sizes; lengths; class (tolerance grade); manner of forming heads (machining, forging, and cold heading); and the manner of forming threads (machining, die cutting, tapping, rolling, and plastic molding).

Thus, you can see that comprehensive treatment of threaded fasteners encompasses extensive data. See the References and Internet sites listed at the end of this chapter.

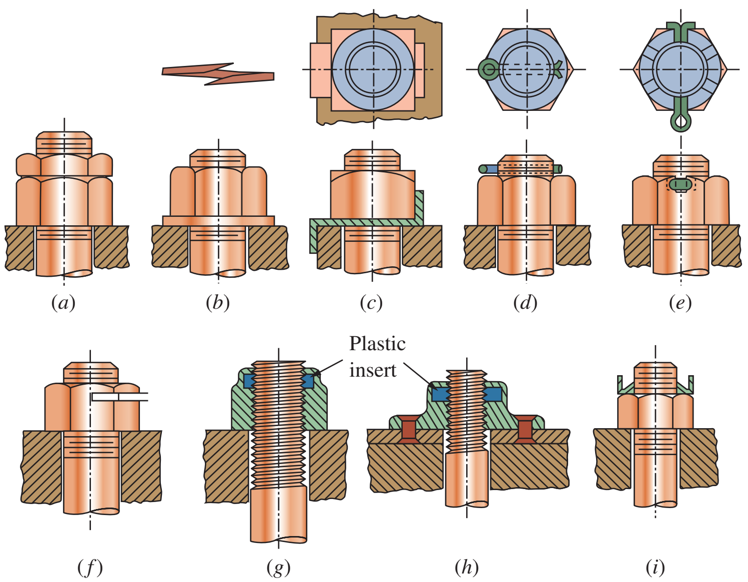


FIGURE 19–6 Locking devices

19–8 OTHER MEANS OF FASTENING AND JOINING

Thus far, this chapter has focused on screws and bolts because of their wide applications. Other types of fastening means will now be discussed.

Rivets are nonthreaded fasteners, usually made of steel or aluminum. They are originally made with one head, and the opposite end is formed after the rivet is inserted through holes in the parts to be joined. Steel rivets are formed hot, whereas aluminum can be formed at room temperatures. Of course, riveted joints are not designed to be assembled more than once. (See Internet sites 3 and 4.)

A large variety of *quick-operating fasteners* is available. Many are of the quarter-turn type, requiring just a 90° rotation to connect or disconnect the fastener. Access panels, hatches, covers, and brackets for removable equipment are attached with such fasteners. Similarly, many varieties of *latches* are available to provide quick action with, perhaps, added holding power. (See Internet sites 3 and 4.)

Welding involves the metallurgical bonding of metals, usually by the application of heat with an electric arc, a gas flame, or electrical resistance heating under heavy pressure. Welding is discussed in Chapter 20.

Brazing and *soldering* use heat to melt a bonding agent that flows into the space between parts to be joined, adhering to both parts and then solidifying as it cools. *Brazing* uses relatively high temperatures, above 840°F (450°C), using alloys of copper, silver, aluminum, silicon, or zinc. Of course, the metals to be joined must have a significantly higher melting temperature. Metals successfully brazed include plain carbon and alloy steels, stainless steels, nickel alloys, copper, aluminum, and magnesium.

Soldering is similar to brazing, except that it is performed at lower temperatures, less than 840°F (450°C). Several soldering alloys of lead-tin, tin-zinc, tin-silver, lead-silver, zinc-cadmium, zinc-aluminum, and others are used. Brazed joints are generally stronger than soldered joints due to the inherently higher strength of the brazing alloys. Most soldered joints are fabricated with interlocking lap joints to provide mechanical strength, and then the solder is used to hold the assembly together and possibly to provide sealing. Joints in piping and tubing are frequently soldered.

Adhesives are seeing wide use. Versatility and ease of application are strong advantages of adhesives used in an array of products from toys and household appliances to automotive and aerospace structures. (See Internet sites 3 and 17.) Some types include the following:

Acrylics: Used for many metals and plastics.

Cyanoacrylates: Very fast curing; flow easily between well-mated surfaces.

Epoxies: Good structural strength; joint is usually rigid. Some require two-part formulations.
A large variety of formulations and properties are available.

Anaerobics: Used for securing nuts and bolts and other joints with small clearances; cures in the absence of oxygen.

Silicones: Flexible adhesive with good high-temperature performance (400°F, 200°C).

Polyester hot melt: Good structural adhesive; easy to apply with special equipment.

Polyurethane: Good bonding; provides a flexible joint.

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INTERNET SITES RELATED TO FASTENERS

1. **Industrial Fasteners Institute (IFI).** An association of manufacturers and suppliers of bolts, nuts, screws, rivets, and special formed parts and the materials and equipment to make them. IFI develops standards, organizes research, and conducts education programs related to the fasteners industry.
2. **Research Council on Structural Connections (RCSC).** An organization that stimulates and supports research on structural connections, prepares and publishes standards, and conducts educational programs. See Reference 12.
3. **Accurate Fasteners, Inc.** A supplier of bolts, cap screws, nuts, rivets, and numerous other types of fasteners for general industry uses. Also provides adhesives for structural and nonstructural applications, thread locking adhesives, and tapes.
4. **The Fastener Group.** A supplier of bolts, cap screws, nuts, rivets, and numerous other types of fasteners for military and general industrial uses.
5. **Haydon Bolts, Inc**. Manufacturer of bolts, nuts, and numerous other types of fasteners for the construction industry.
6. **Nucor Fastener Division.** Manufacturer of hex head cap screws in ASME, ASTM, and metric grades, hex nuts, and structural bolts, nuts and washers.
7. **Nylok Fastener Corporation.** Manufacturer of Nylok® self-locking fasteners for automotive, aerospace, consumer products, agricultural, industrial, furniture, and many other applications.
8. **Phillips Screw Company.** Developer of the Phillips® screwdriver. Manufacturer of related fasteners for the aerospace, automotive, and industrial markets.
9. **Unbrako Group.** Manufacturer of high-strength socket head cap screws and related products for industrial, aerospace, automotive, and other applications under the Unbrako and Durlok brands. Part of Deepak Fasteners, Ltd. Engineering guide available on the website.
10. **St. Louis Screw & Bolt Company.** Manufacturer of bolts, nuts, and washers to ASTM standards for the construction industry.
11. **Acument® Global Technologies.** Provider of fasteners and related products to the automotive, electronics, and other manufacturing industries under the Camrail®, Camcar® and Ring Screw brands. Products include cap screws, thread forming self-tapping screws, the Torx Plus® drive system, and other related products.
12. **Sensor Products, Inc.** Provider of tactile thin film surface pressure and force sensors useful in designing and verifying the effectiveness of bolted joints, gaskets, seals, and related applications. Also offers consulting services to clients.
13. **Alcoa Fastening Systems/Huck Fasteners.** Manufacturer of Huck LockBolts and structural blind fasteners for the truck, trailer, agricultural, HVAC, aerospace, industrial equipment, and other markets, often replacing welding or traditional nut/bolt fastening methods.
14. **ASTM International.** A standards-setting organization with numerous standards for engineering materials, fasteners, and associated products.